

Fundamentals of Neurobiology

4BI036

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 Centre for Neuroscience - NeuroSU, IBPS

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regis.lambert@sorbonne-universite.fr

ALL informations will be on the MOODLE page

- *Neuroscience: Exploring the Brain* – Bear, Connors & Paradiso
<https://seti.net/Neuron%20Lab/NeuronReferences/Neuroscience%20-%20Bear.pdf>
- *Principles of Neural Science* – Kandel et al.
https://archive.org/details/isbn_9780838577011
- *Fundamental Neuroscience* – Squire, Berg, Bloom, et al.
- *Neuroscience* – Purves, Augustine, Fitzpatrick, et al.
- *Harvard’s Fundamentals of Neuroscience* (free edX course)

Date	Time	Topics	Speaker	Room/Amphi
18/09/2026	8.30-10.30			24.34.207
	10.45-12.45	Learning & Memory	S.Daumas	24.34.207
25/09/2026	8.30-10.30	Human Memory	P.Fossati	24.34.207
	10.45-12.45	NeuroPhy-1	R.Lambert	24.34.207
2/10/2026	8.30-10.30	NeuroPhy-2	R.Lambert	24.34.207
	10.45-12.45	NeuroPhy-3	R.Lambert	24.34.207
9/10/2026	8.30-10.30	NeuroPhy-4	R.Lambert	AMPHI.45B
	10.45-12.45	Modelling pathologies	H. Cheval	AMPHI.DURAND
16/10/2026	8.30-10.30	NeuroPhy-5	R.Lambert	24.34.207
	10.45-12.45	Decision Making	J.Naudé	24.34.207
23/10/2026	8.30-10.30	Neurodeg diseases	N.Villain	24.34.207
	10.45-12.45	Social cognition	M.Giurfa	24.34.207

TD & Practicals:

Date	Time
Thursday 15/10/2026	1.45-6 pm
Thursday 22/10/2026	1.45-6 pm
Friday 6/11/2026	8.30-12.45 am

Exact date to be determined depending on the groups of students

Memories & Emotions

4BI036- *Fundamentals of Neurobiology* - 2026/27



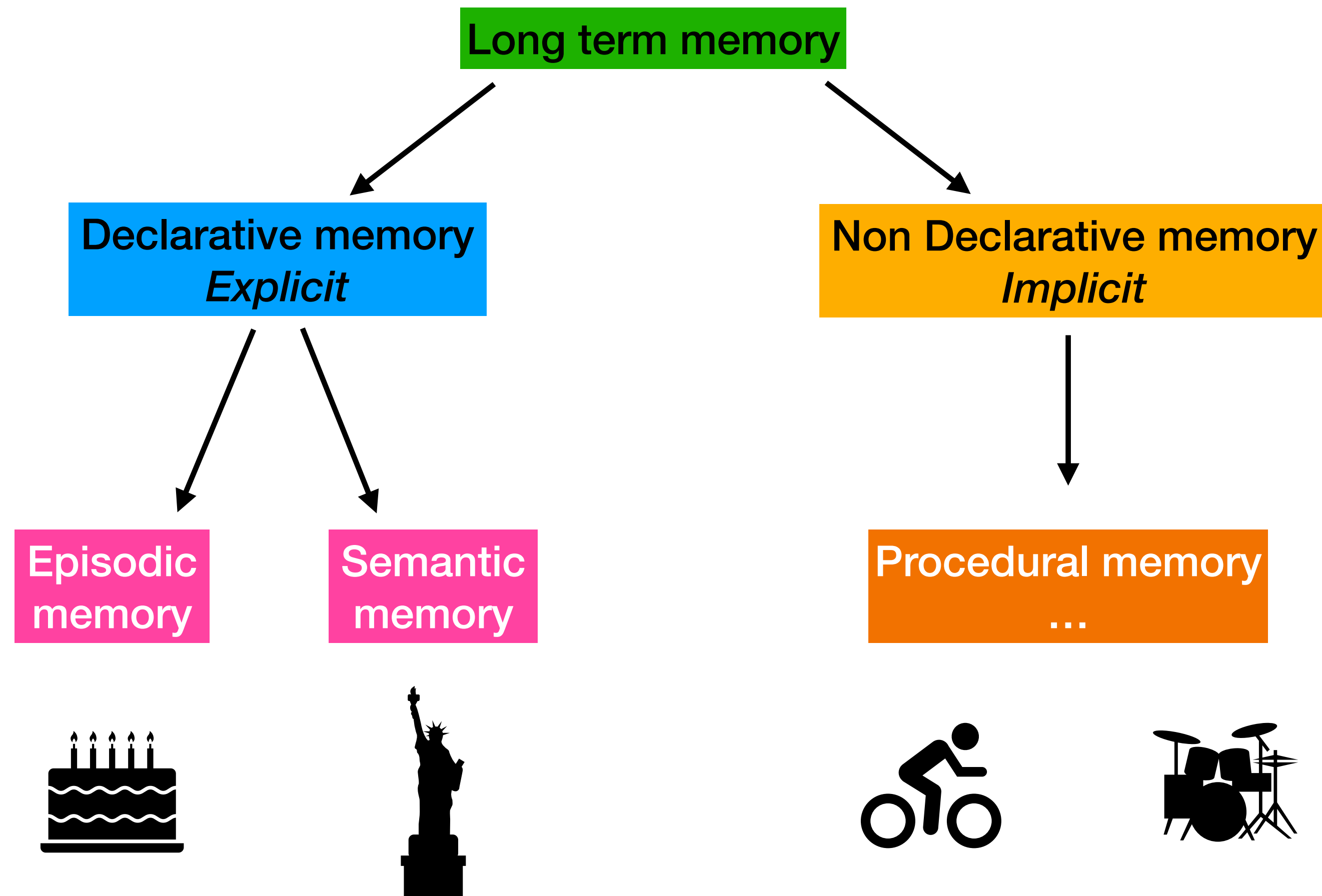
Prof. Stéphanie Daumas

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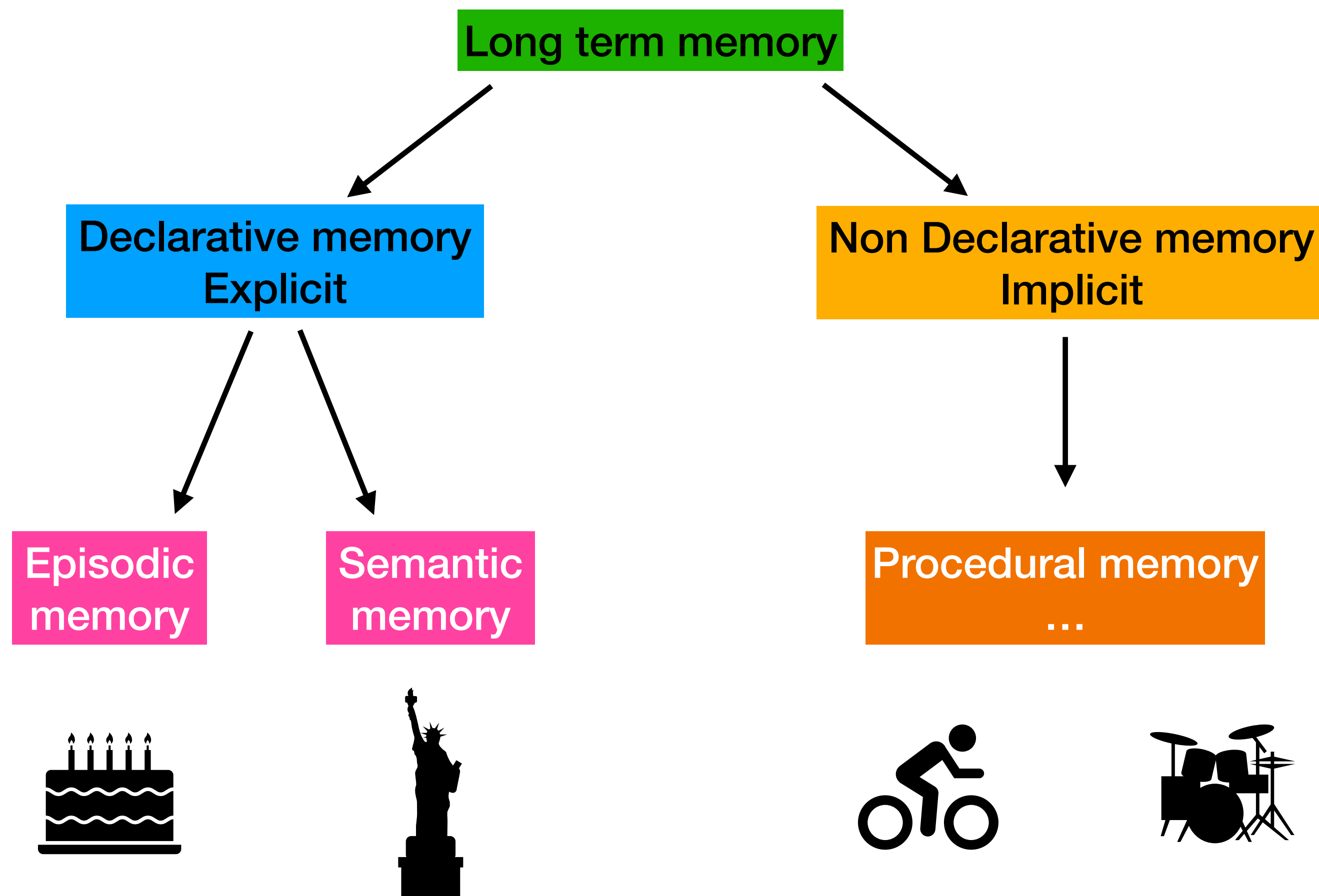
Memory or Memories

Classification



Memory or Memories

Classification



Long-term memory

- Declarative - Explicit: **Knowing that**
(conscious awareness)

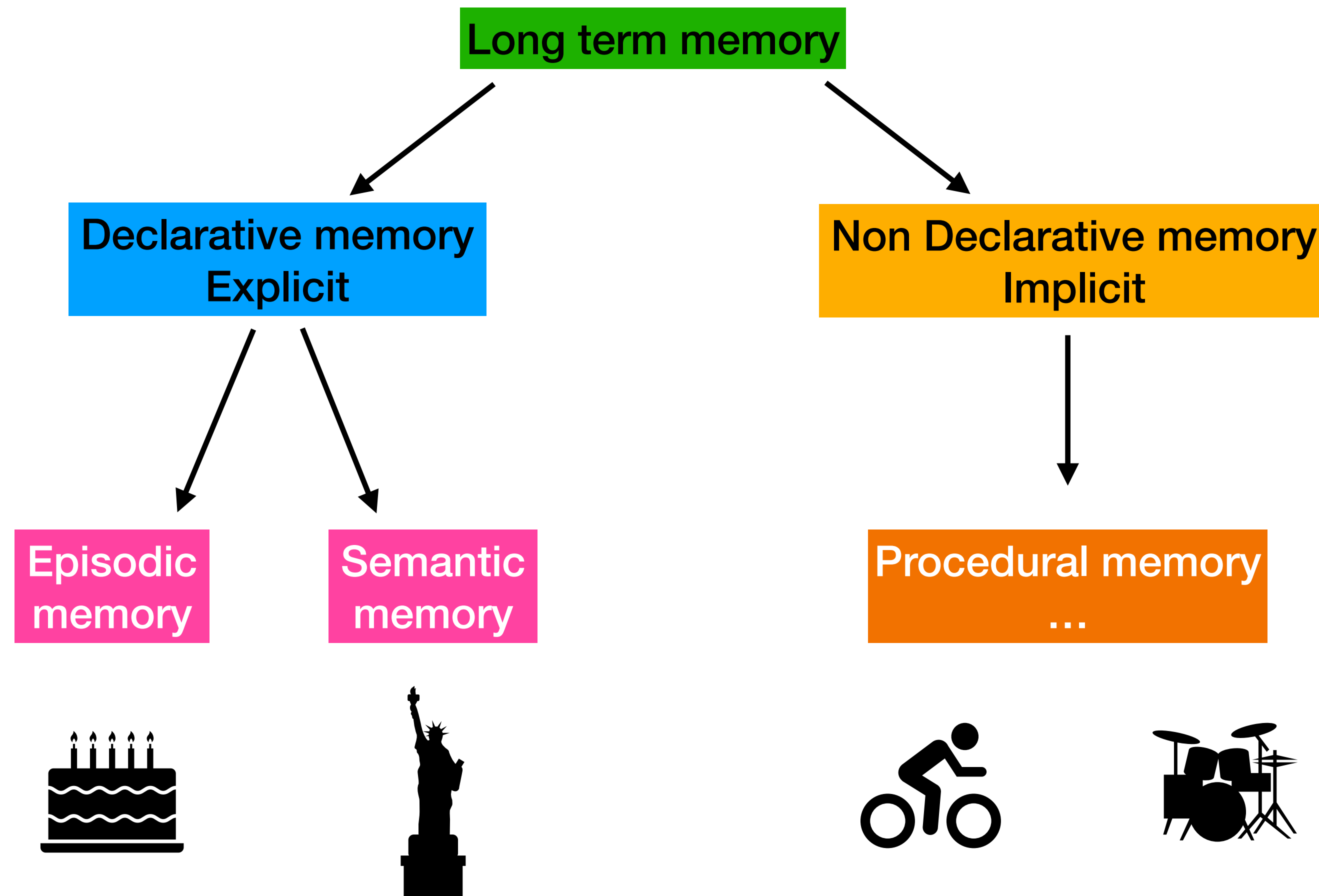
- **Semantic memory**: general knowledge
- **Episodic memory**: episodes of your life, embedded in a given spatio-temporal context

- Non-Declarative - Implicit: **Knowing how**
(unconscious awareness)

- **Procedural**: skills and motor movements...

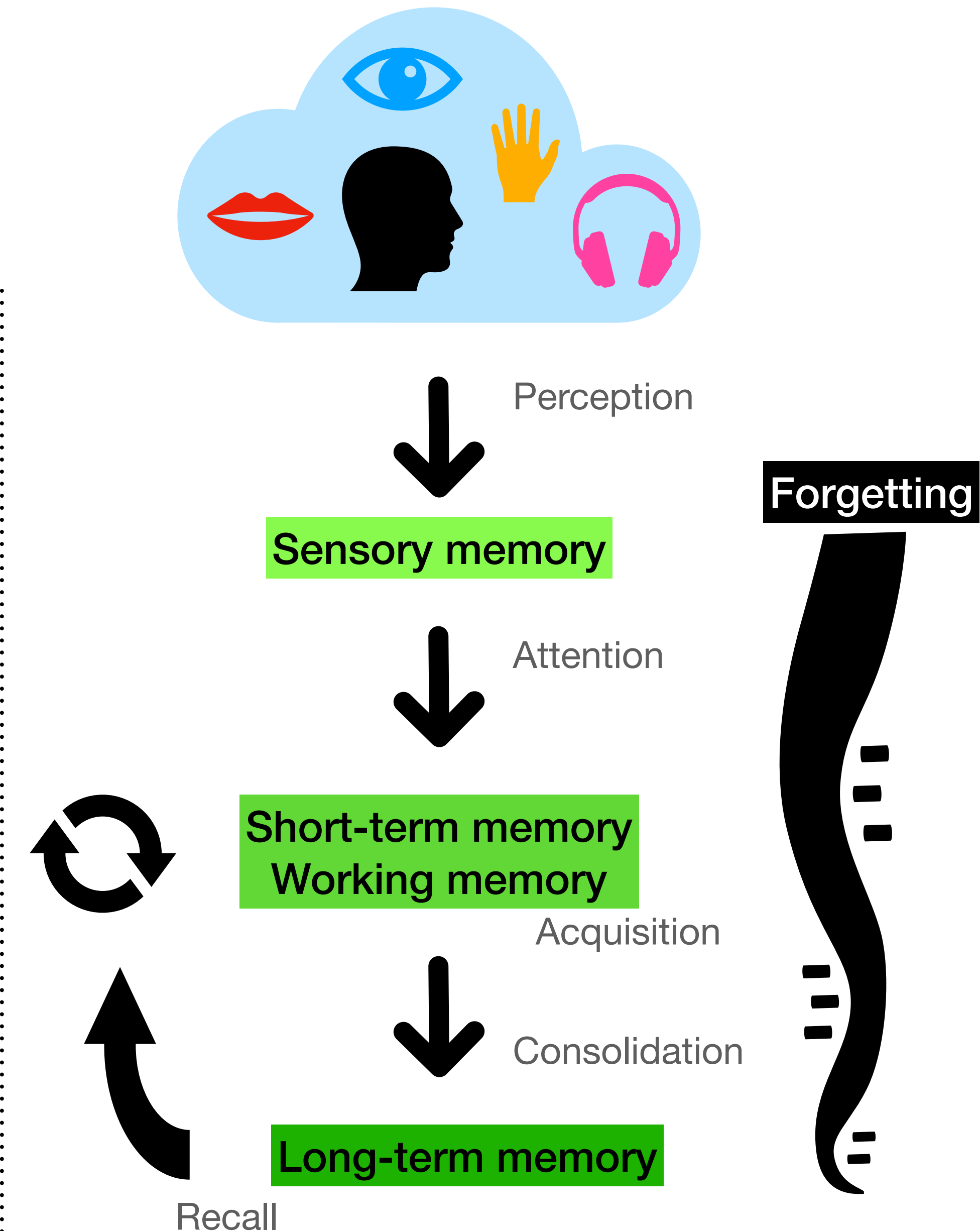
Memory or Memories

Classification



From Squire

5



From Atkinson & Shiffrin

Memory or Memories

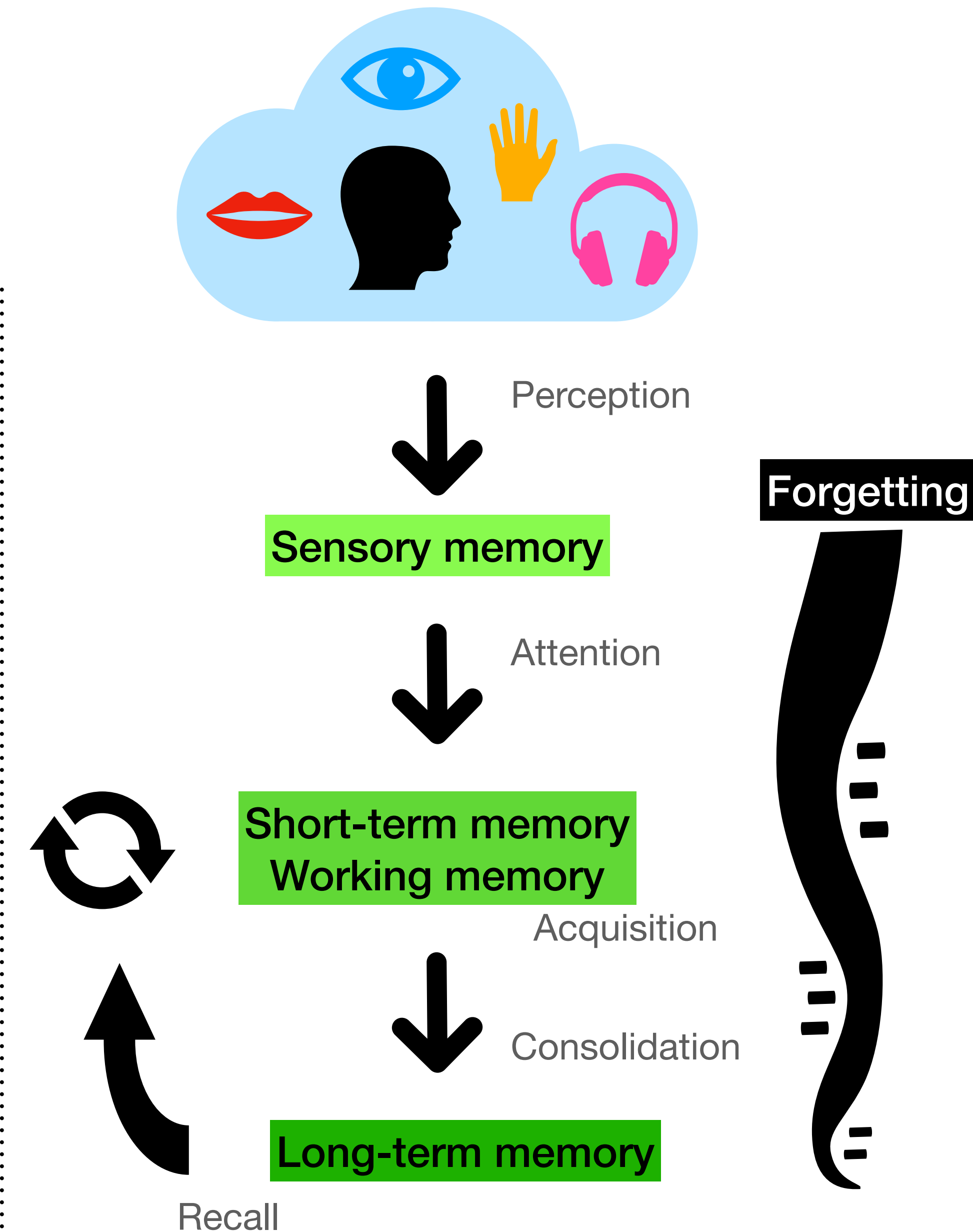
Classification

Short-term memory labile vs. long-term memory stable



3 steps to long-term memory:

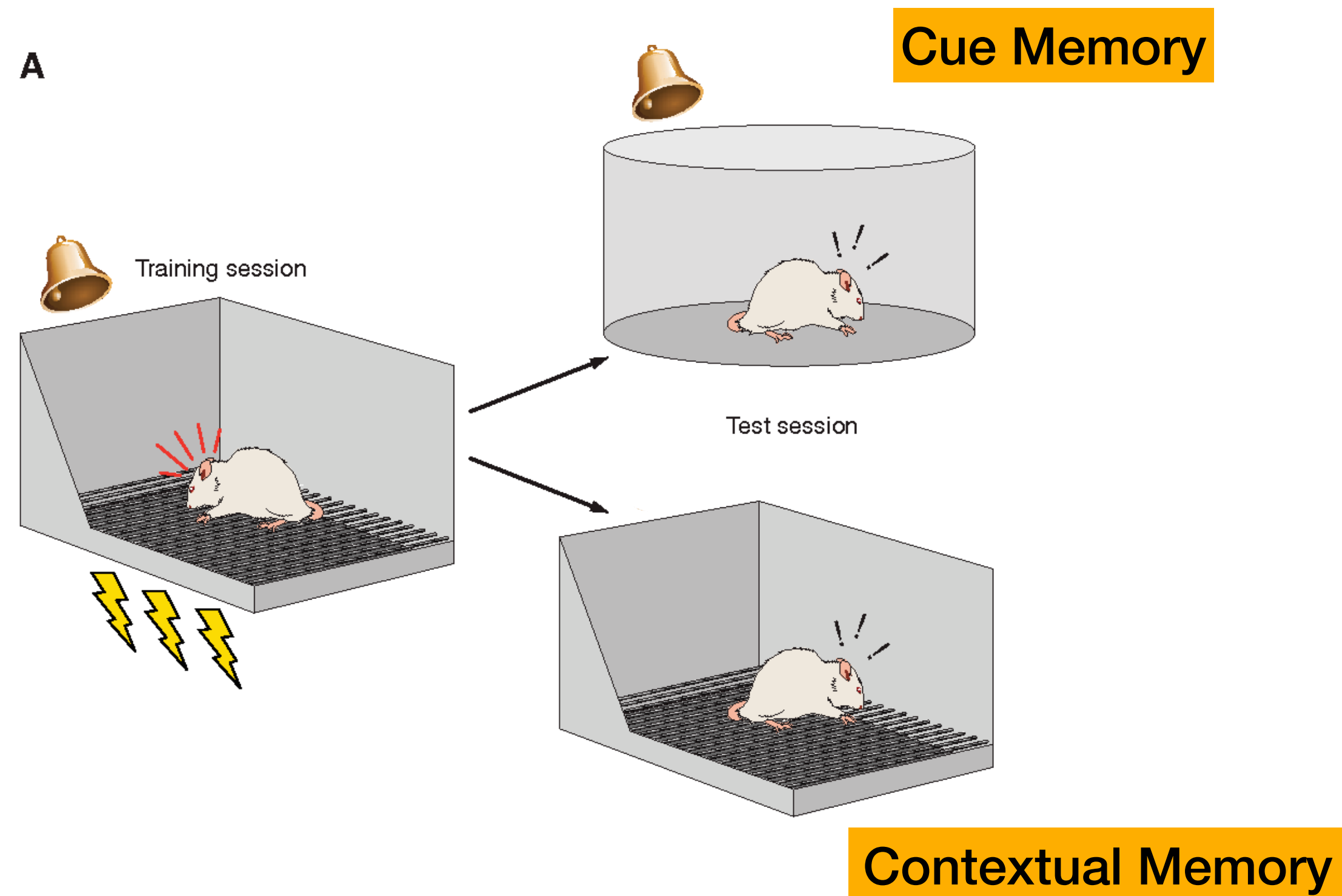
- Acquisition (Encoding)
- Consolidation
- Recall



Aversive memory

Fear memory

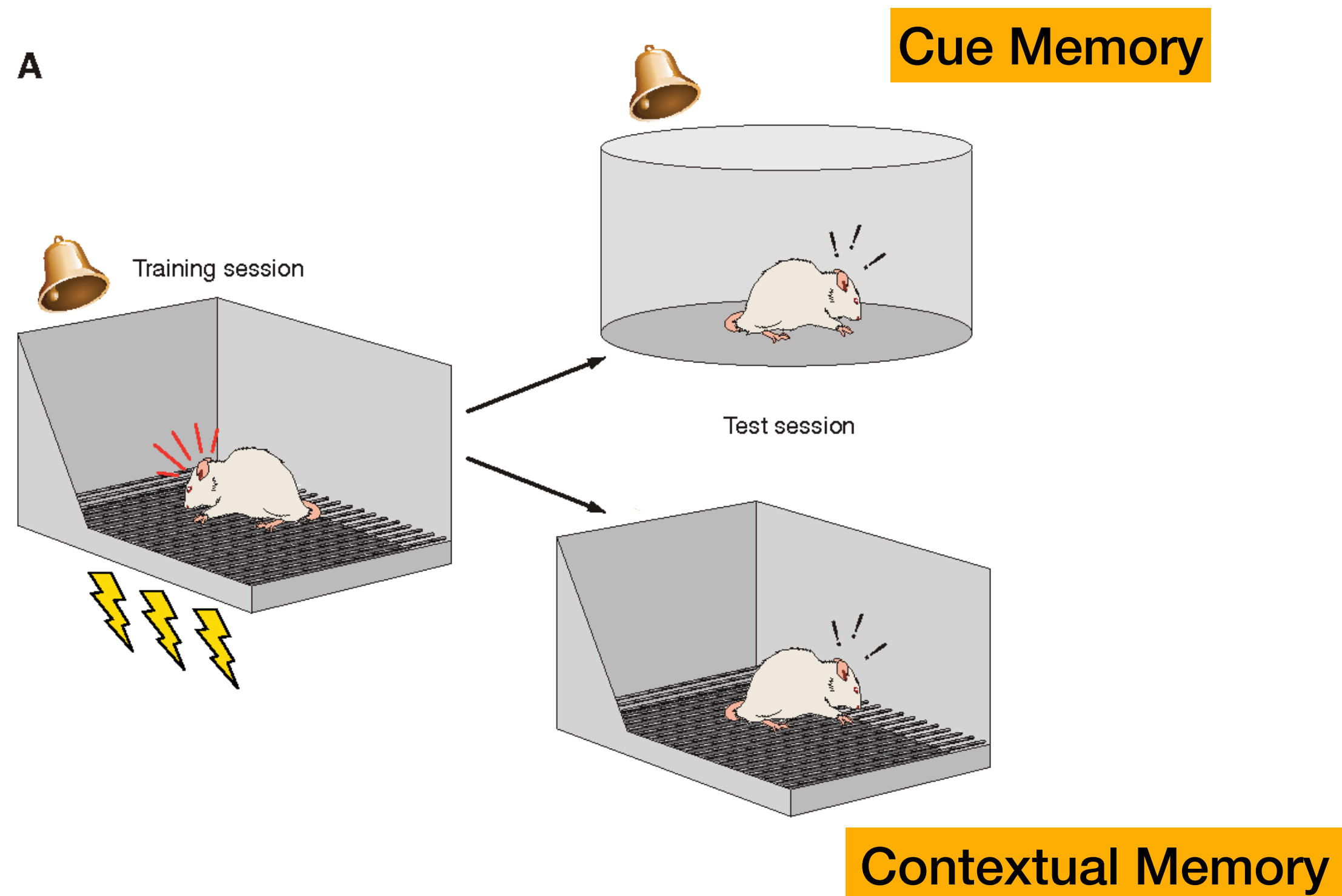
Fear conditioning in rodents



Aversive memory

Fear memory

Fear conditioning in rodents



Pavlovian/classical conditioning:

innocuous Conditional Stimulus: CS

- Sound : unimodal stimulus
- Context : multimodal stimulus

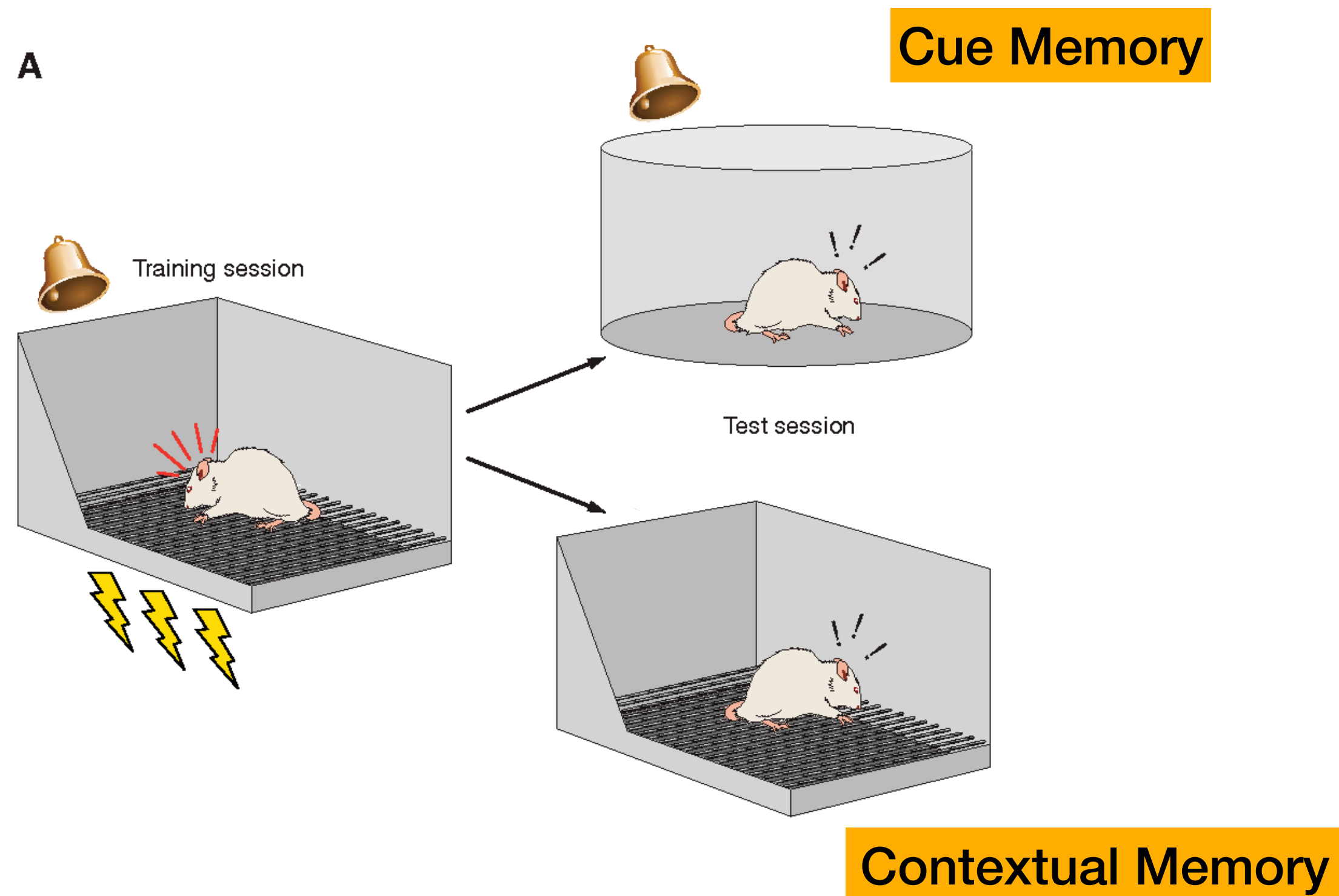
noxious Unconditional Stimulus : US

- Electric shock: aversive/traumatic stimulus *per se*

Aversive memory

Fear memory

Fear conditioning in rodents



Pavlovian/classical conditioning:

innocuous Conditional Stimulus: CS

- Sound : unimodal stimulus
- Context : multimodal stimulus

noxious Unconditional Stimulus : US

- Electric shock: aversive/traumatic stimulus *per se*

Unconditional Response: UR

- Freezing : innate fear response

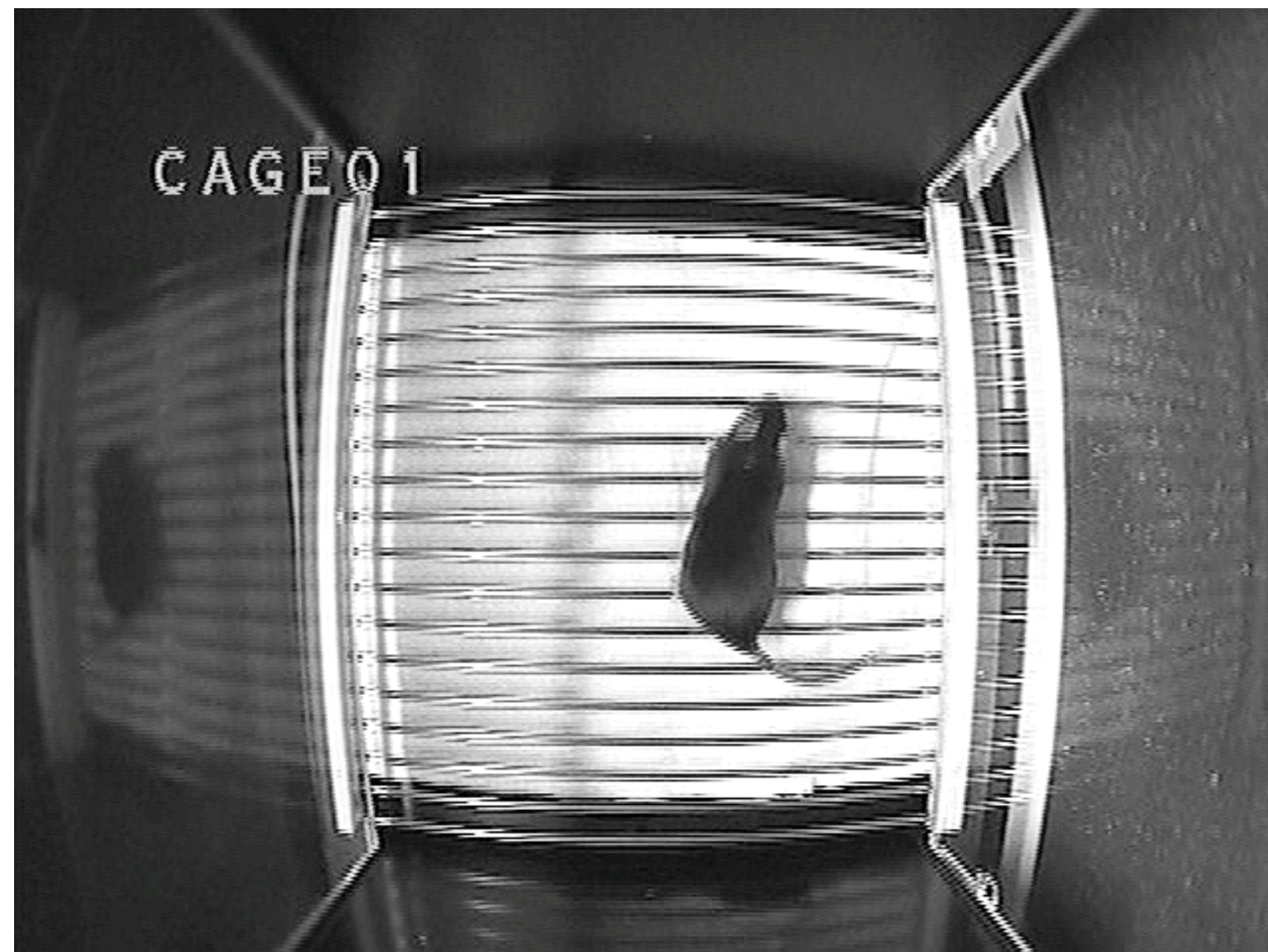
Conditioned Response: CR

- Freezing : acquired fear (to sound or context)
 - *Cue memory*
 - *Contextual memory*

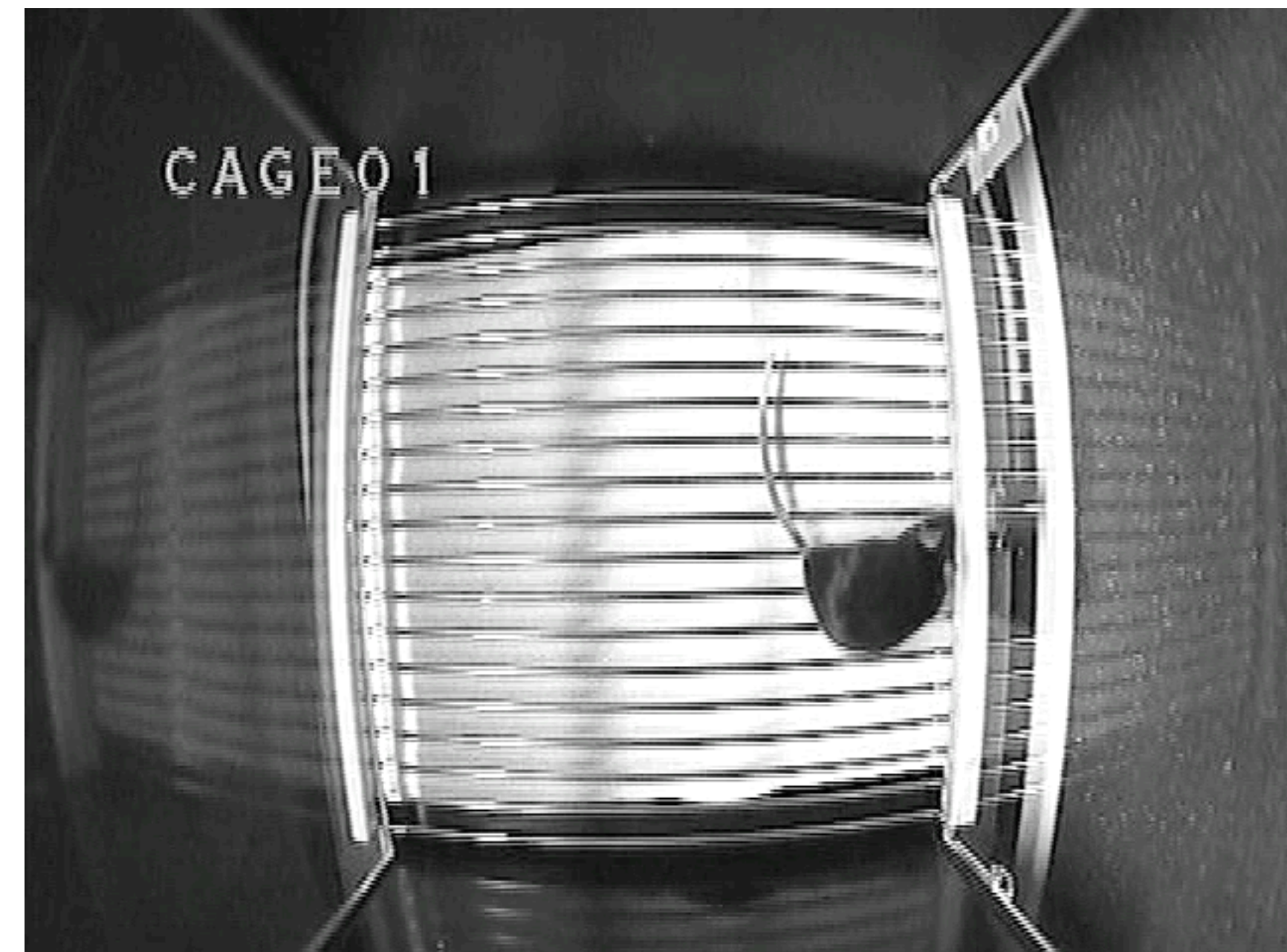
Aversive memory

Fear memory

Before Conditioning

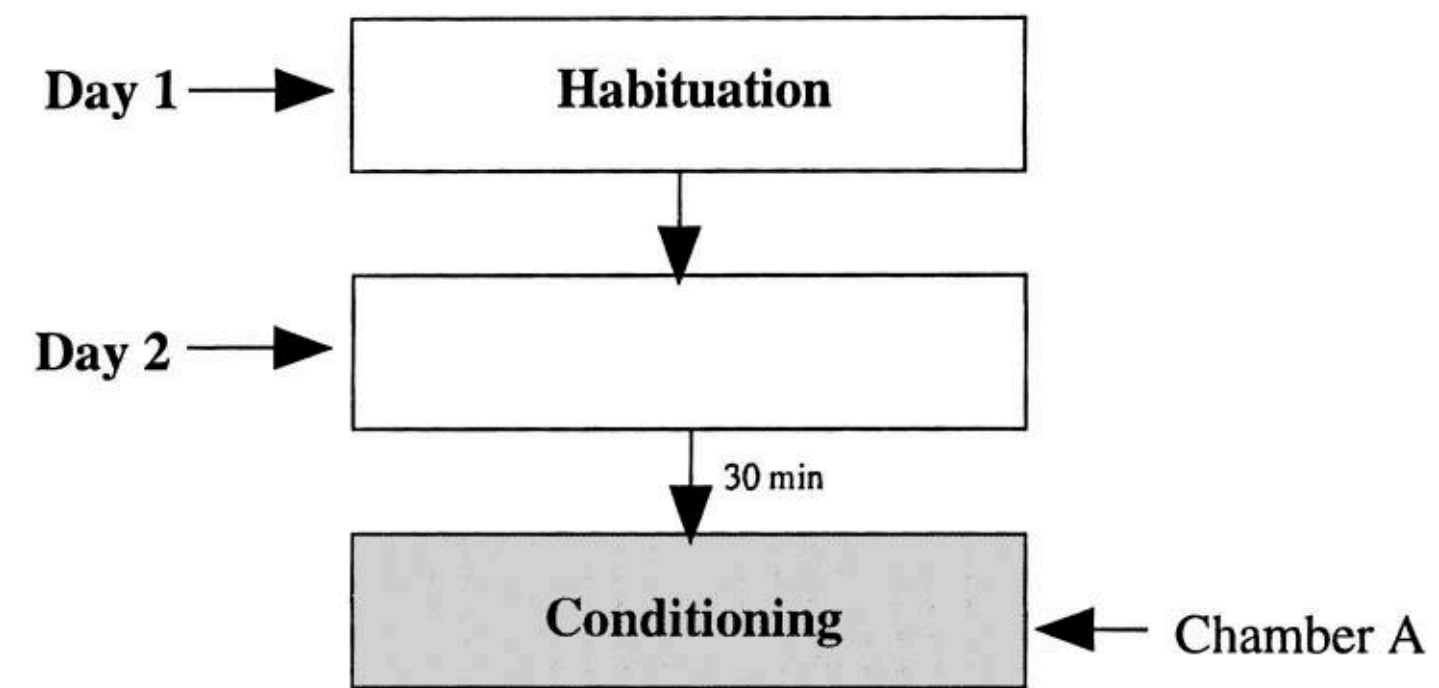


After Conditioning

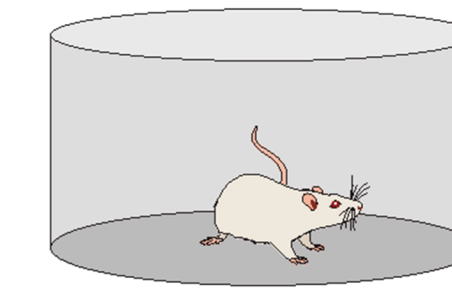


From Schafe et al, 1999

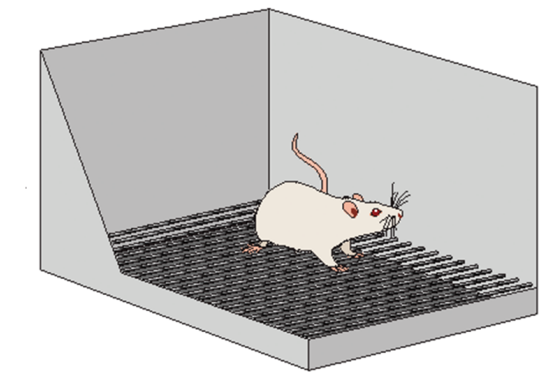
The role of *protein synthesis* in fear conditioning to sound and context in rats



Day 1: Habituation

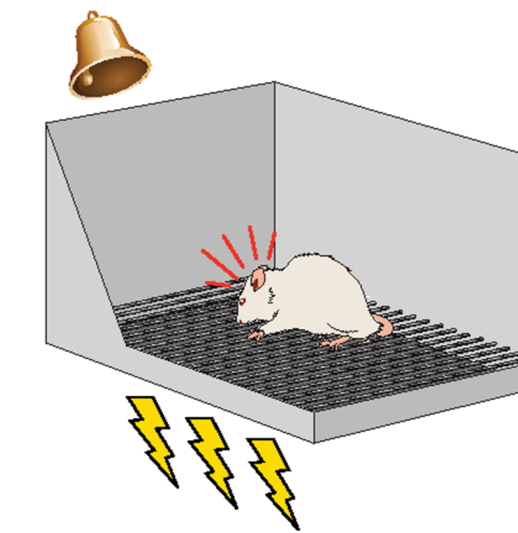


Chamber B



Chamber A

Day 2: Conditioning



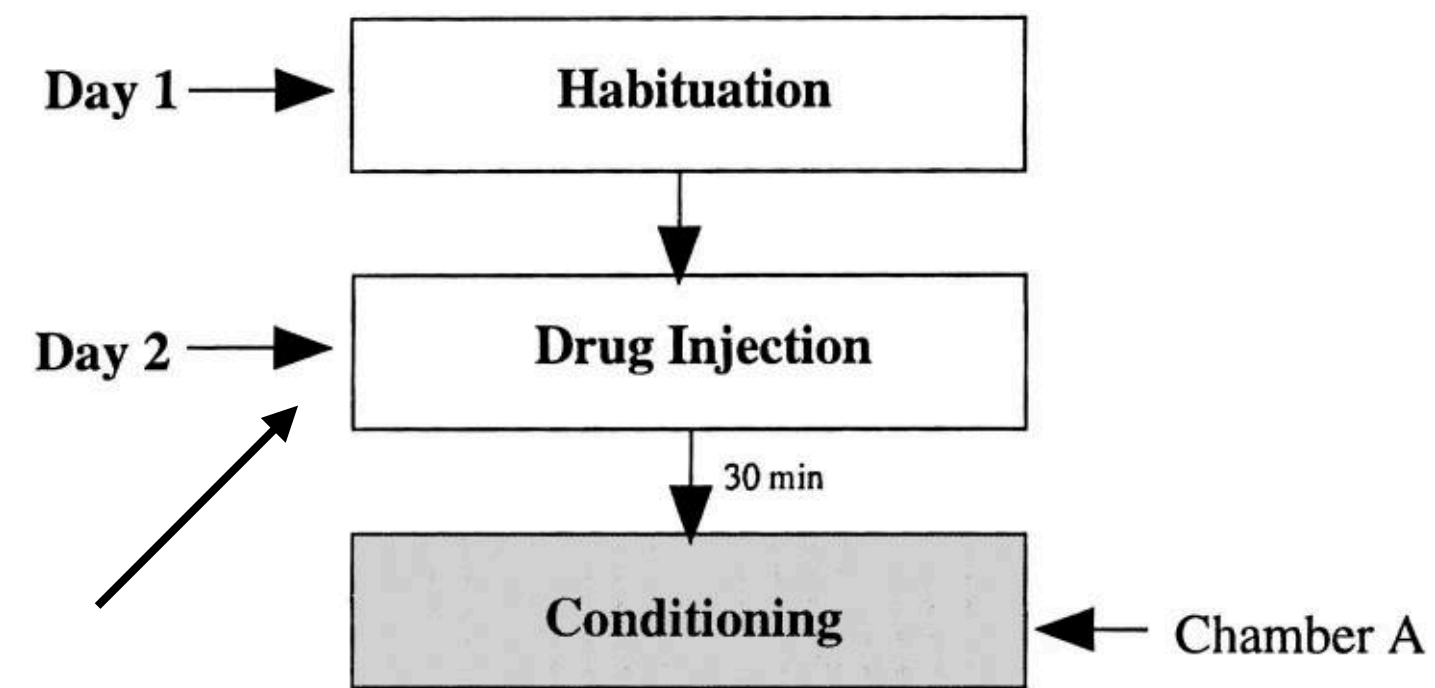
Son: 30sec, 70dB 5kHz
Choc: 1sec, 1,5mA

From Schafe et al, 1999

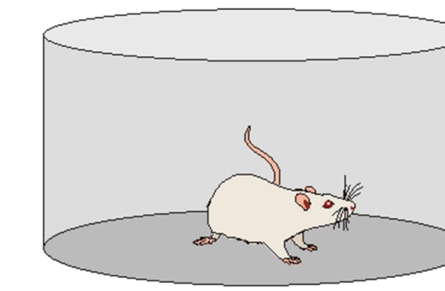
The role of *protein synthesis* in fear conditioning to sound and context in rats

*intra-ventricular injection of **anisomycin***

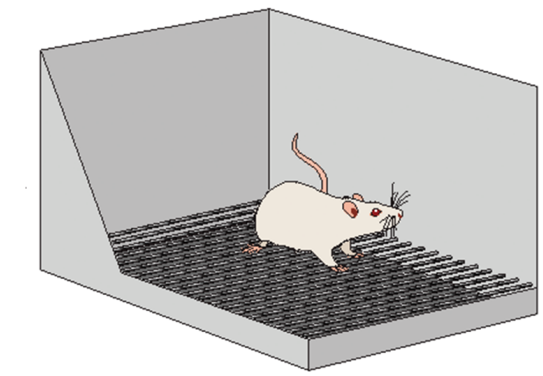
Anisomycin : *protein synthesis inhibitor*



Day 1: Habituation

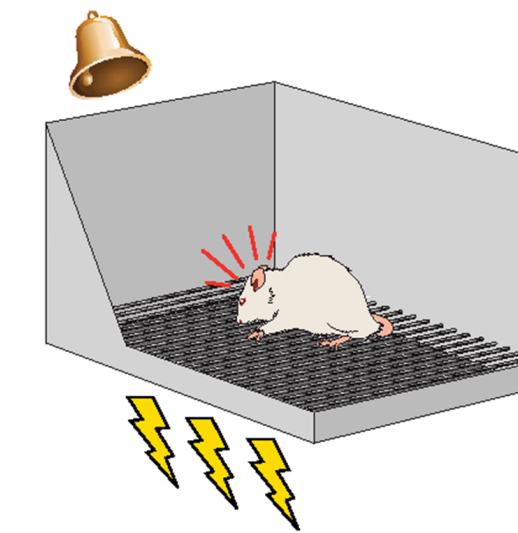


Chamber B



Chamber A

Day 2: Conditioning



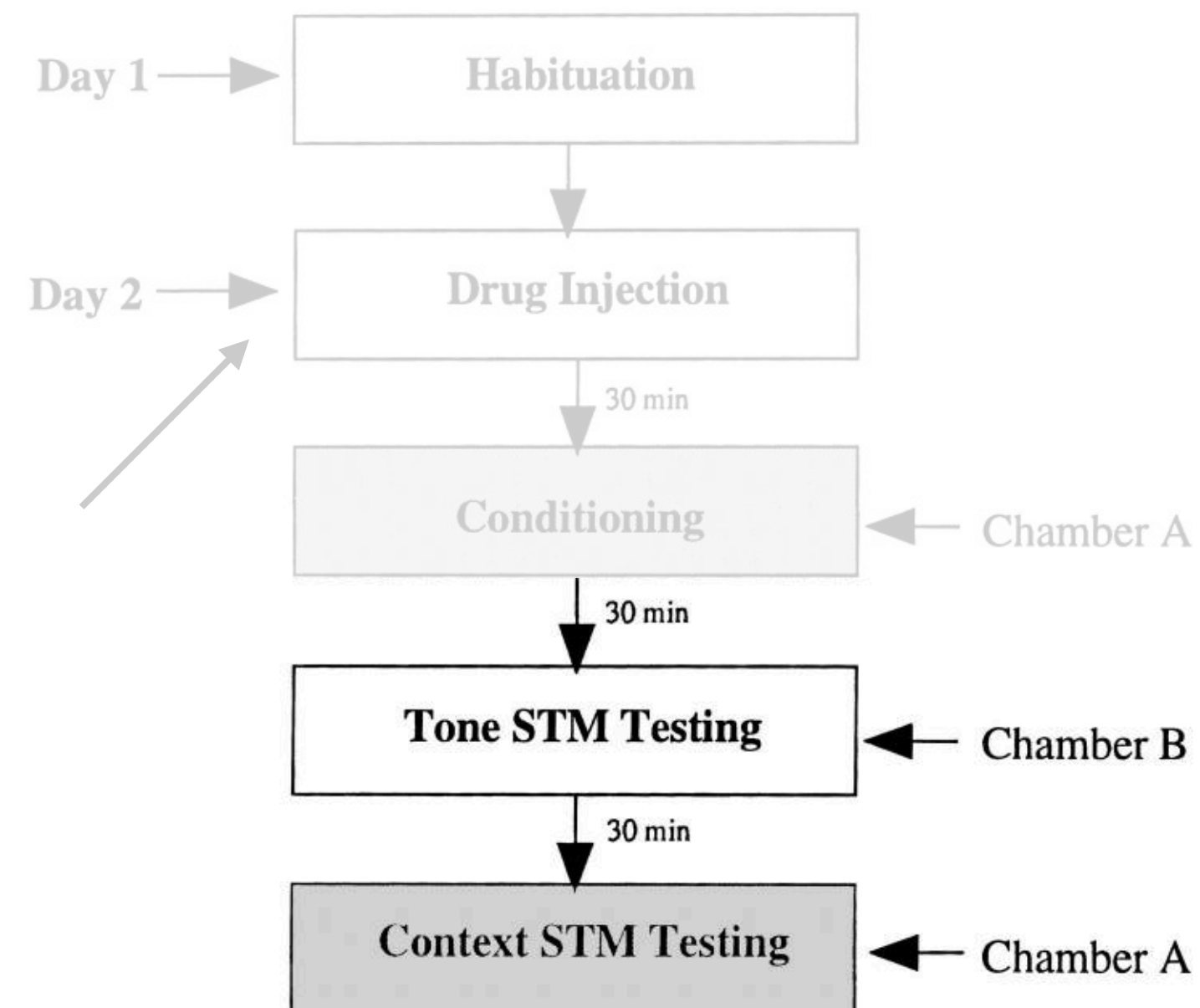
Son: 30sec, 70dB 5kHz
Choc: 1sec, 1,5mA

From Schafe et al, 1999

The role of *protein synthesis* in fear conditioning to sound and context in rats

*intra-ventricular injection of **anisomycin***

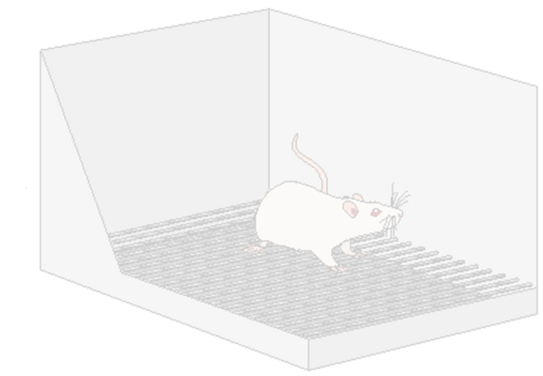
Anisomycin : protein synthesis inhibitor



Day 1: Habituation

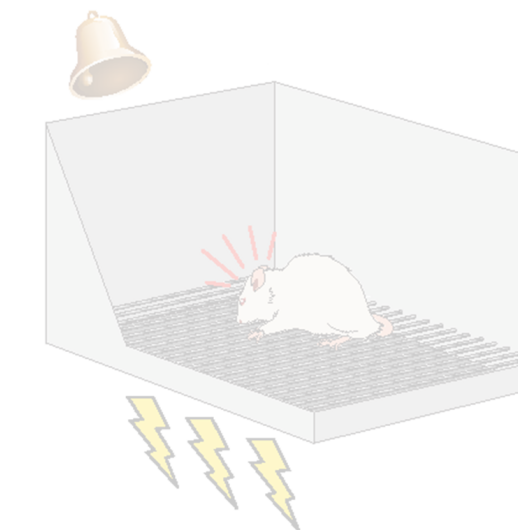


Chamber B



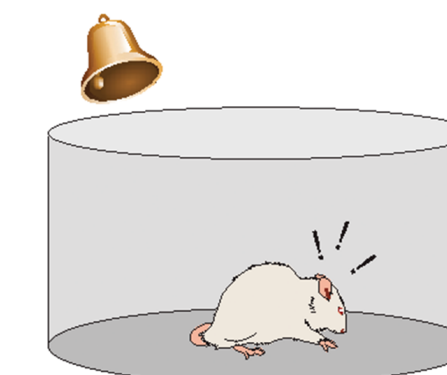
Chamber A

Day 2: Conditioning

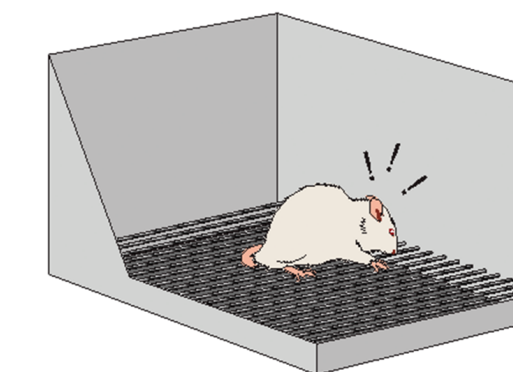


Son: 30sec, 70dB 5kHz
Choc: 1sec, 1,5mA

Tests (Day 2 STM; Day 3 LTM)



Tone/ Cue Memory



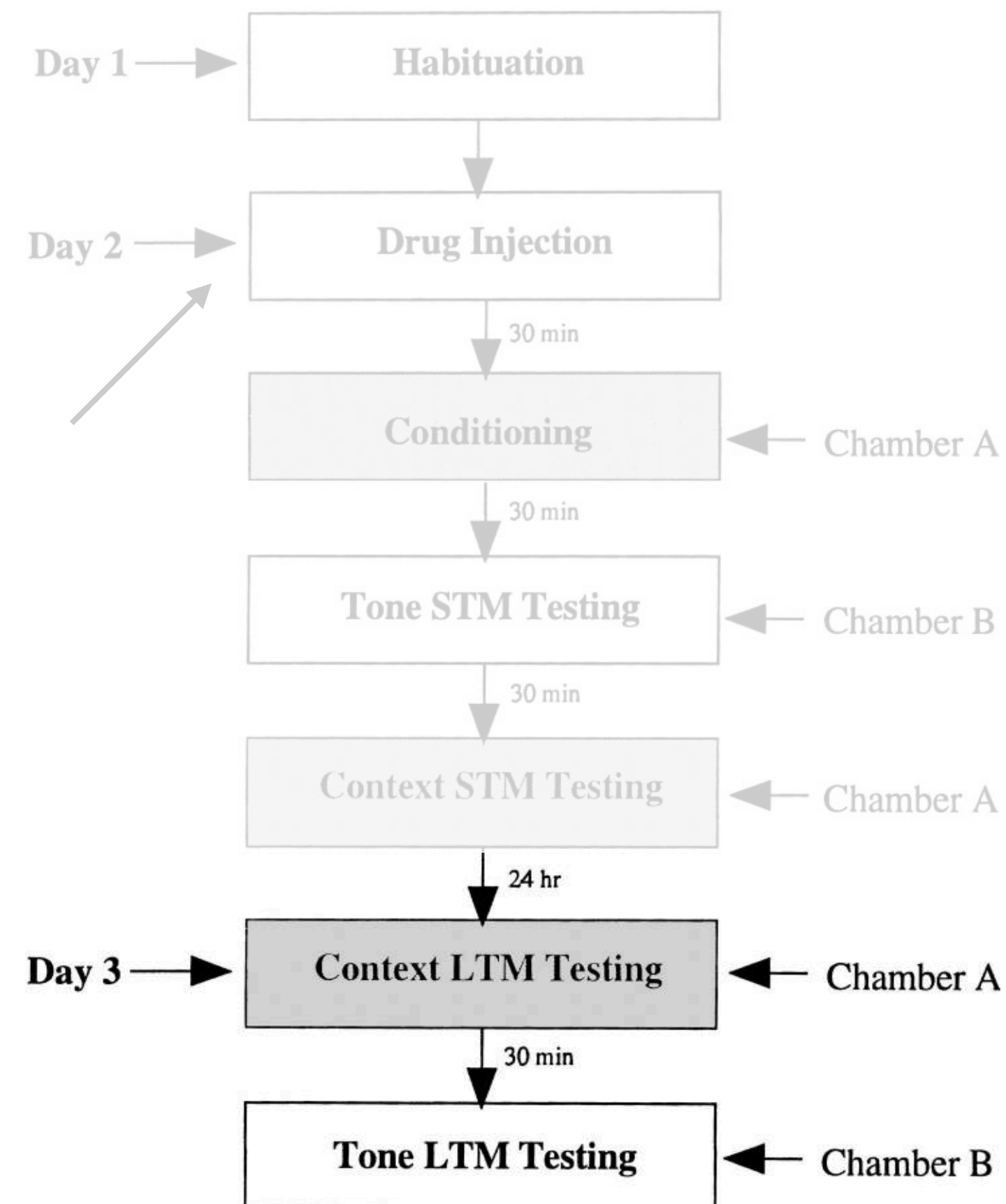
Contextual Memory

From Schafe et al, 1999

The role of *protein synthesis* in fear conditioning to sound and context in rats

*intra-ventricular injection of **anisomycin***

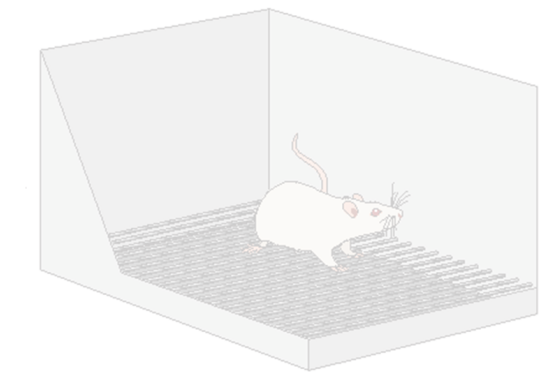
Anisomycin : protein synthesis inhibitor



Day 1: Habituation

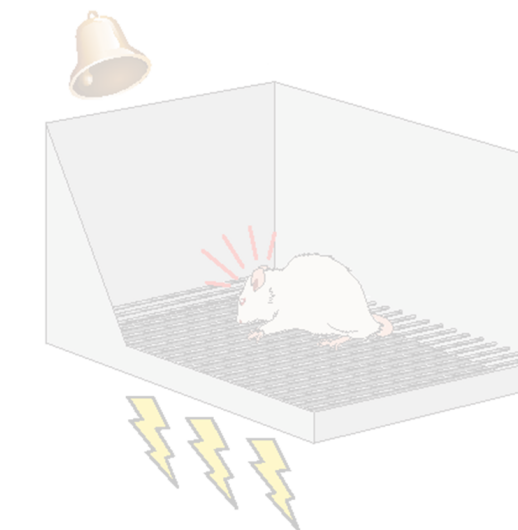


Chamber B



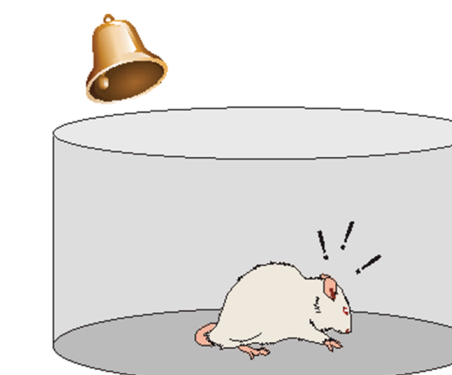
Chamber A

Day 2: Conditioning

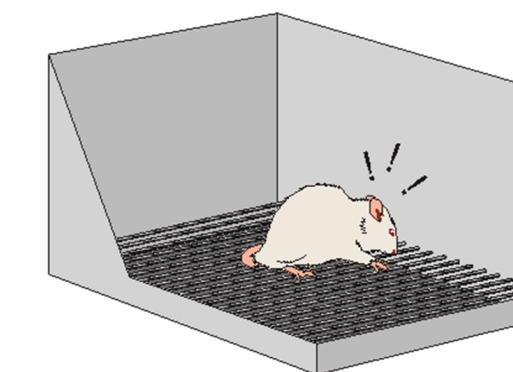


Son: 30sec, 70dB 5kHz
Choc: 1sec, 1,5mA

Tests (Day 2 STM; Day 3 LTM)

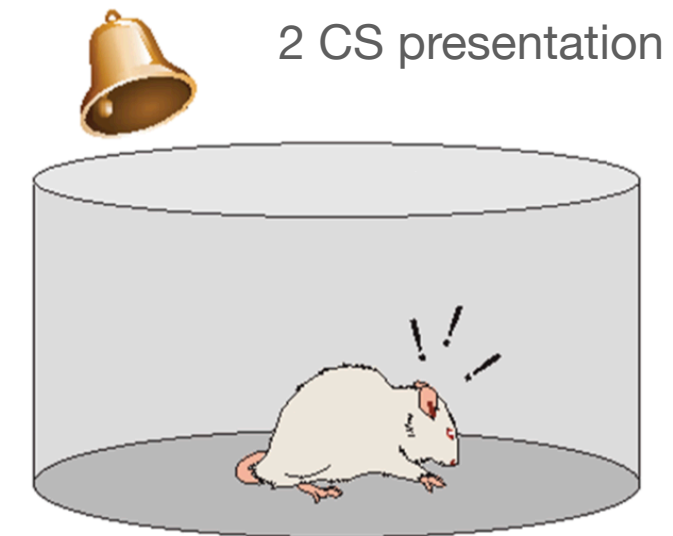
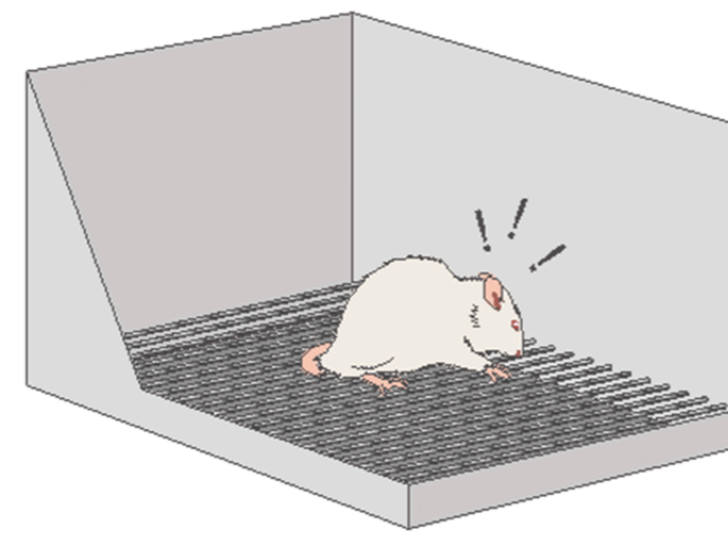


Tone/ Cue Memory



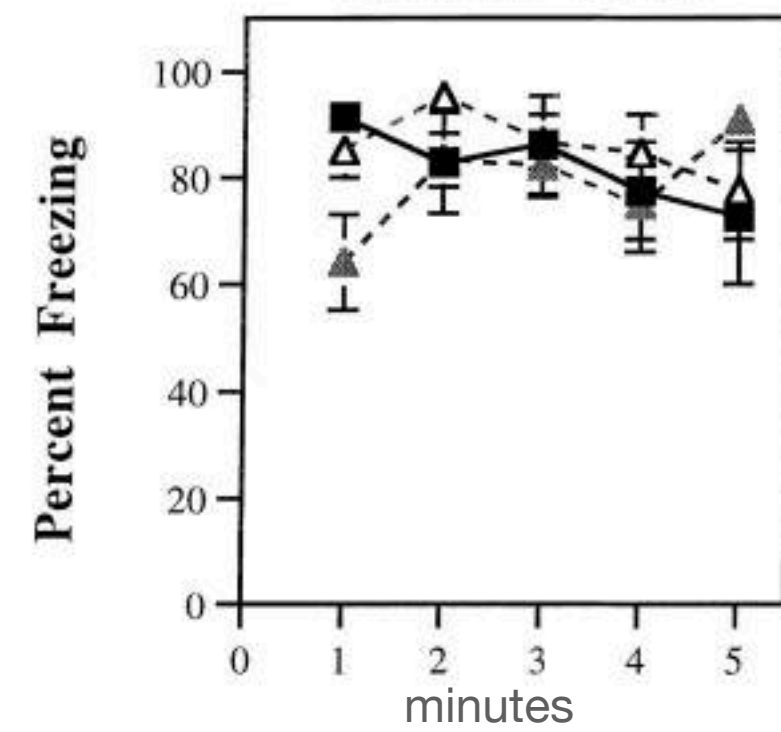
Contextual Memory

Short term memory - STM

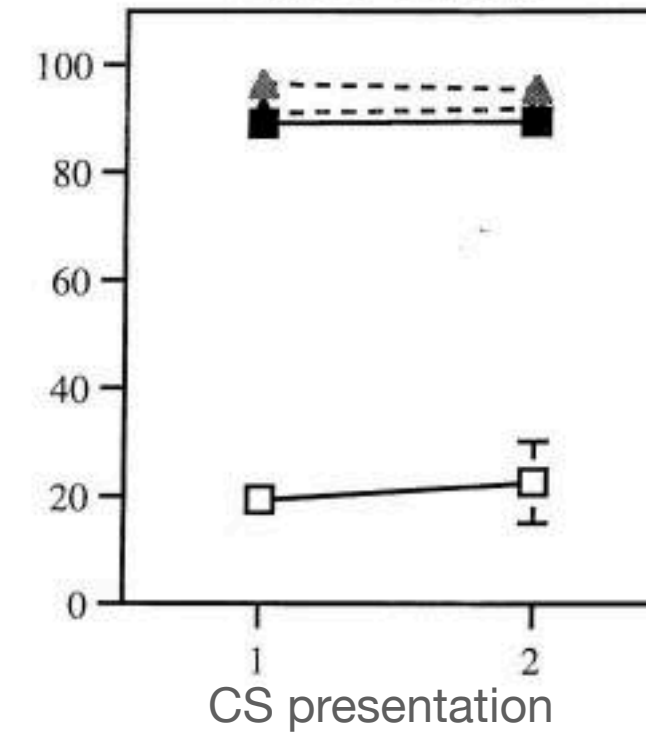


2 CS presentation

Context STM



Tone STM



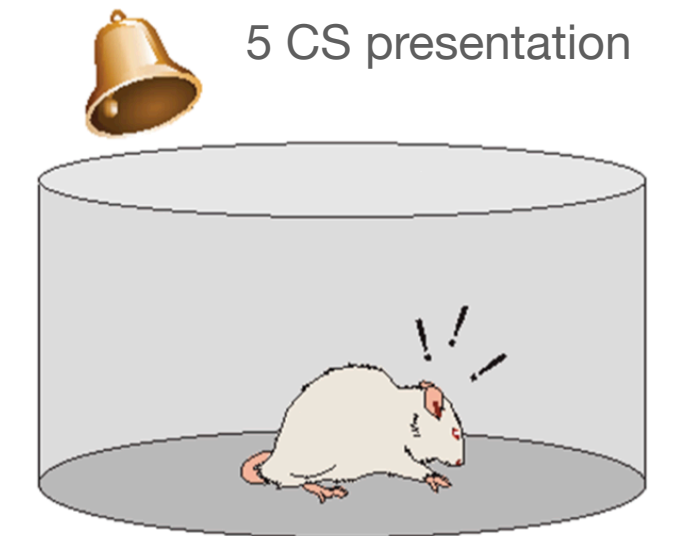
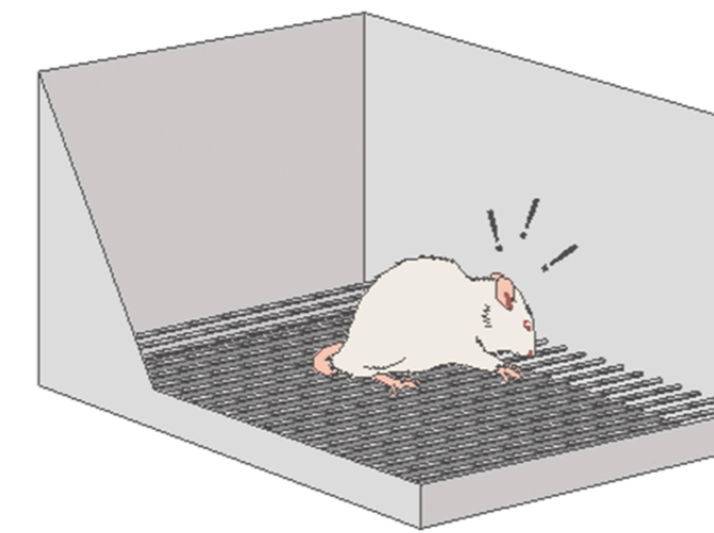
■ Vehicle 5uL
 △ Anisomycin 100 ug/5uL
 ▲ Anisomycin 300 ug/5uL

□ unpaired controls

Conclusion

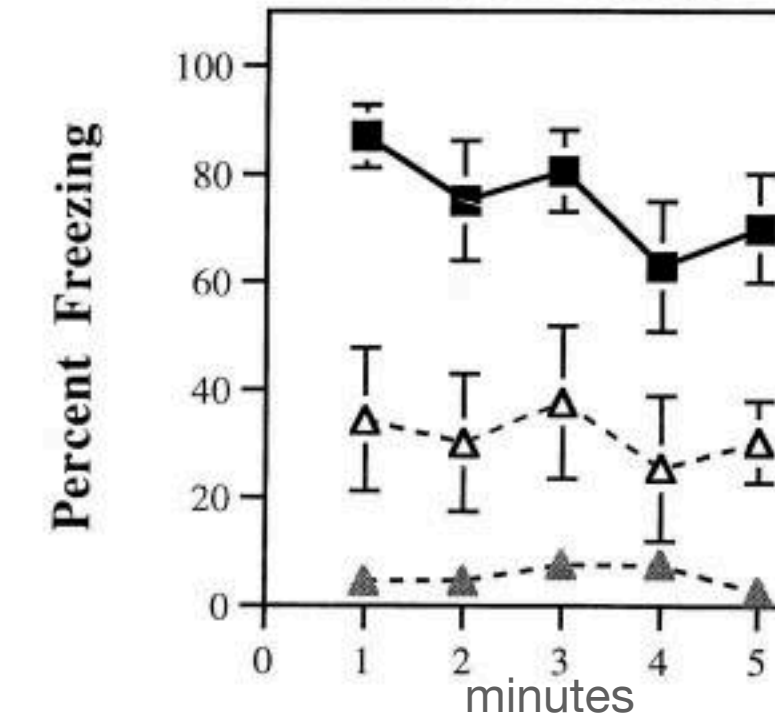
Contextual and cue STM are intact after anisomycin treatment

Long Term memory - LTM

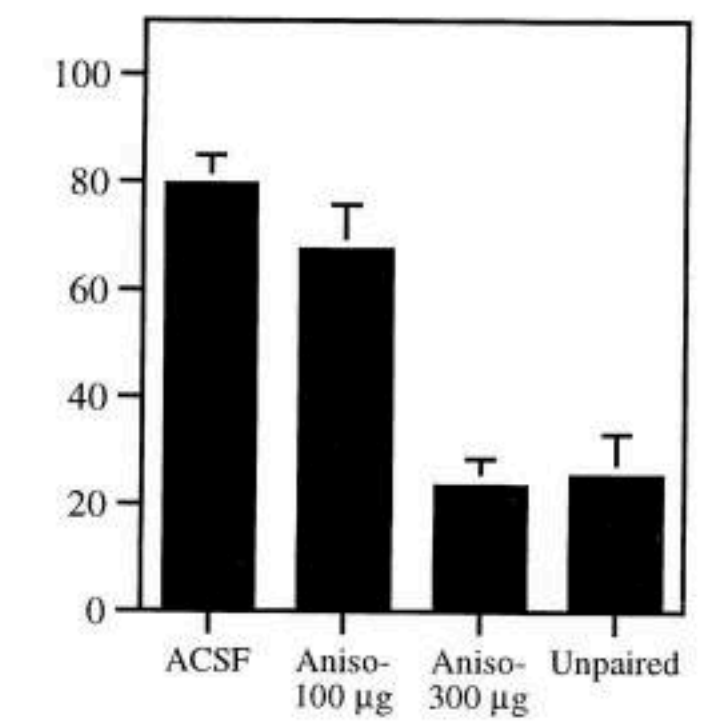
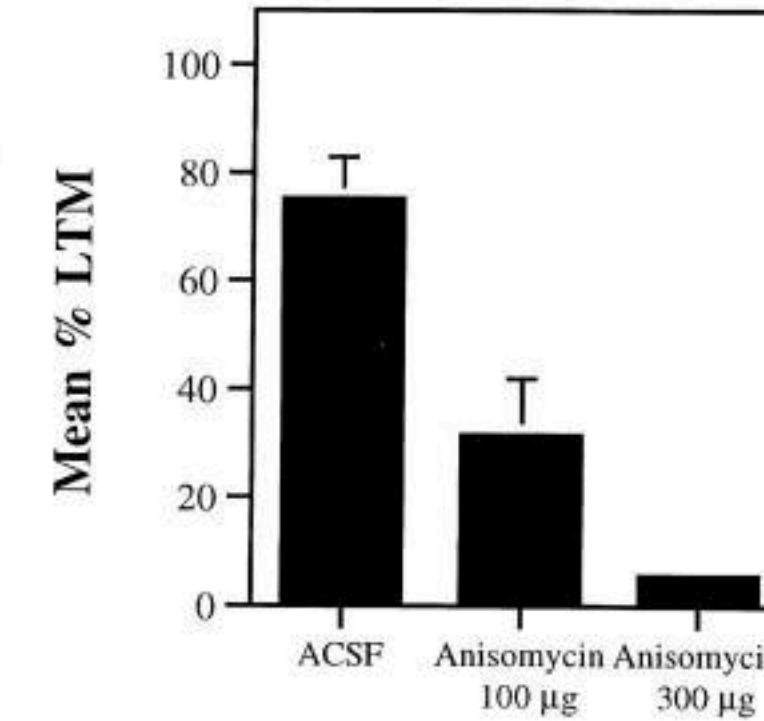
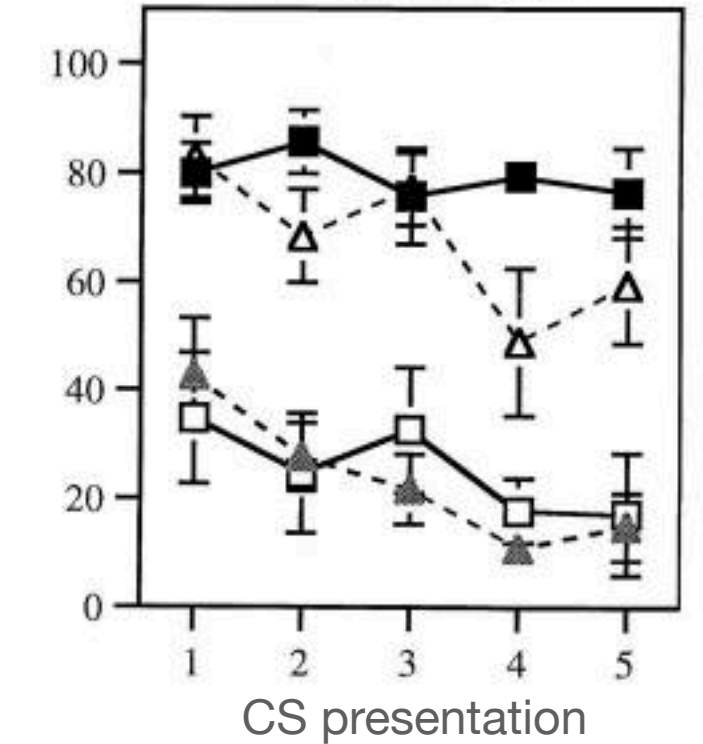


5 CS presentation

Context LTM



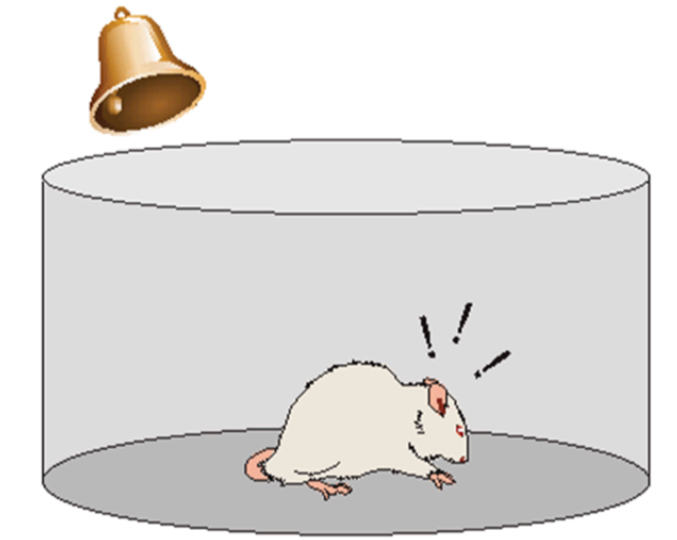
Tone LTM



Conclusion

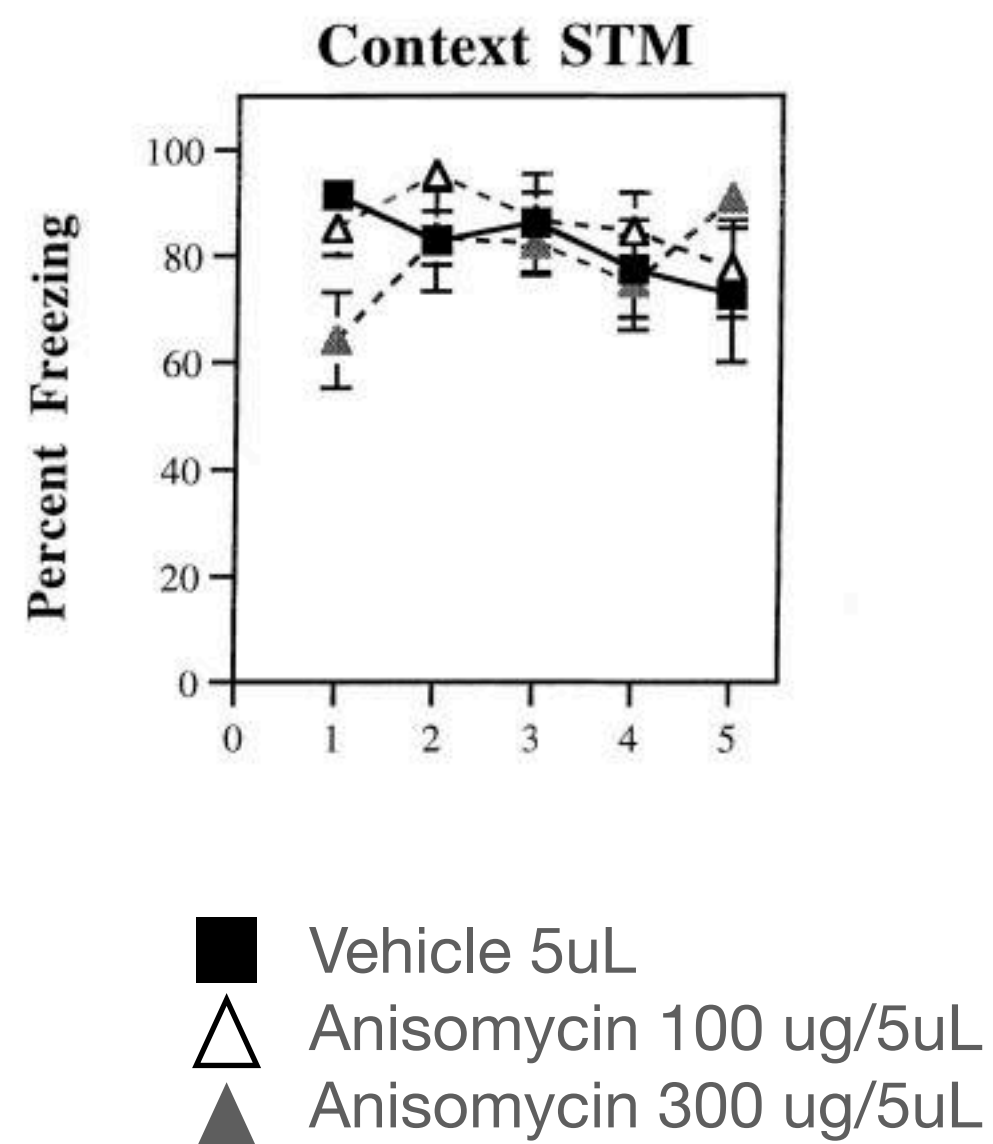
Dose-dependent impairment of contextual LTM by anisomycin

Contextual and cue memory consolidation in rat is protein synthesis dependent

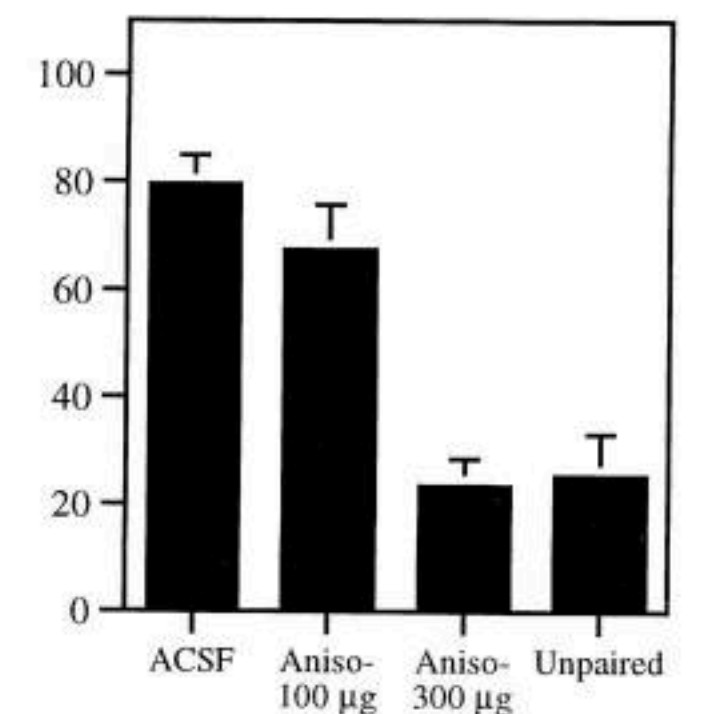
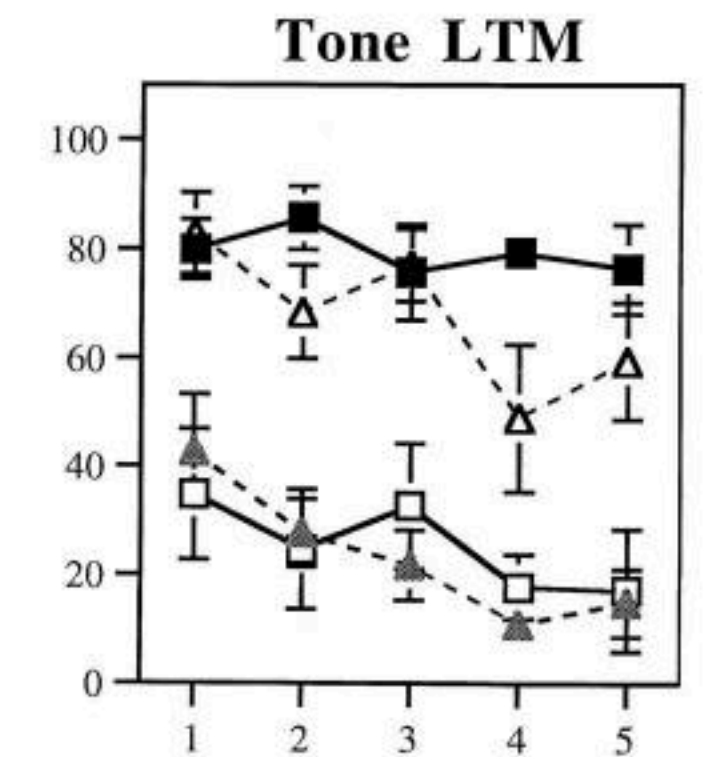
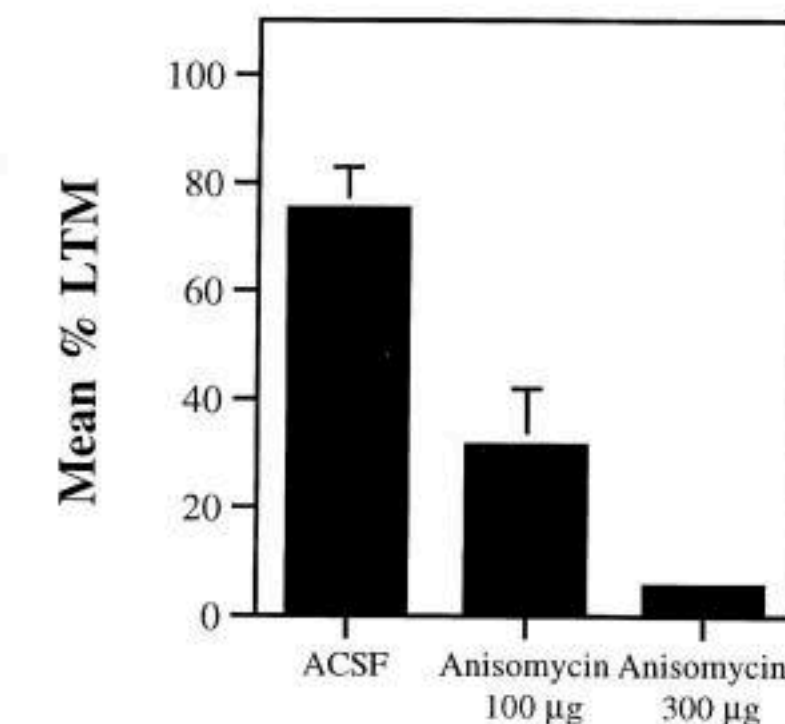
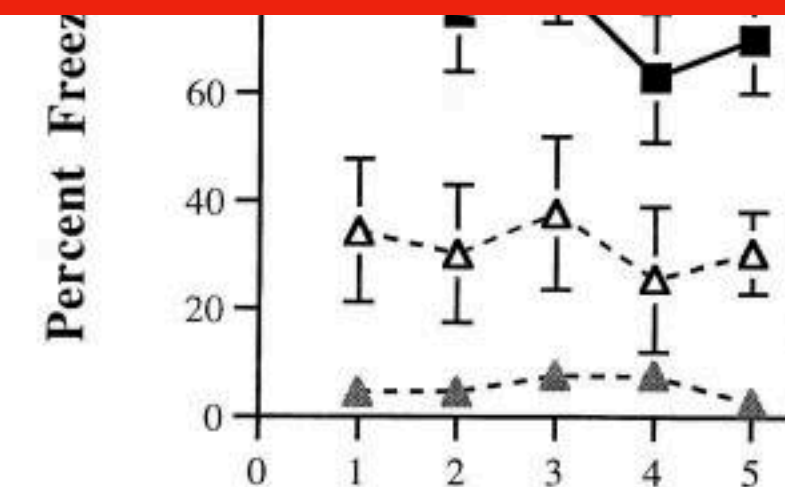
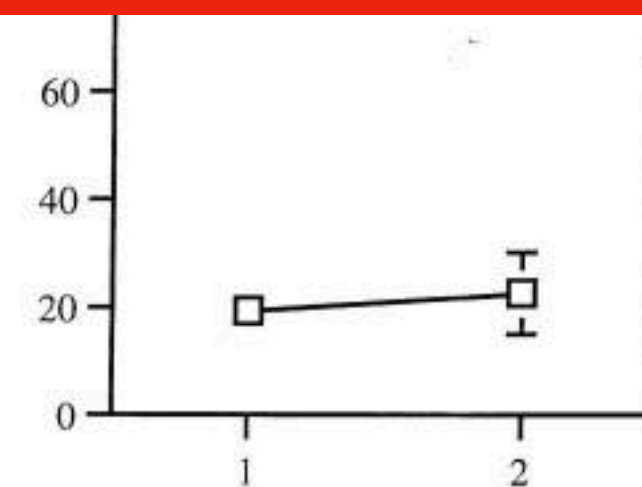


Limitation:

INTRA VENTRICULAR INJECTION



□ unpaired controls



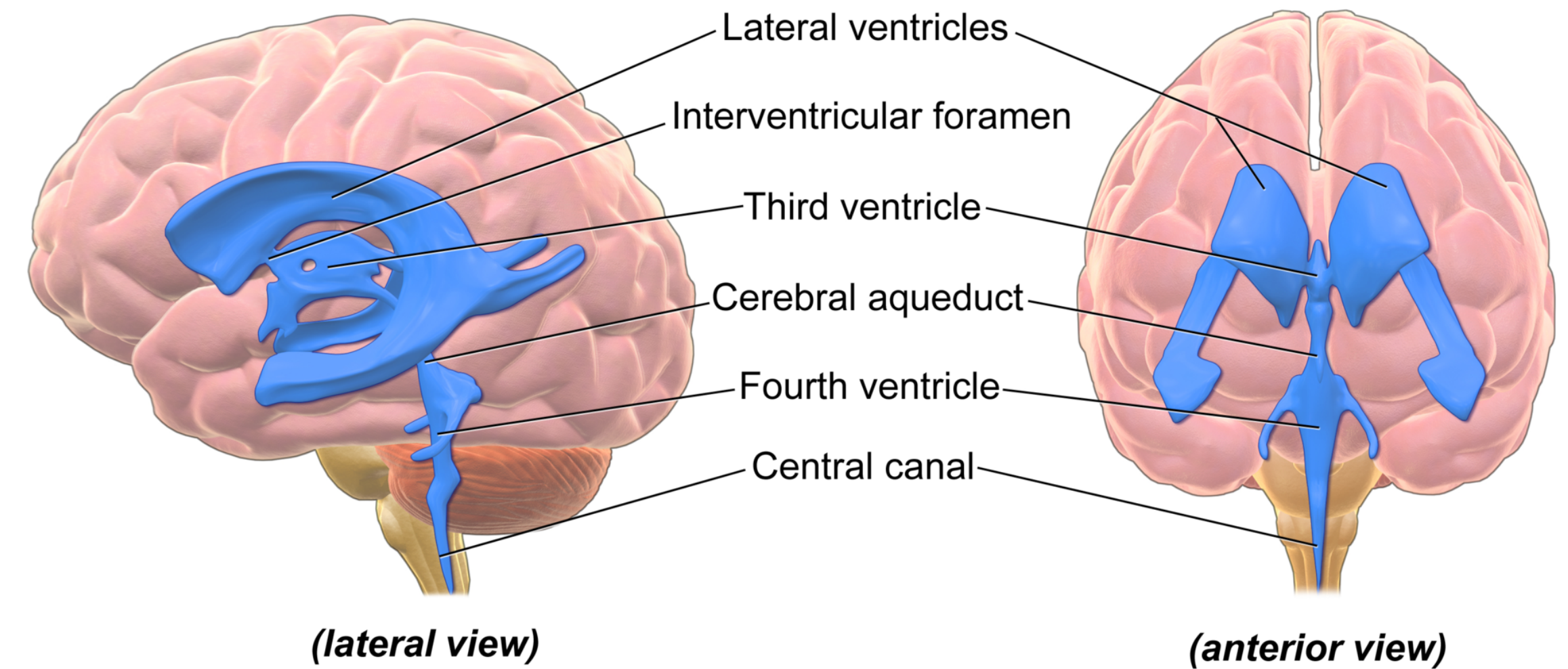
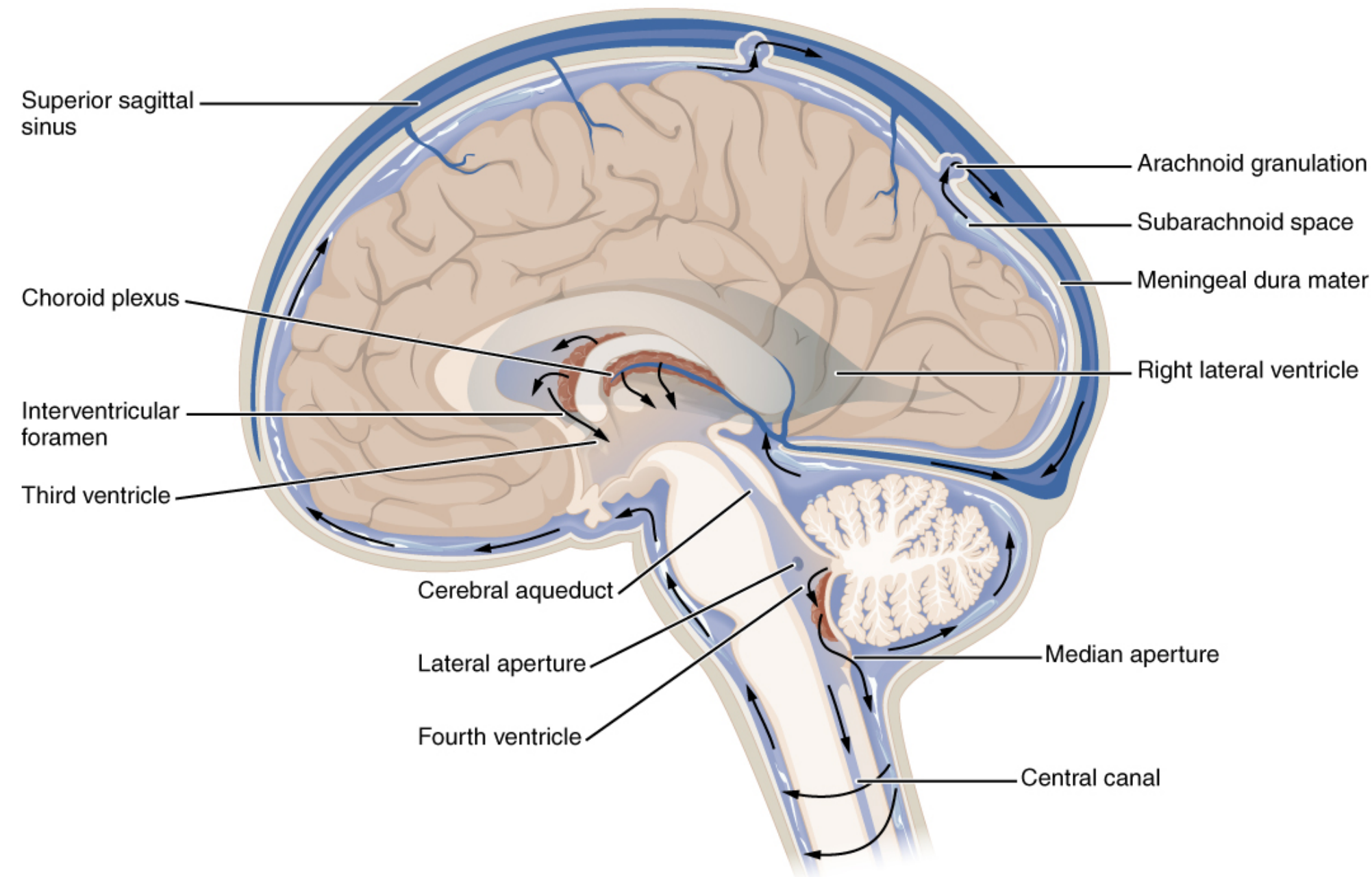
Conclusion

Contextual and cue STM are intact after anisomycin treatment

Conclusion

Dose-dependent impairment of contextual LTM by anisomycin

back to basic - Ventricular system

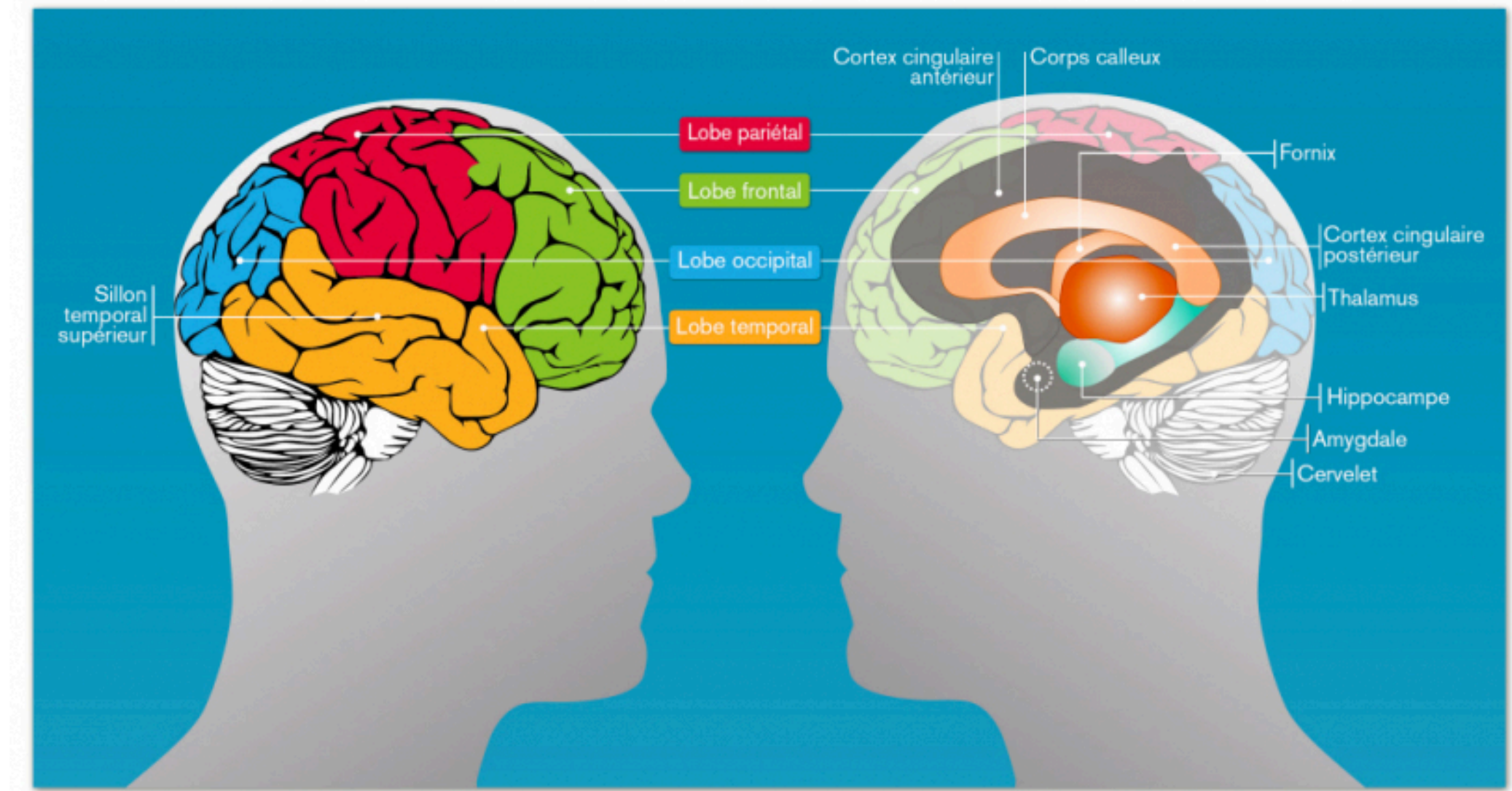


Functions of cerebrospinal fluid (CSF) :

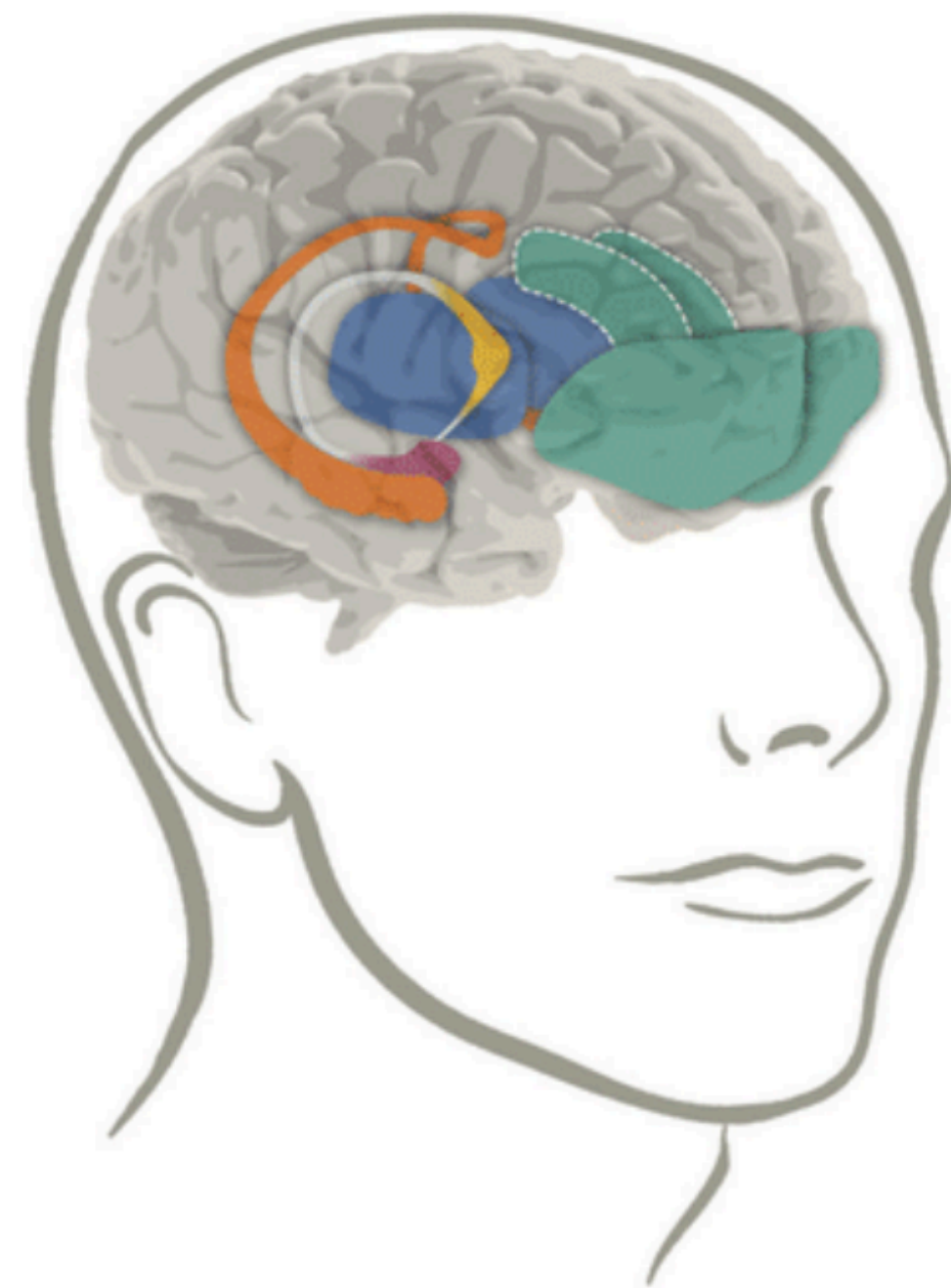
- Protective
- Nourishing
- Cleaning

Emotional Memory

Brain regions involved



Zones du cerveau impliquées dans la mémoire © Inserm, F. Koulikoff



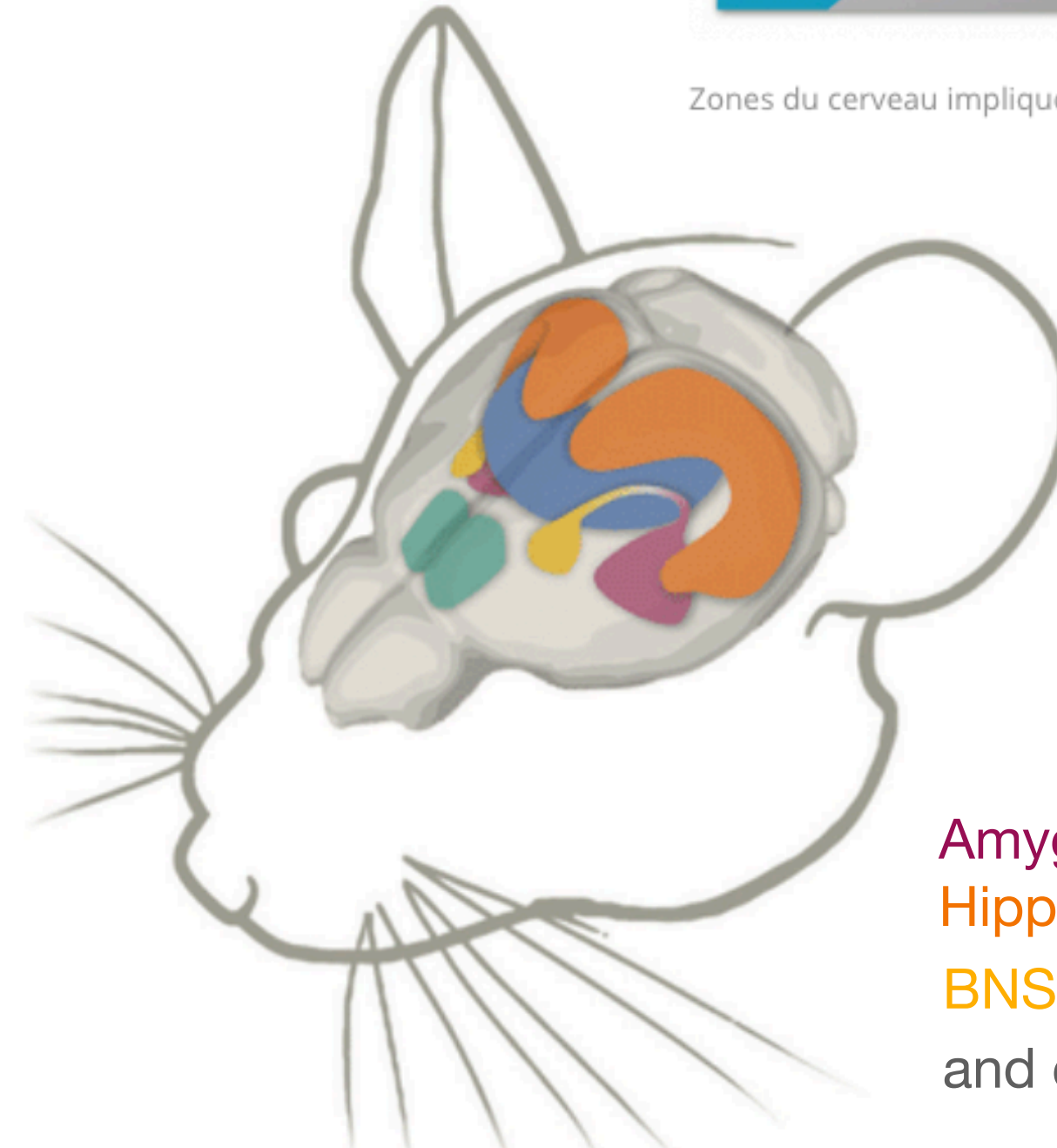
Pre Frontal Cortex

Hippocampus

Amygdala

BNST

Thalamus



Amygdala : emotions (fear...)

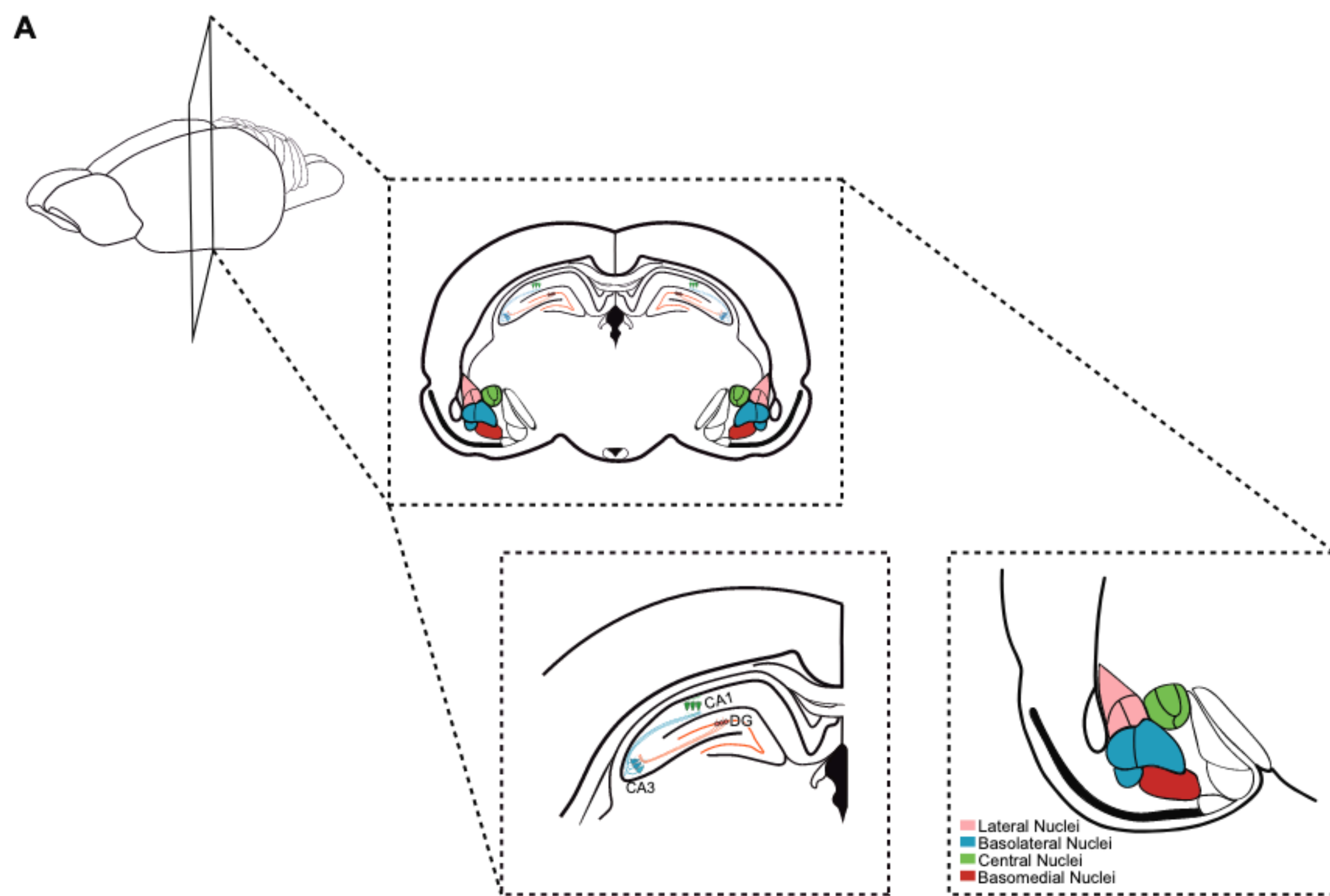
Hippocampus : explicit memory (spatial, episodic...)

BNST: bed nucleus of the stria terminalis is a heterogeneous and complex limbic forebrain structure

Adapted from <https://www.smallpocketlibrary.com/2019/10/why-do-some-drugs-work-when-tested-on.html>

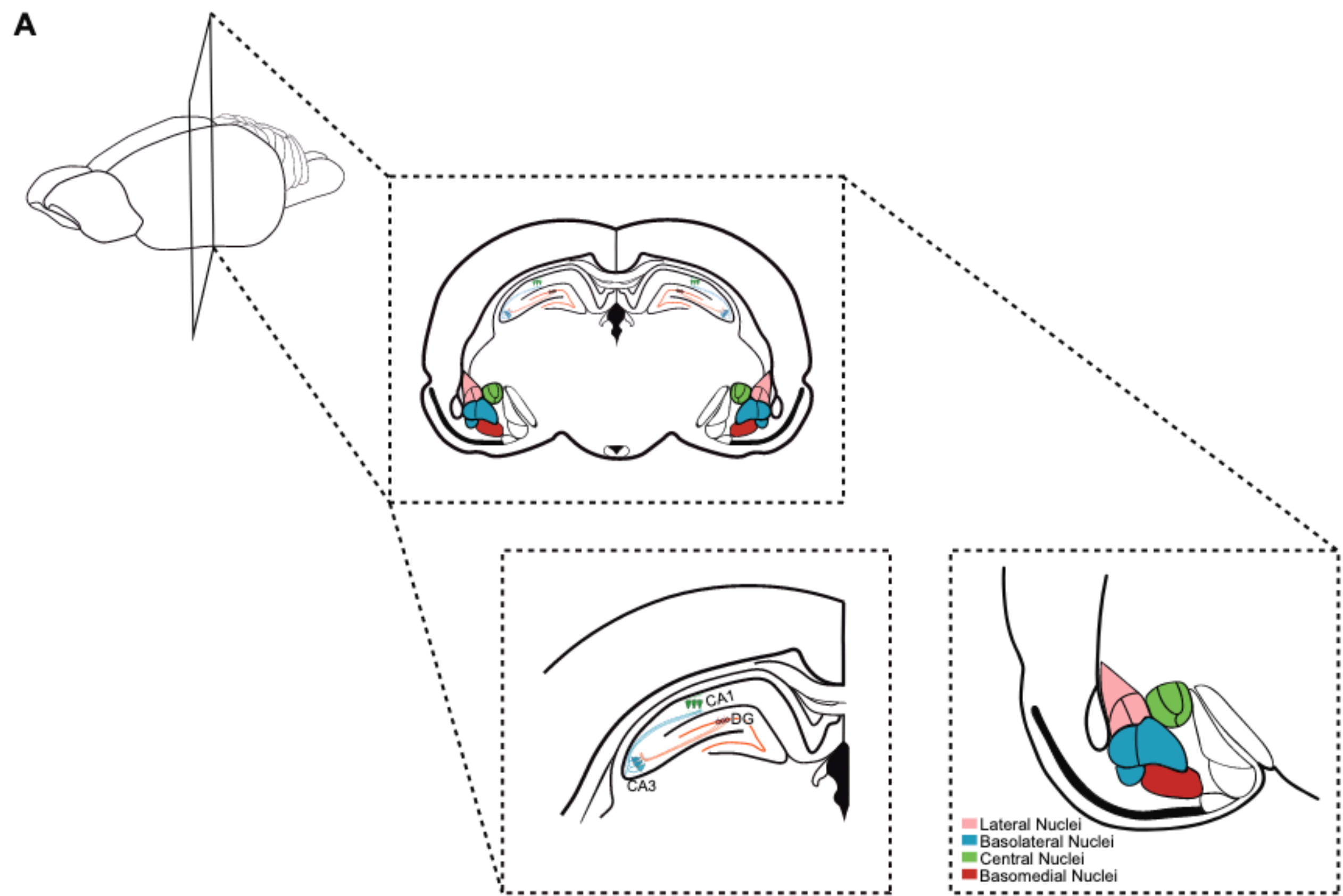
Emotional Memory

Brain regions involved



Emotional Memory

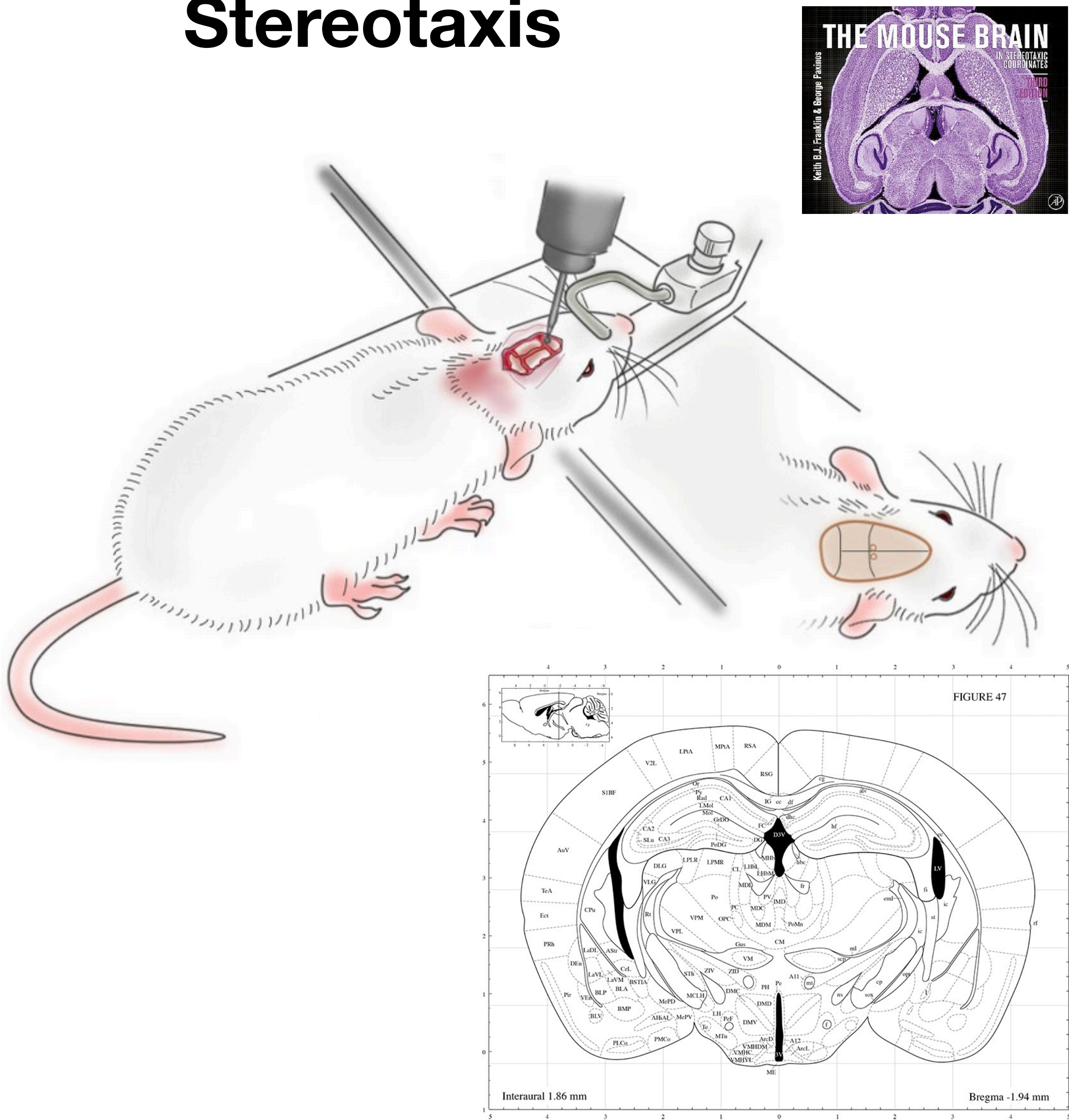
Brain regions involved



from Izquierdo et al, 2016, Physiol Rev 96: 695-750

20

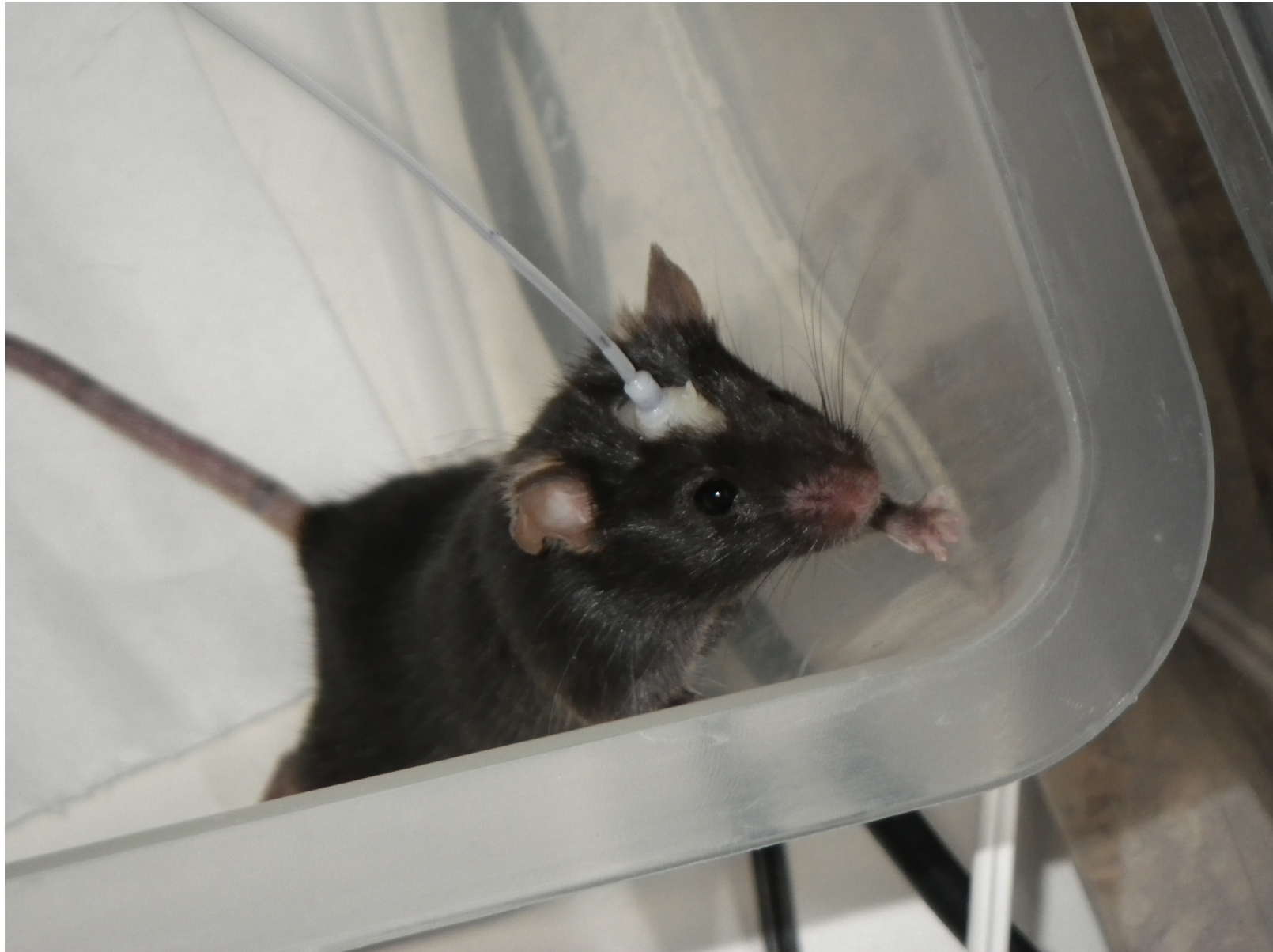
Stereotaxis



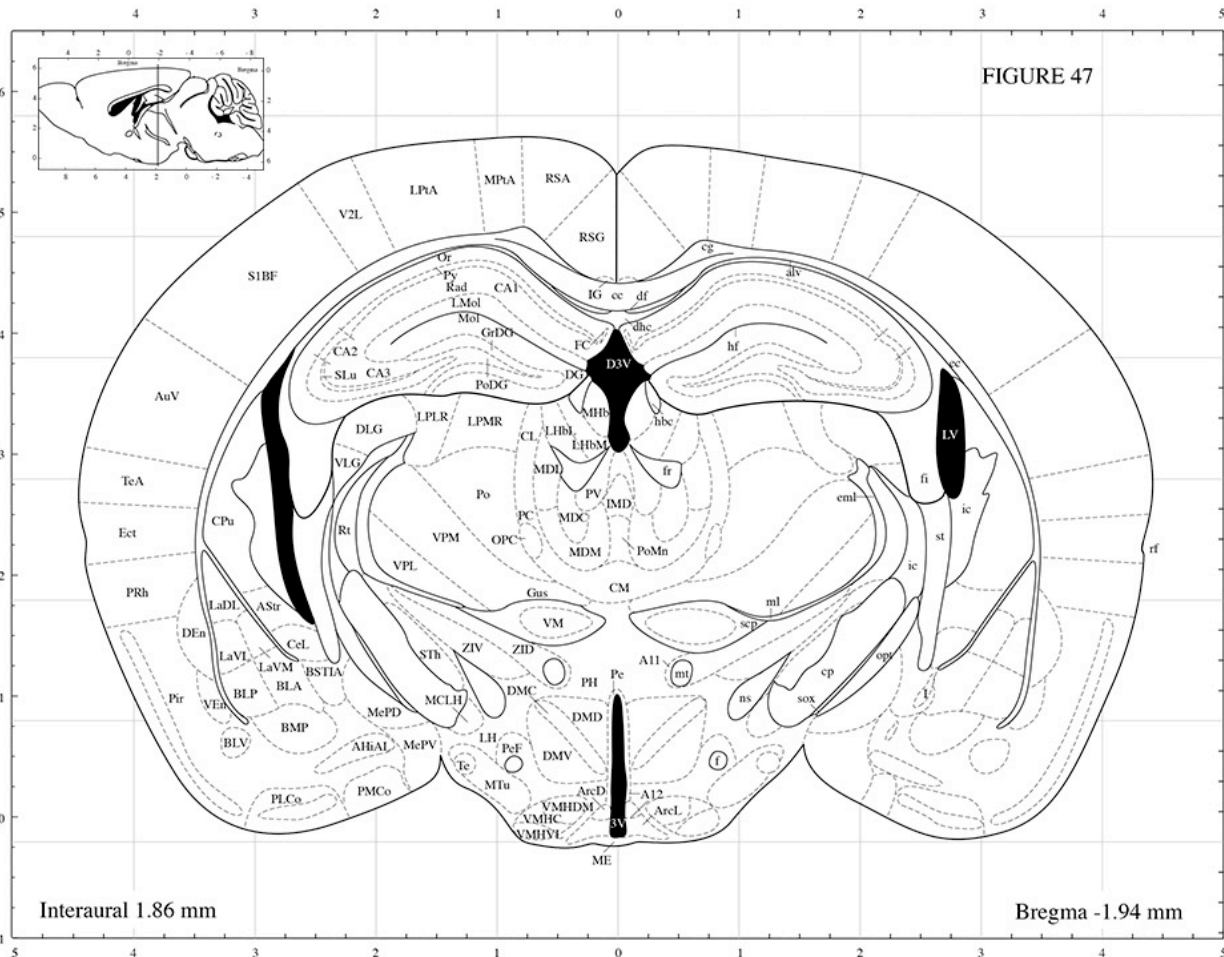
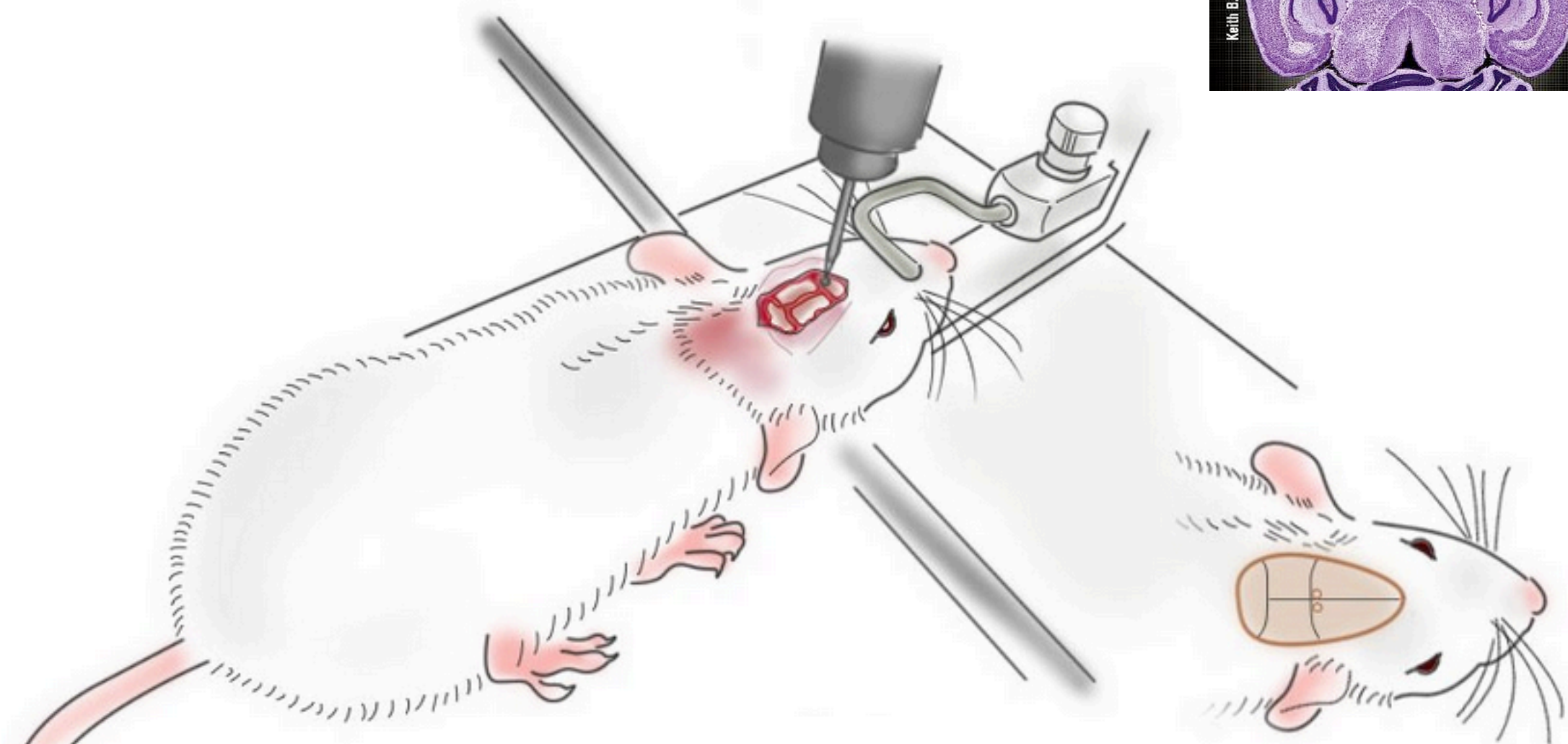
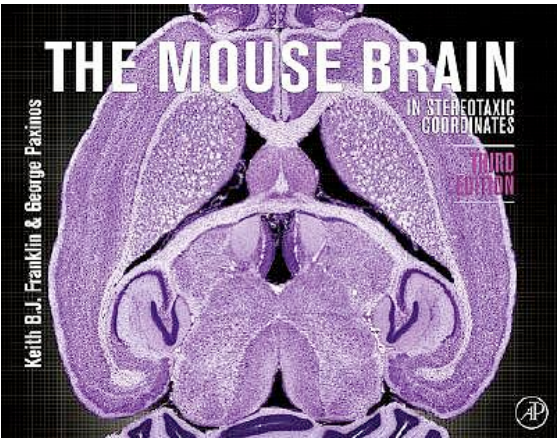
from <https://encapsula.com/stereotaxic-administration/>

Emotional Memory

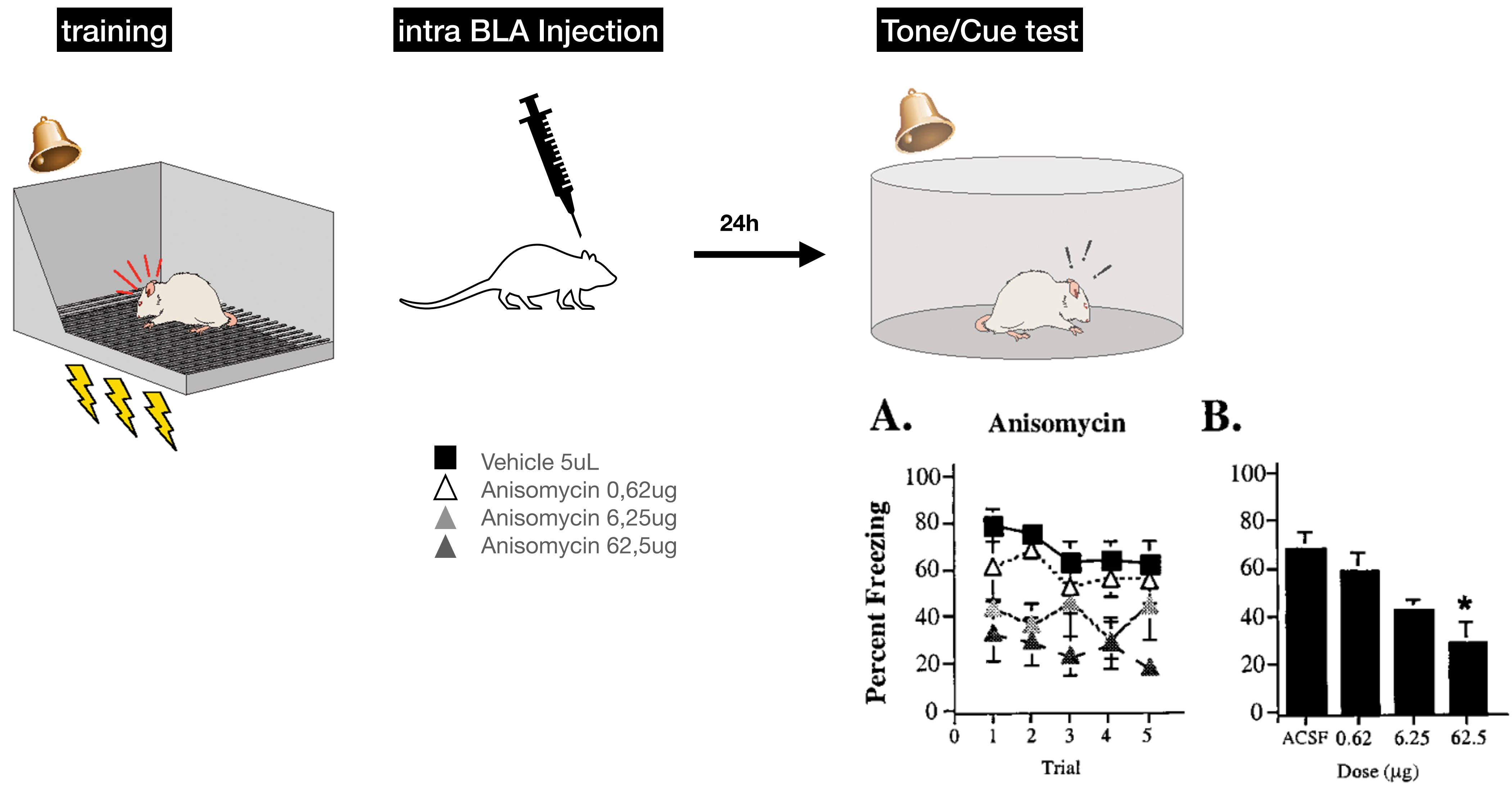
Brain regions involved



Stereotaxis



Emotional memory - Brain regions involved

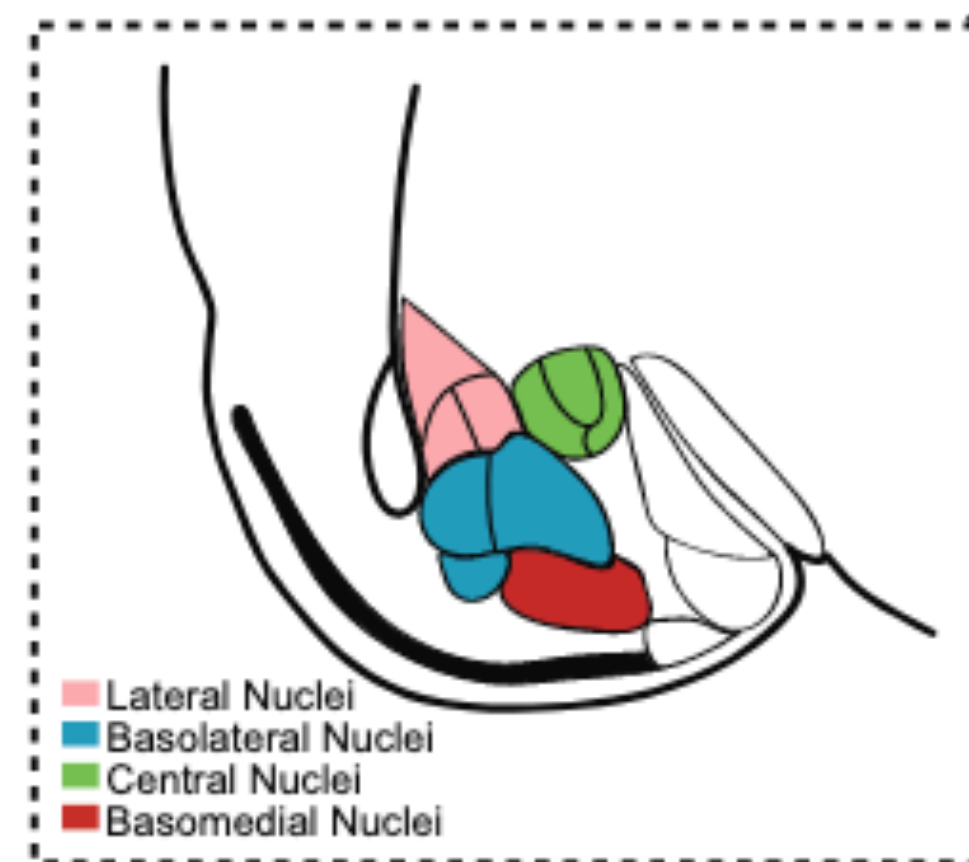


Cue memory consolidation is dependent on protein synthesis in the BLA

Emotional Memory

Brain regions involved : the amygdala

Amygdala



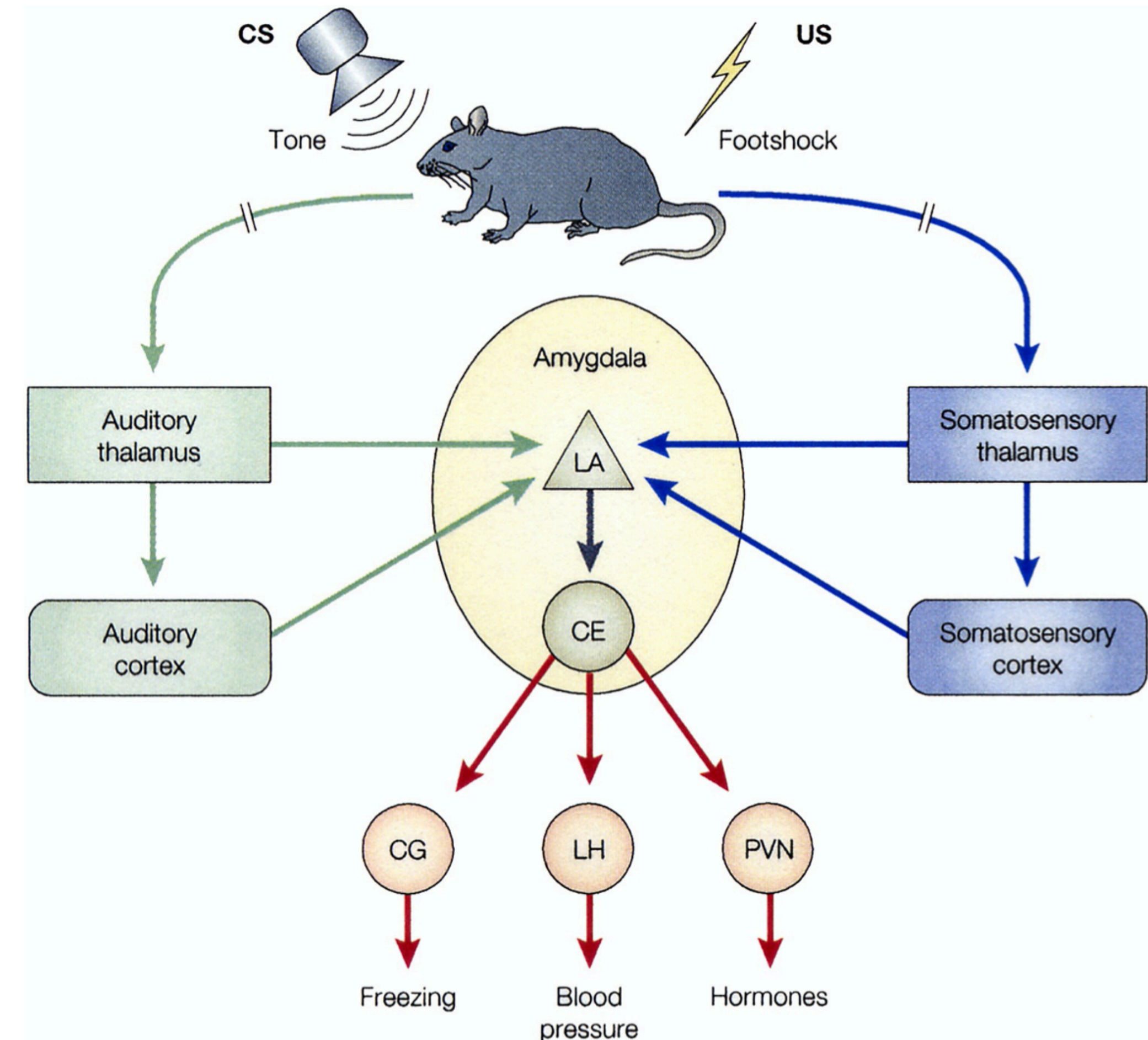
Lateral Amygdala (LA)

BasoLateral Amygdala (BLA)

Central Amygdala (CE)

BasoMedial Amygdala (BMA)

- is the **acquisition** of tone/cue fear memory is dependent upon:
- the amygdala?
- synaptic plasticity mechanisms occurring in the amygdala?



Emotional Memory

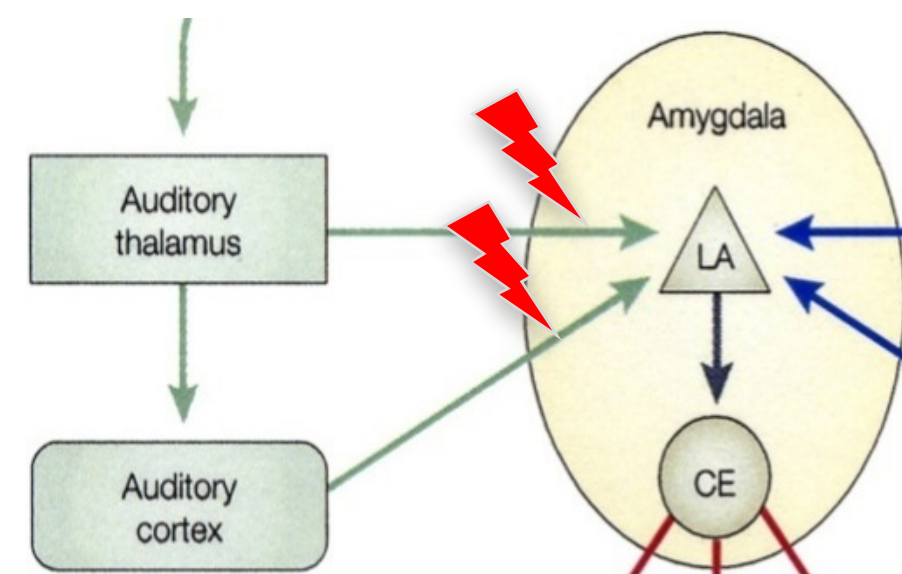
Brain regions involved : the amygdala

US-CS conditioning
(shock-sound)



Optogenetics

Stimulation of auditive fibres
connecting the lateral amygdala



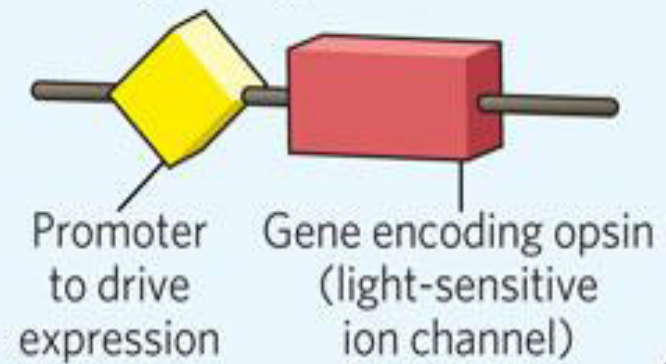
Technique - Optogenetics

SIX STEPS TO OPTOGENETICS

With optogenetic techniques, researchers can modulate the activity of targeted neurons using light.

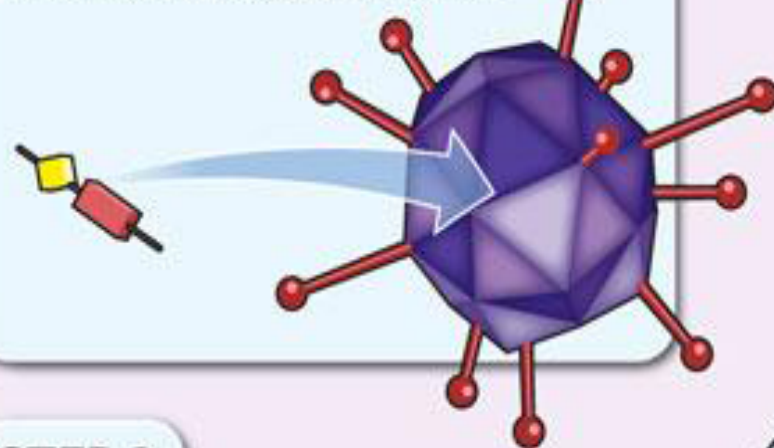
STEP 1

Piece together genetic construct.



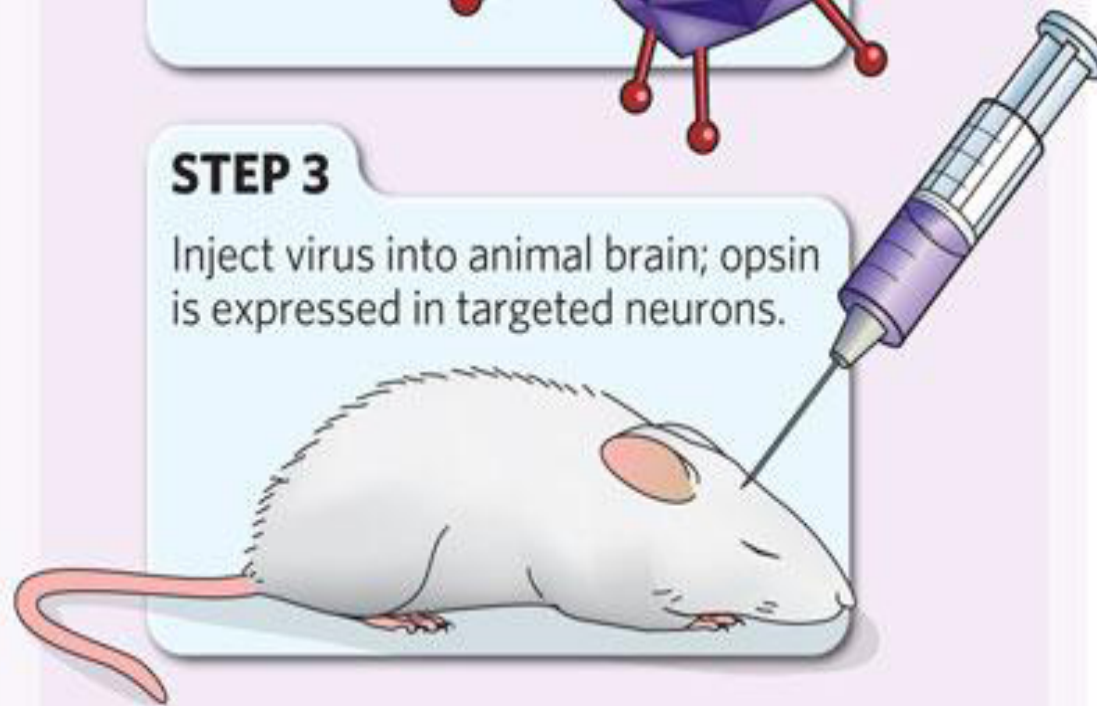
STEP 2

Insert construct into virus.



STEP 3

Inject virus into animal brain; opsin is expressed in targeted neurons.



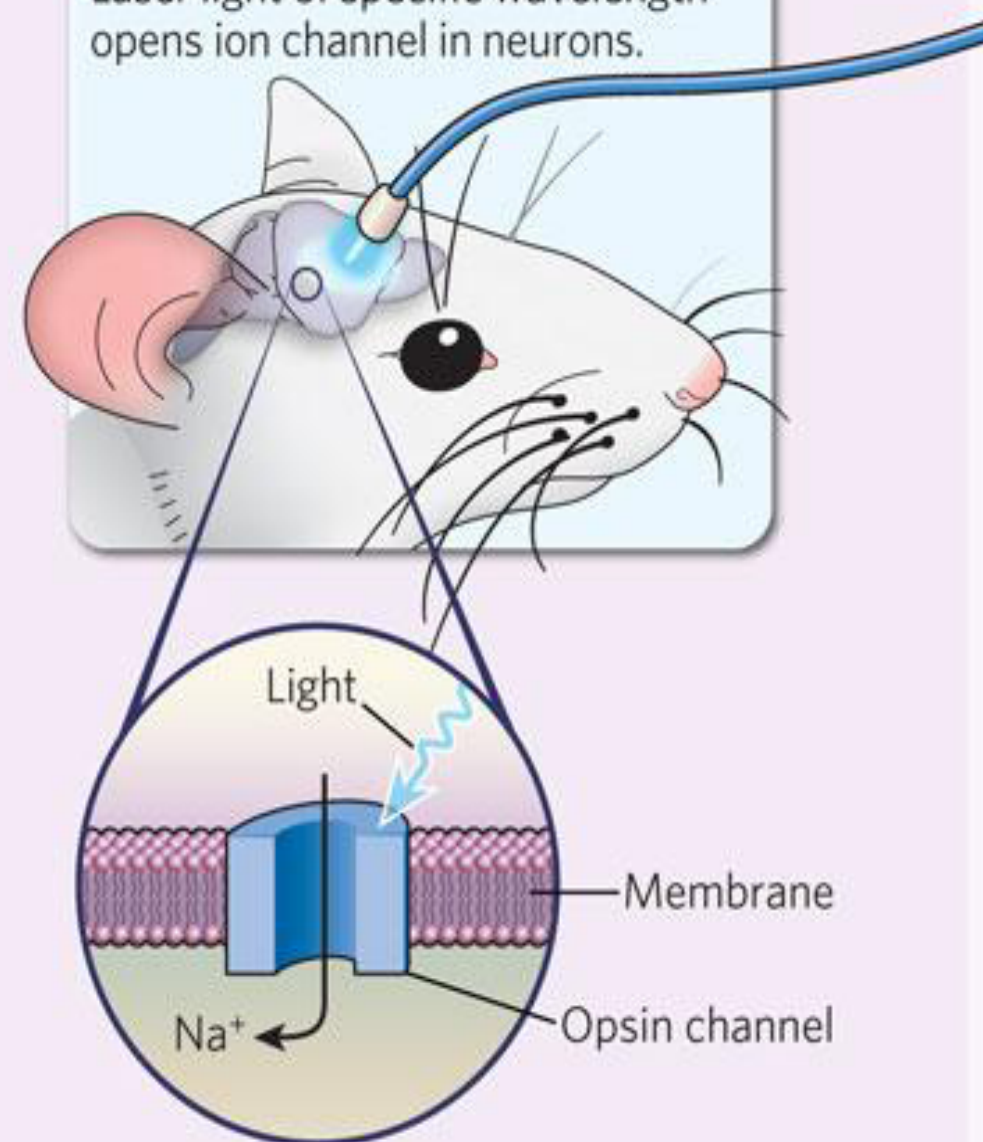
STEP 4

Insert 'optrode', fibre-optic cable plus electrode.



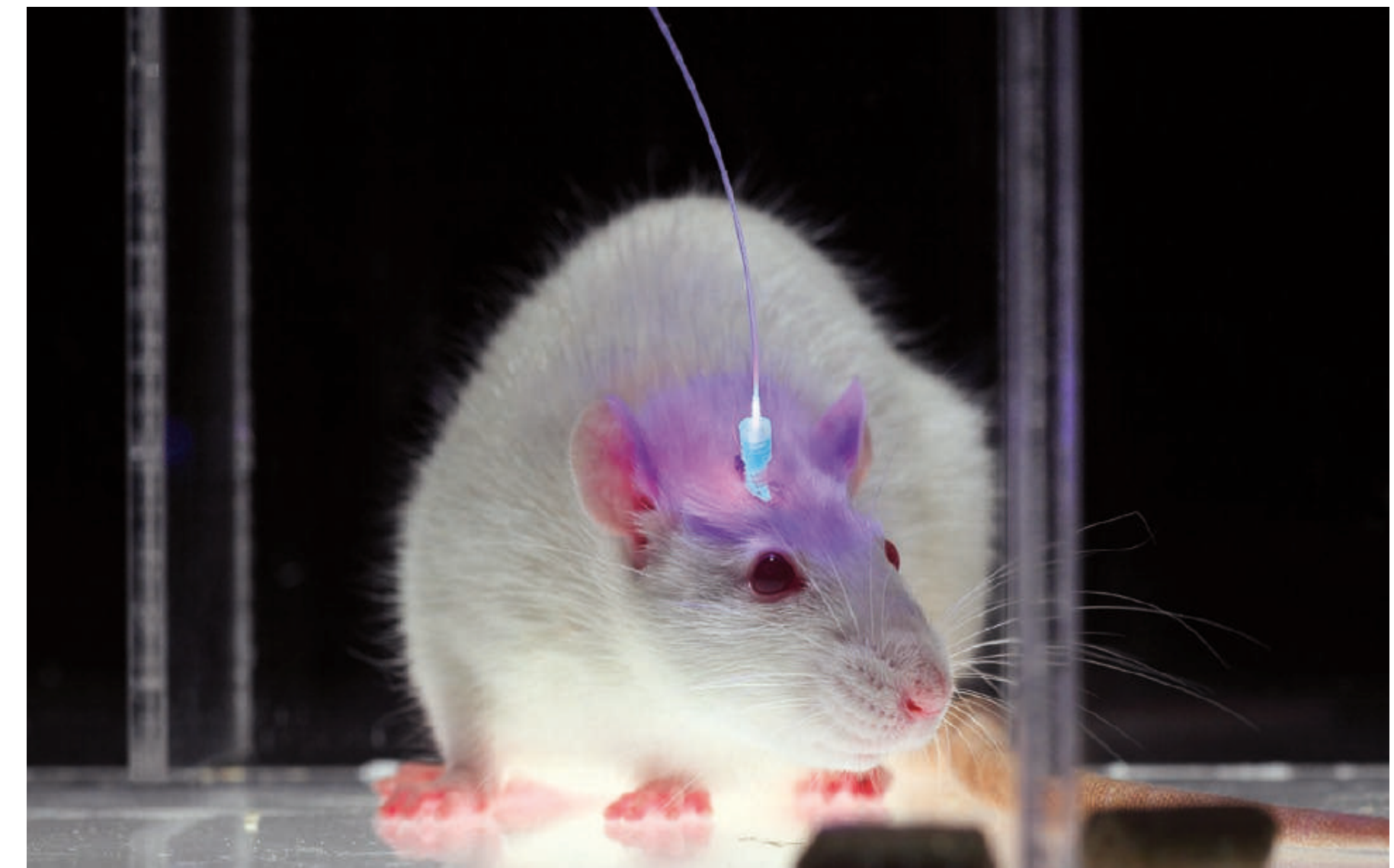
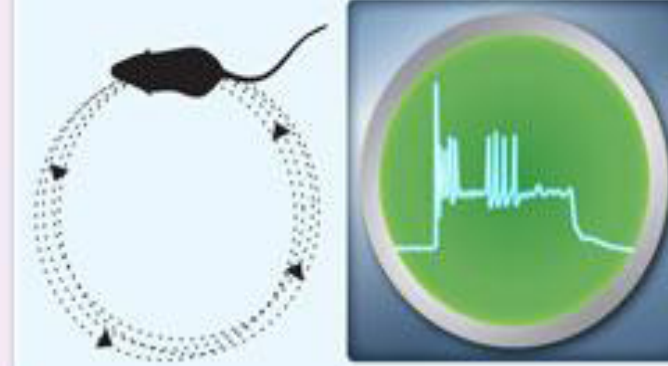
STEP 5

Laser light of specific wavelength opens ion channel in neurons.



STEP 6

Record electrophysiological and behavioural results.



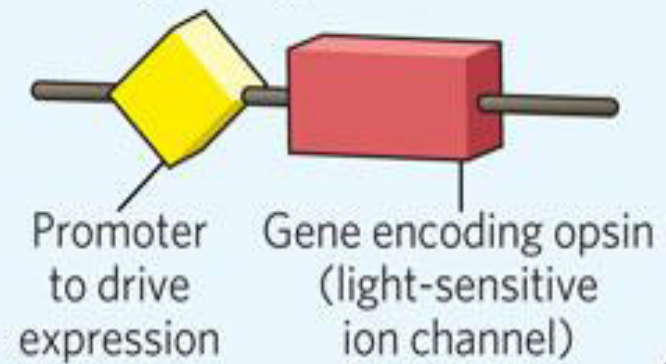
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SIX STEPS TO OPTOGENETICS

With optogenetic techniques, researchers can modulate the activity of targeted neurons using light.

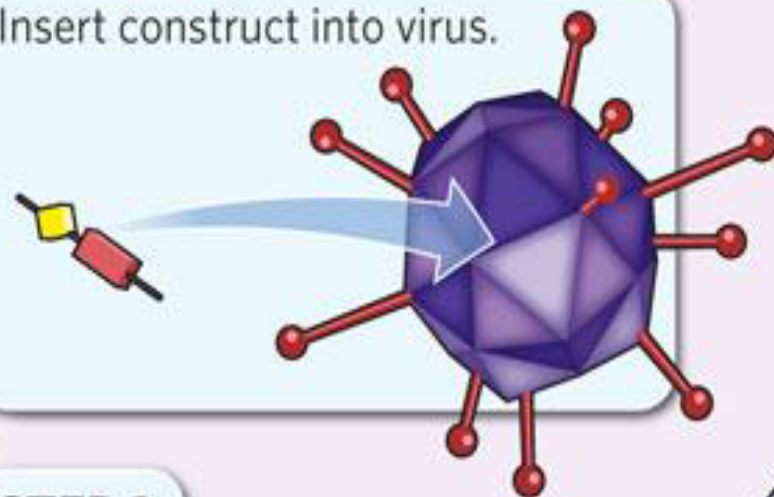
STEP 1

Piece together genetic construct.



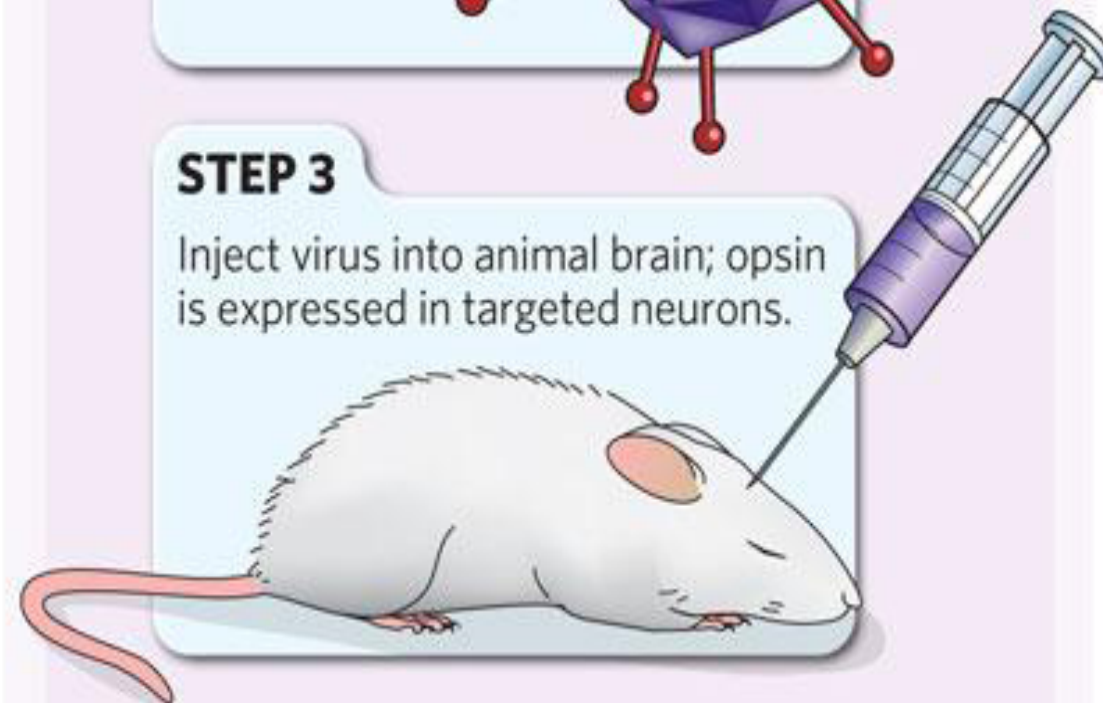
STEP 2

Insert construct into virus.



STEP 3

Inject virus into animal brain; opsin is expressed in targeted neurons.



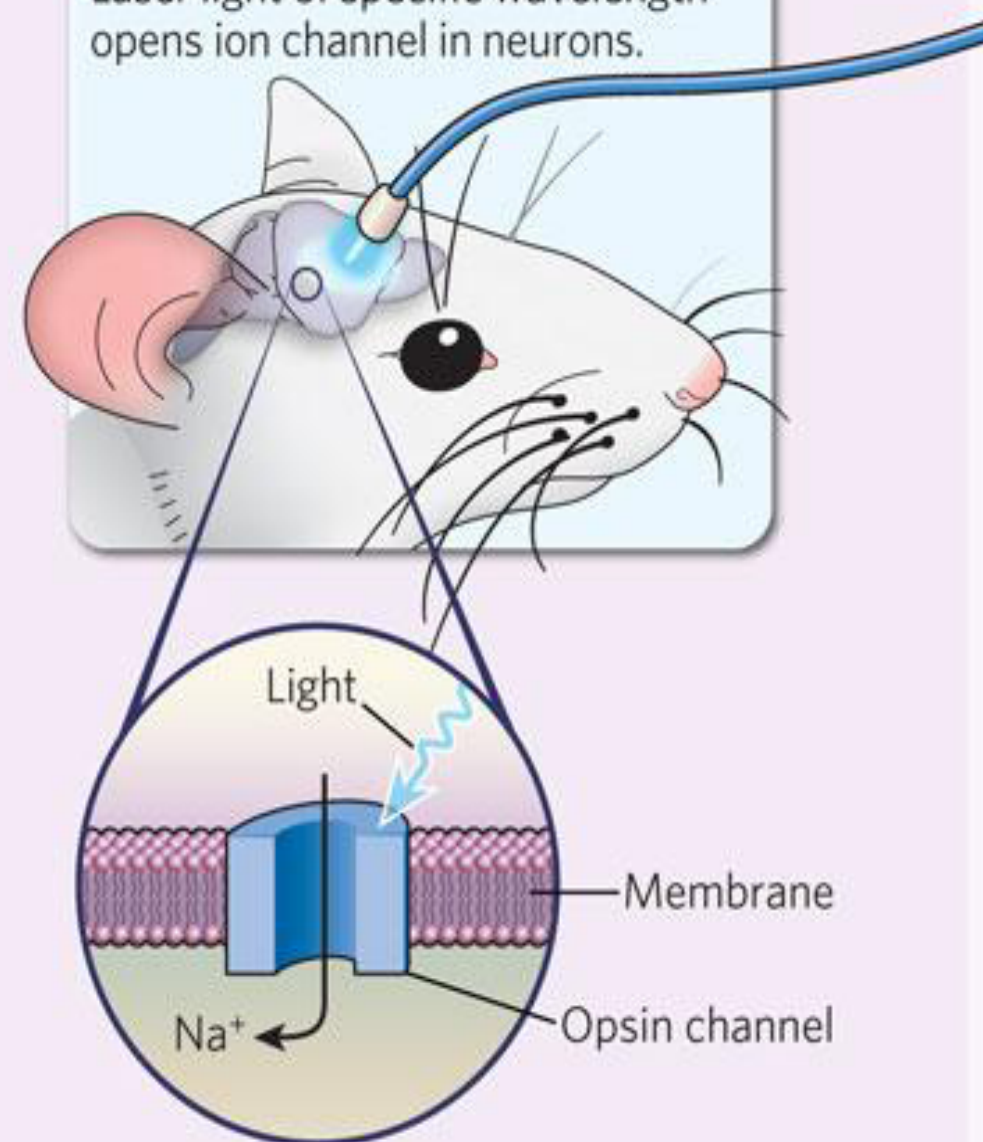
STEP 4

Insert 'optrode', fibre-optic cable plus electrode.



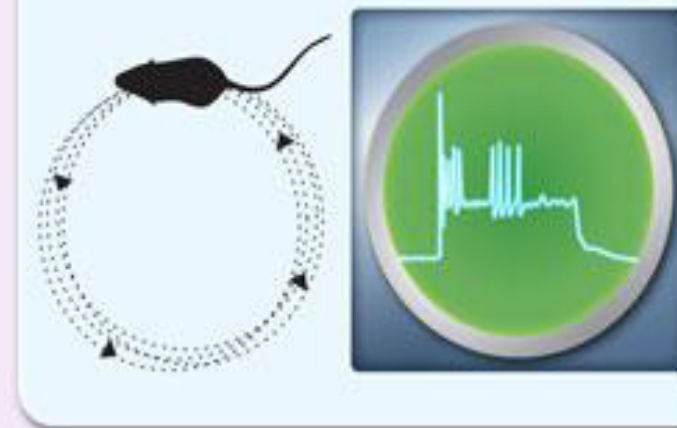
STEP 5

Laser light of specific wavelength opens ion channel in neurons.



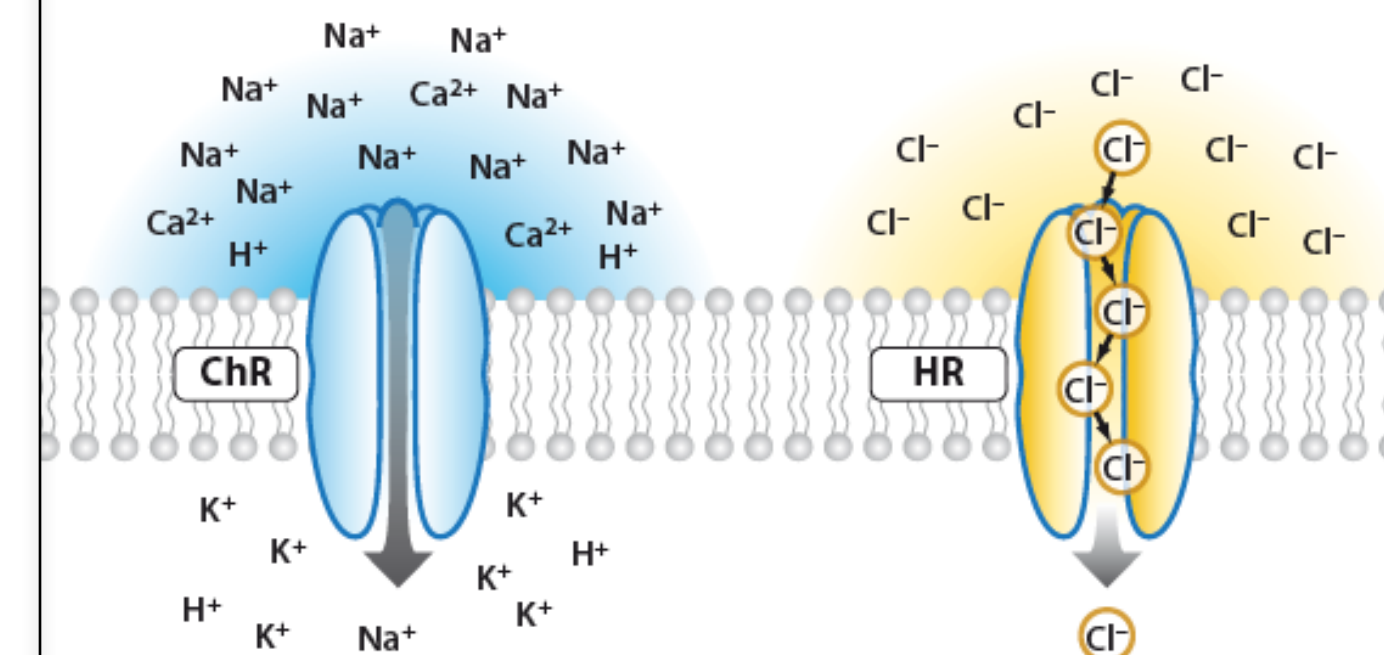
STEP 6

Record electrophysiological and behavioural results.



various opsins could be used:

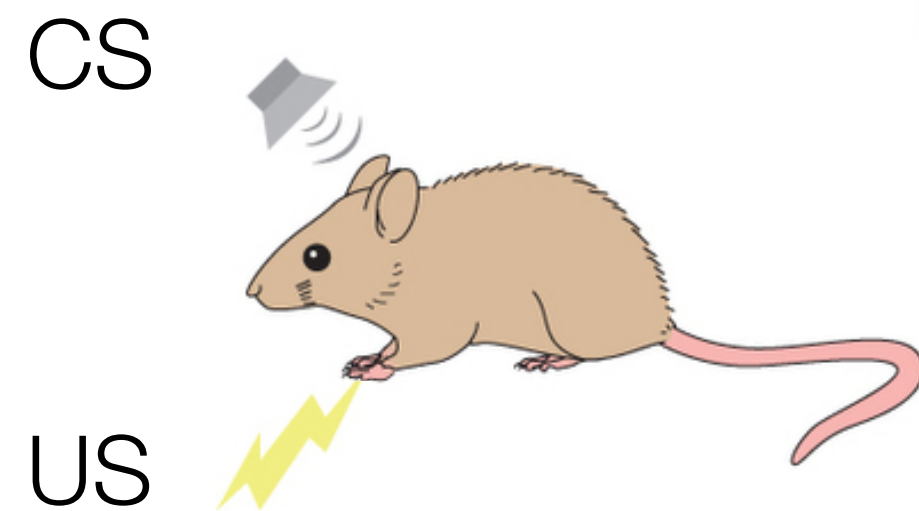
Channelrhodopsins / Halorhodo.



Emotional Memory

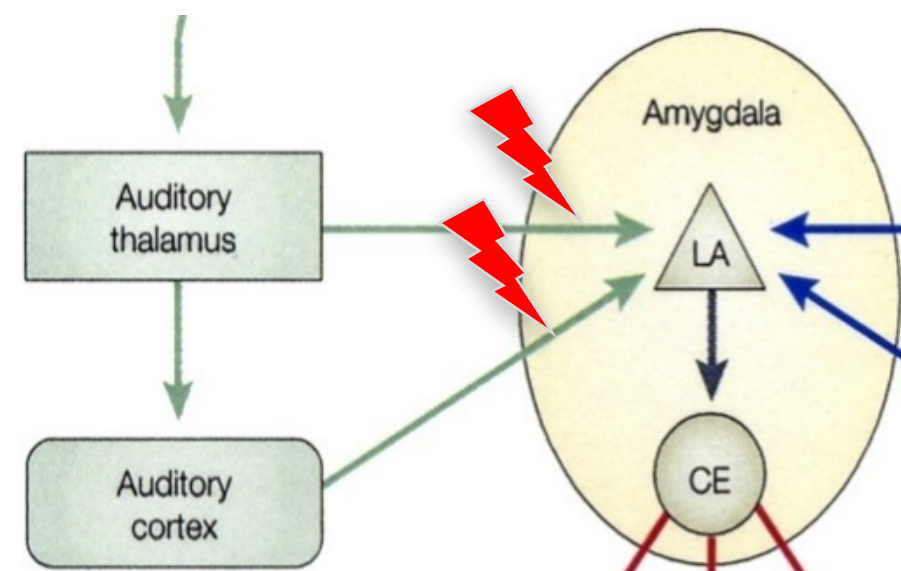
Brain regions involved : the amygdala

CS-US conditioning
(sound-shock)

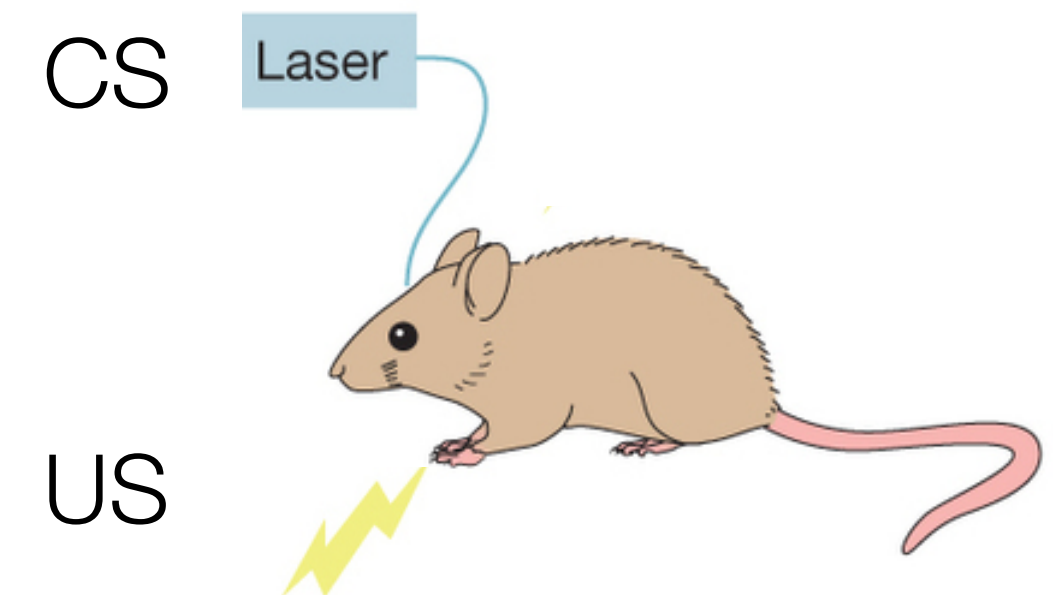


Optogenetics

Stimulation of auditory fibres
connecting the lateral amygdala



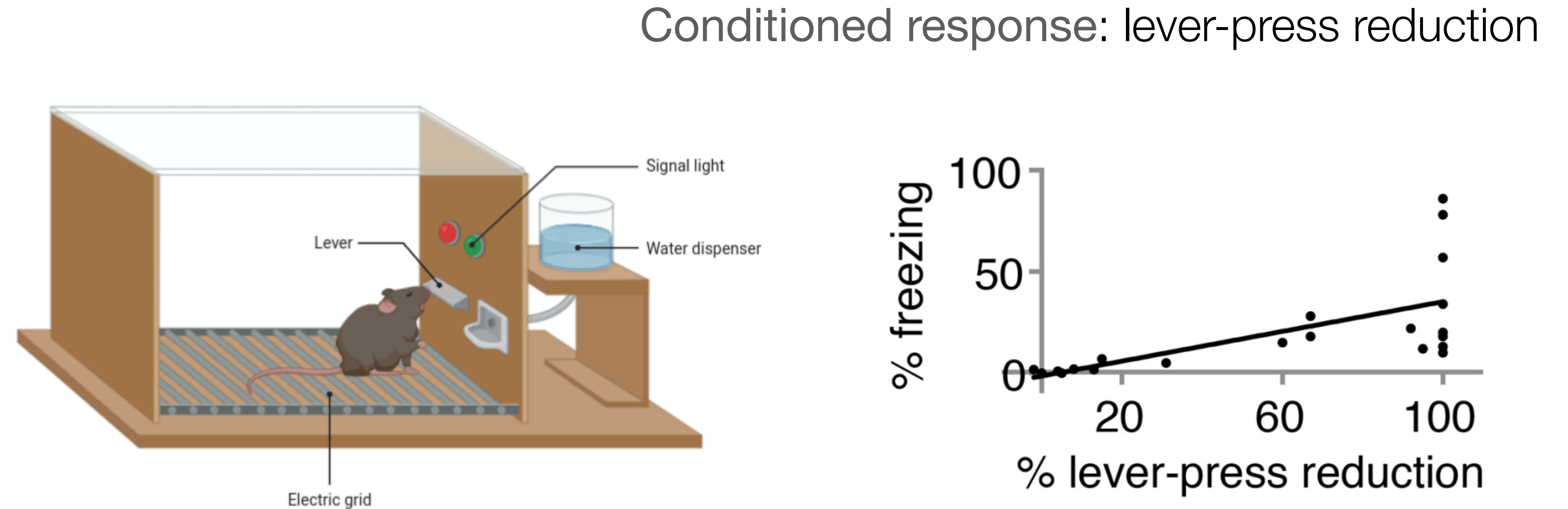
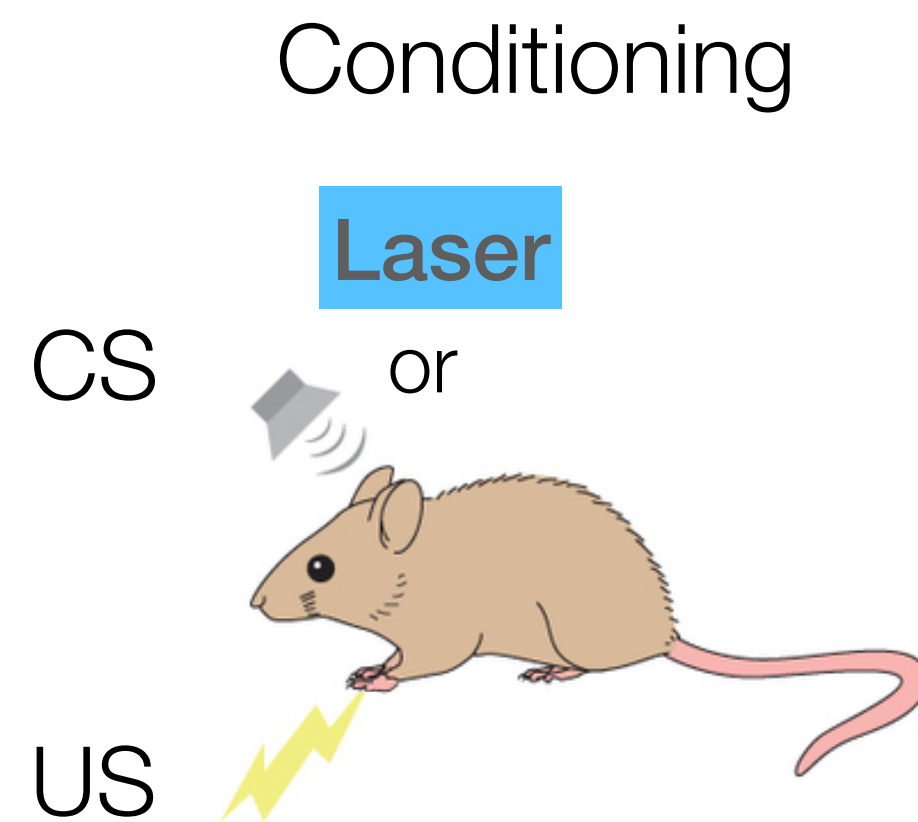
Optogenetics conditioning
(stim fibre-shock)



- Associative conditioning / fear conditioning (amygdala dependent)
- 1) Classical : association sound (CS) – shock (US)
- 2) Optogenetics : association stim fibre (CS) – shock (US)

Emotional Memory

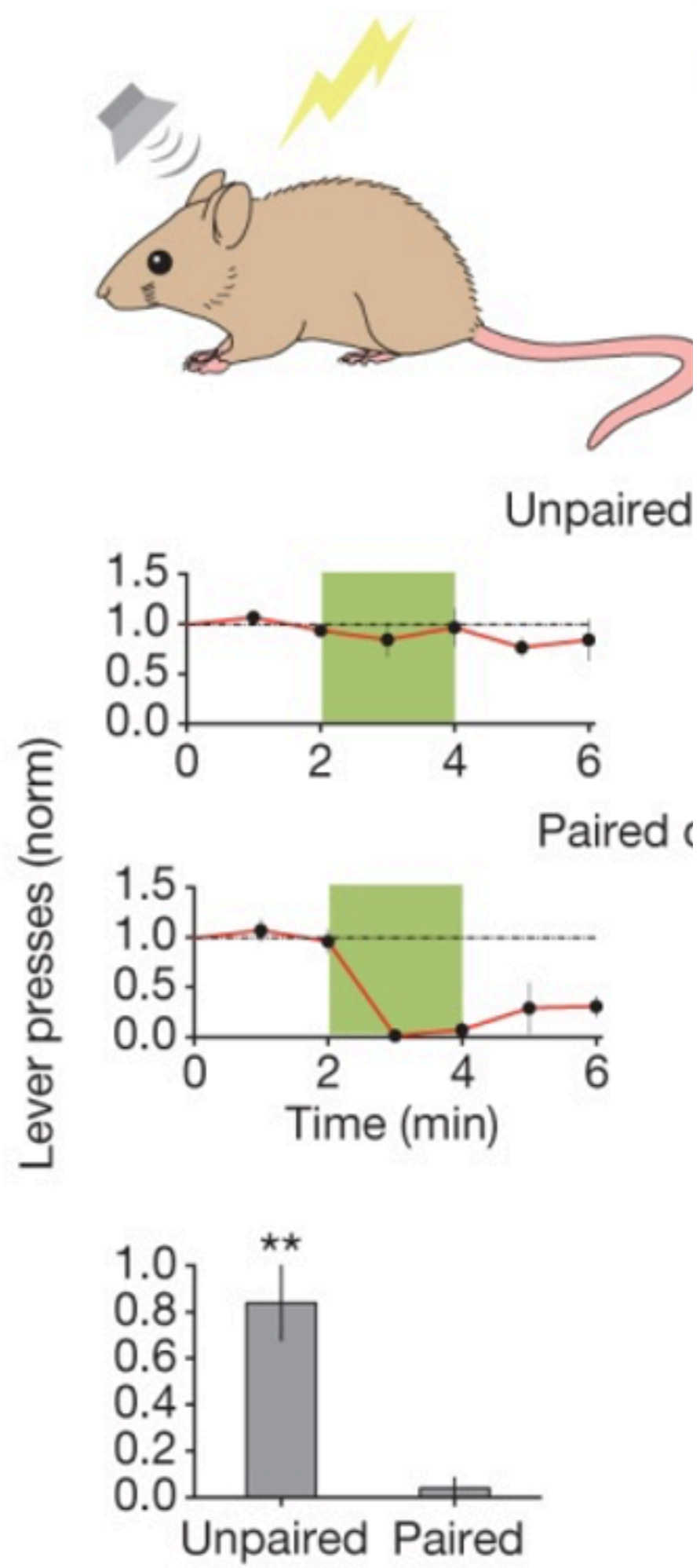
Brain regions involved : the amygdala



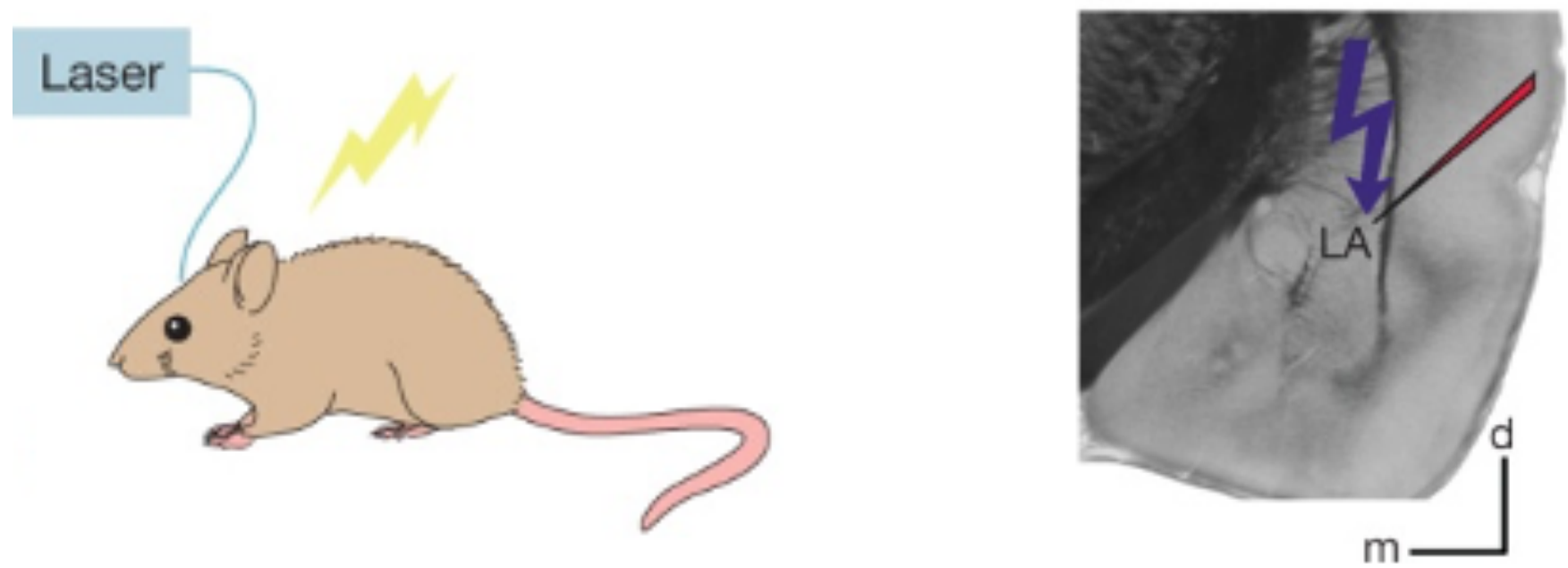
- Associative conditioning / fear conditioning (amygdala dependent)
 - 1) Classical : association sound (CS) – shock (US)
 - 2) Optogenetics : association stim fibre (CS) – shock (US)
- ➡ Conditioned response: lever-press reduction

Emotional memory - Brain regions involved: the amygdala

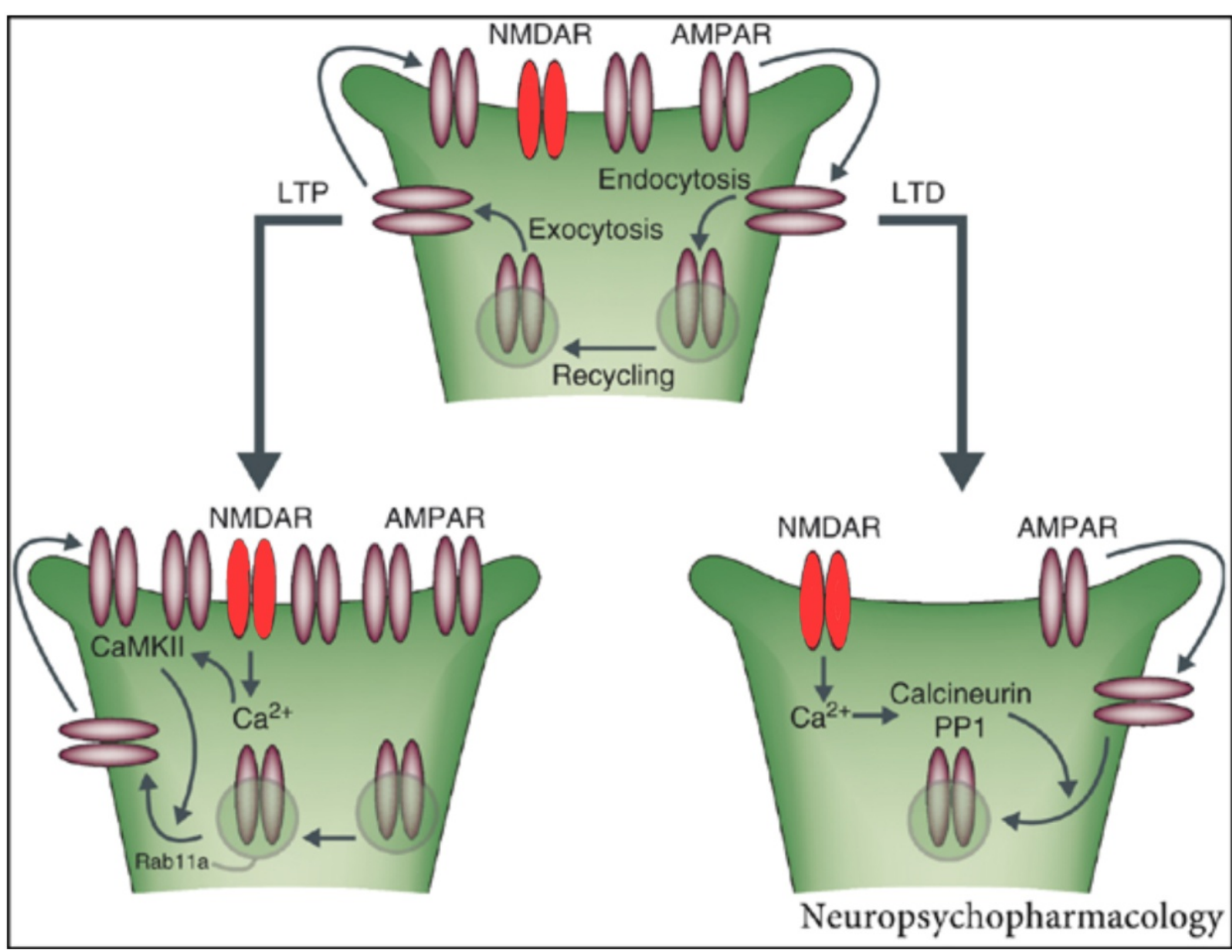
Classic (Sound)



Optogenetics (ODI: *optically driven input*)



back to basic - *Synaptic plasticity*

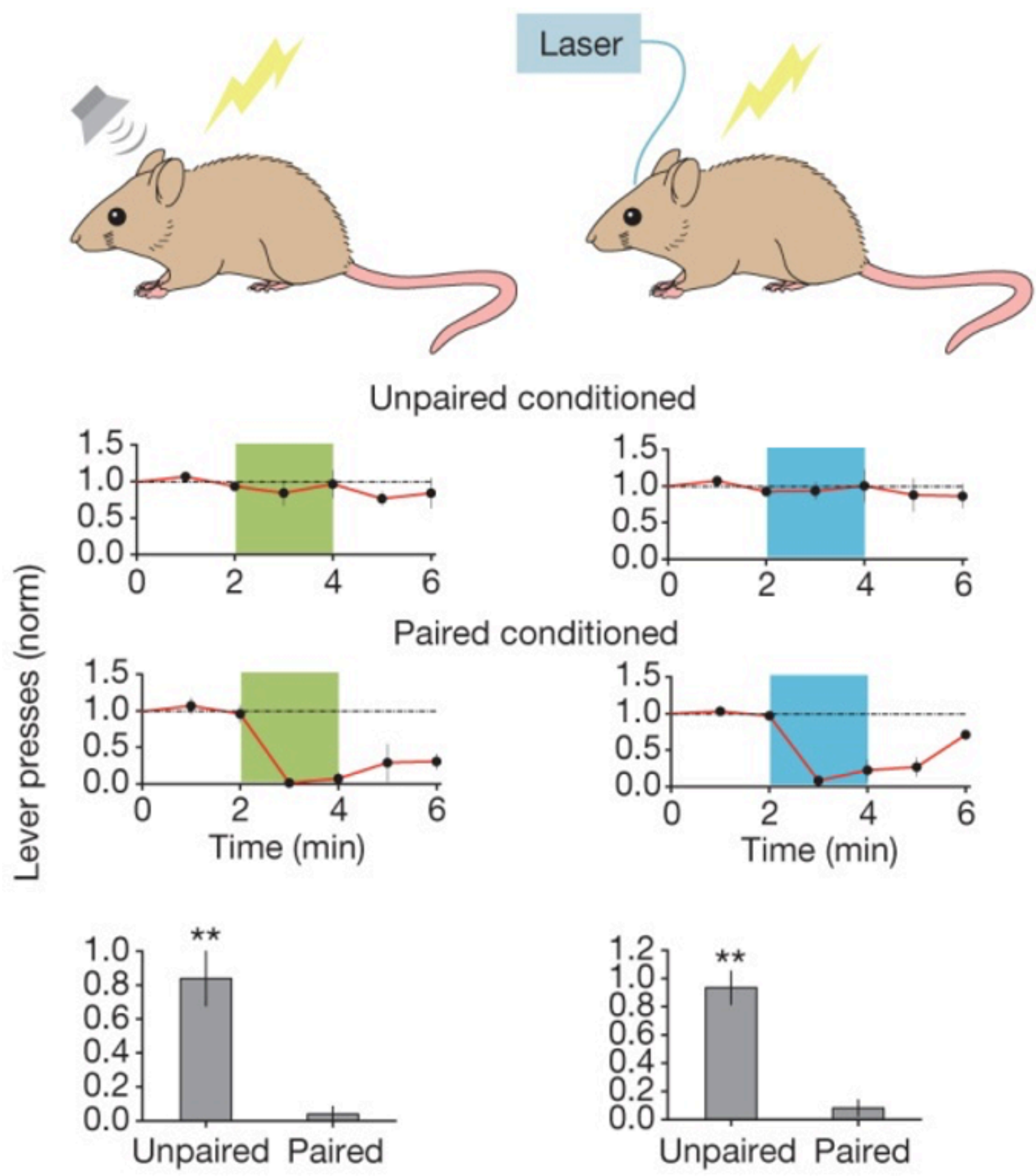


stim opto-shock association learnt

Emotional memory - Brain regions involved: the amygdala

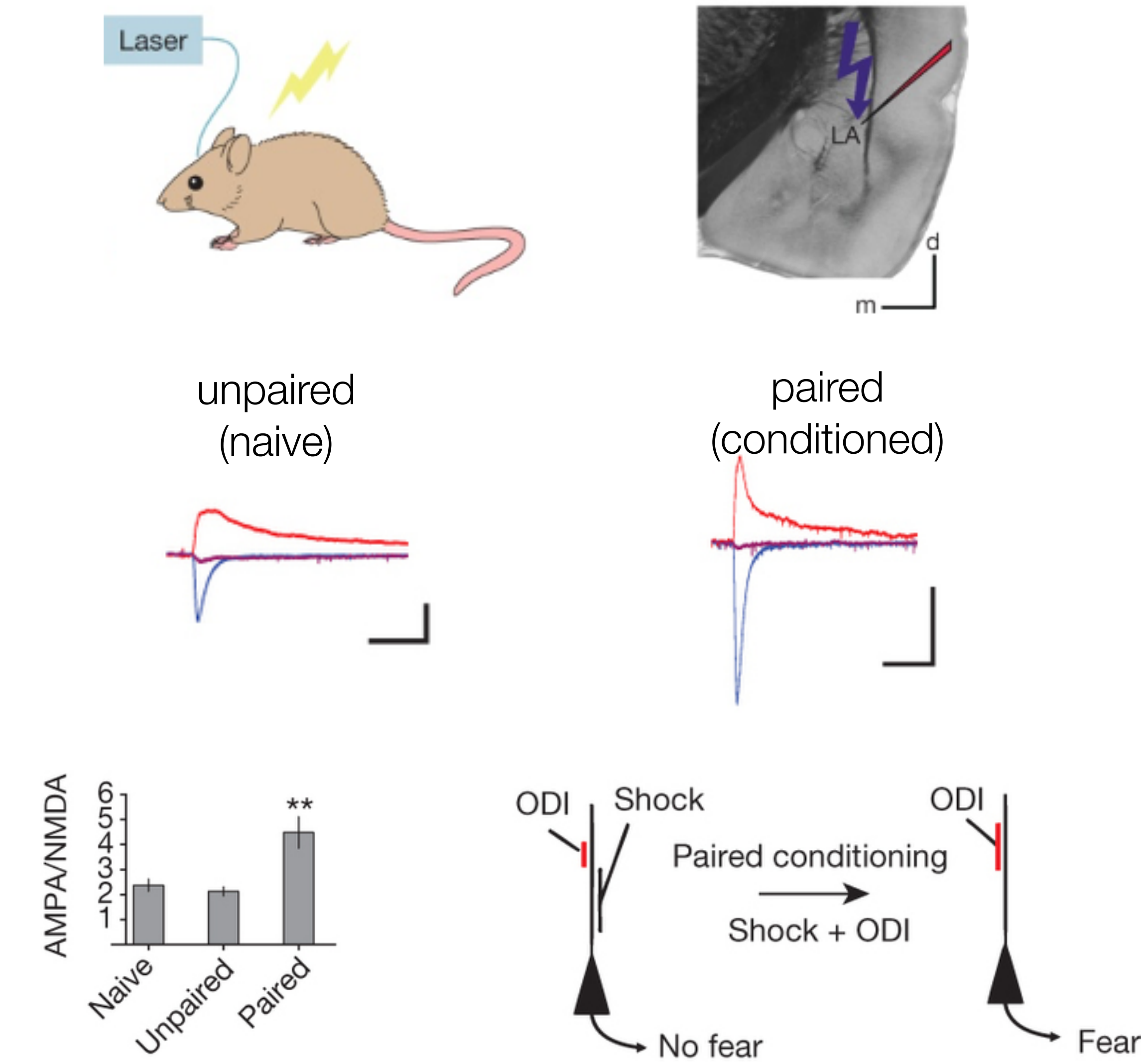
Classic (Sound)

Optogenetics (ODI)



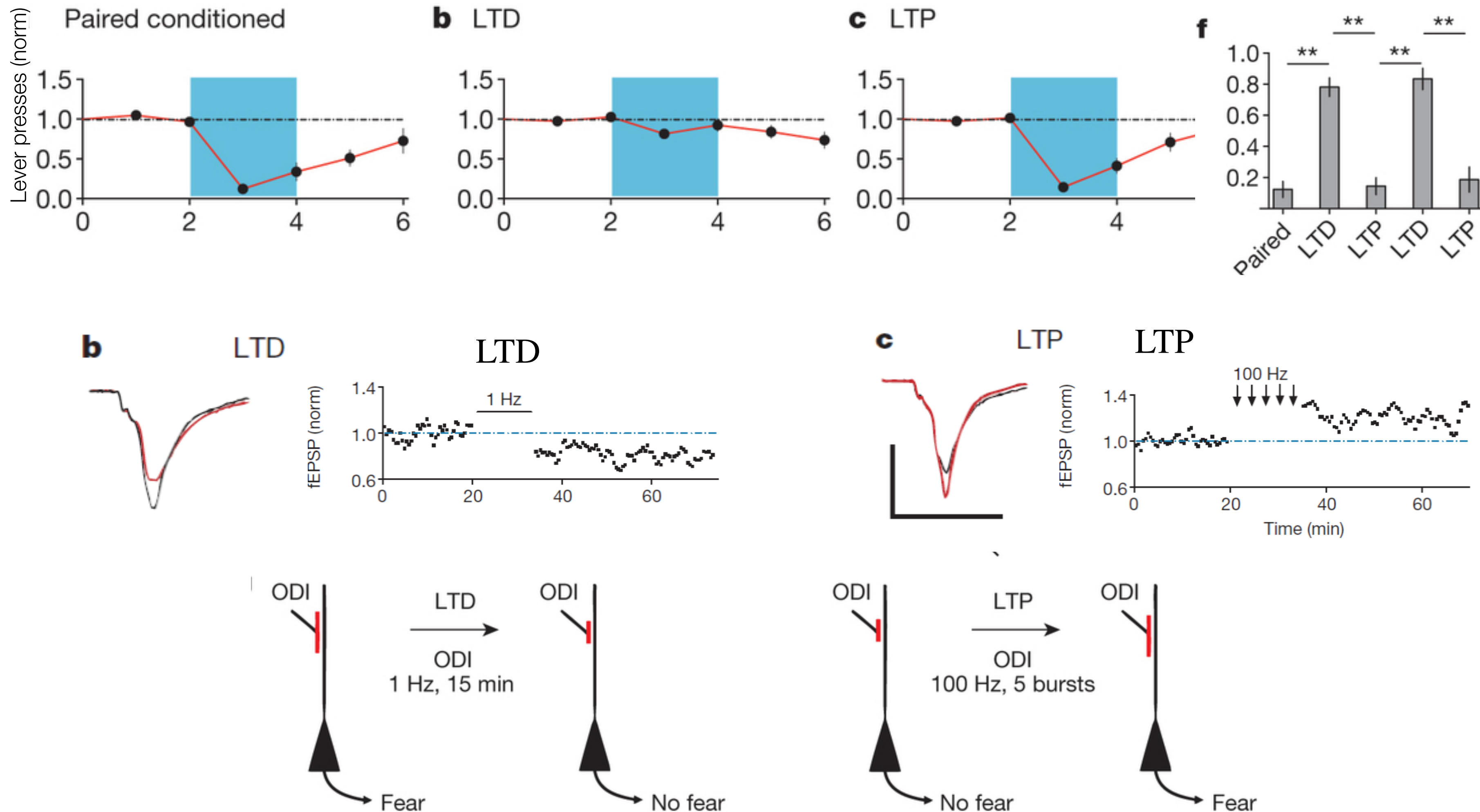
stim opto-shock association learnt

Optogenetics (ODI)



after conditioning, synaptic responses are potentiated

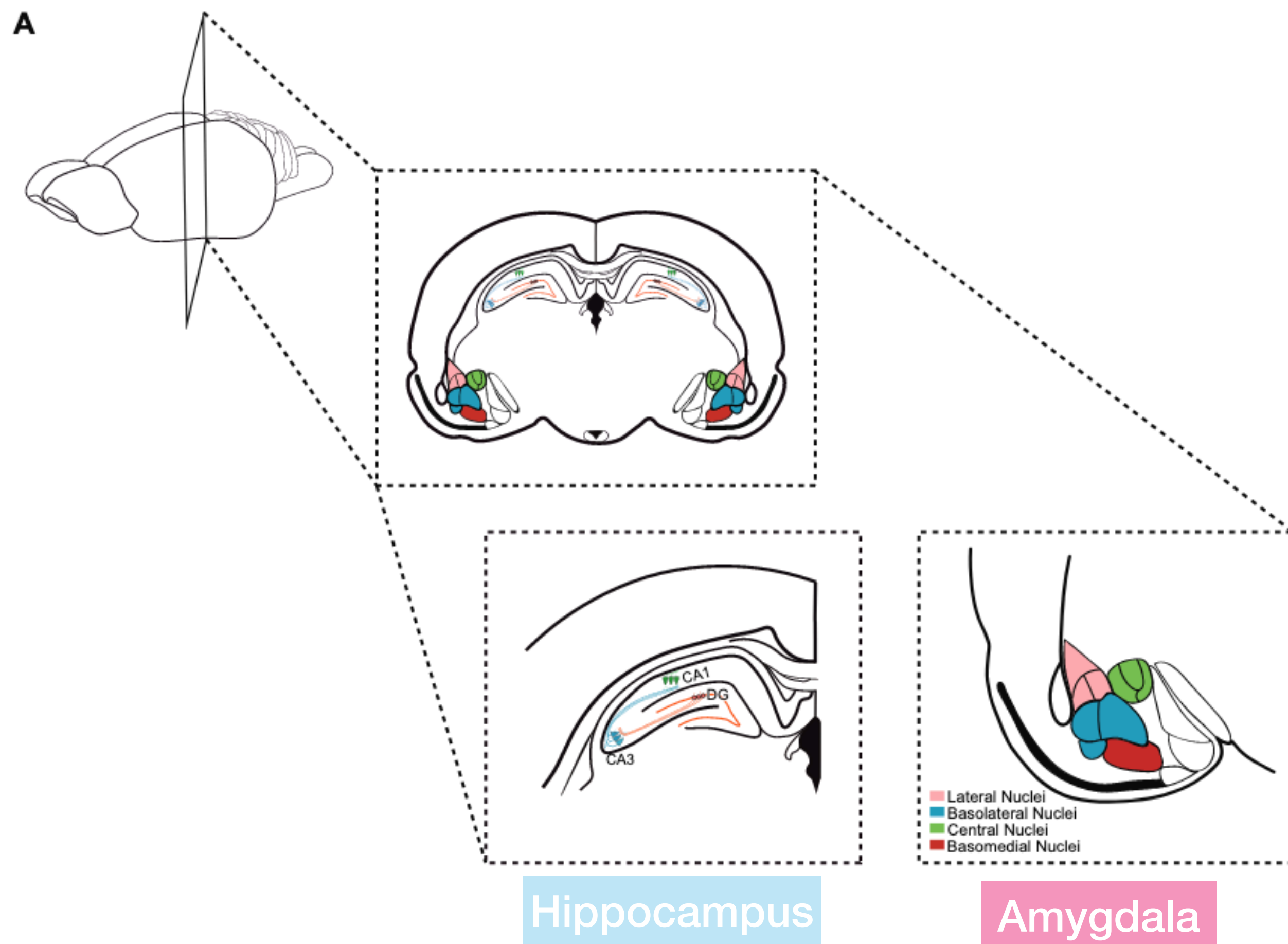
Emotional memory - Brain regions involved: the amygdala



LTD induction leads to memory inactivation whereas LTP induction reactivates the memory

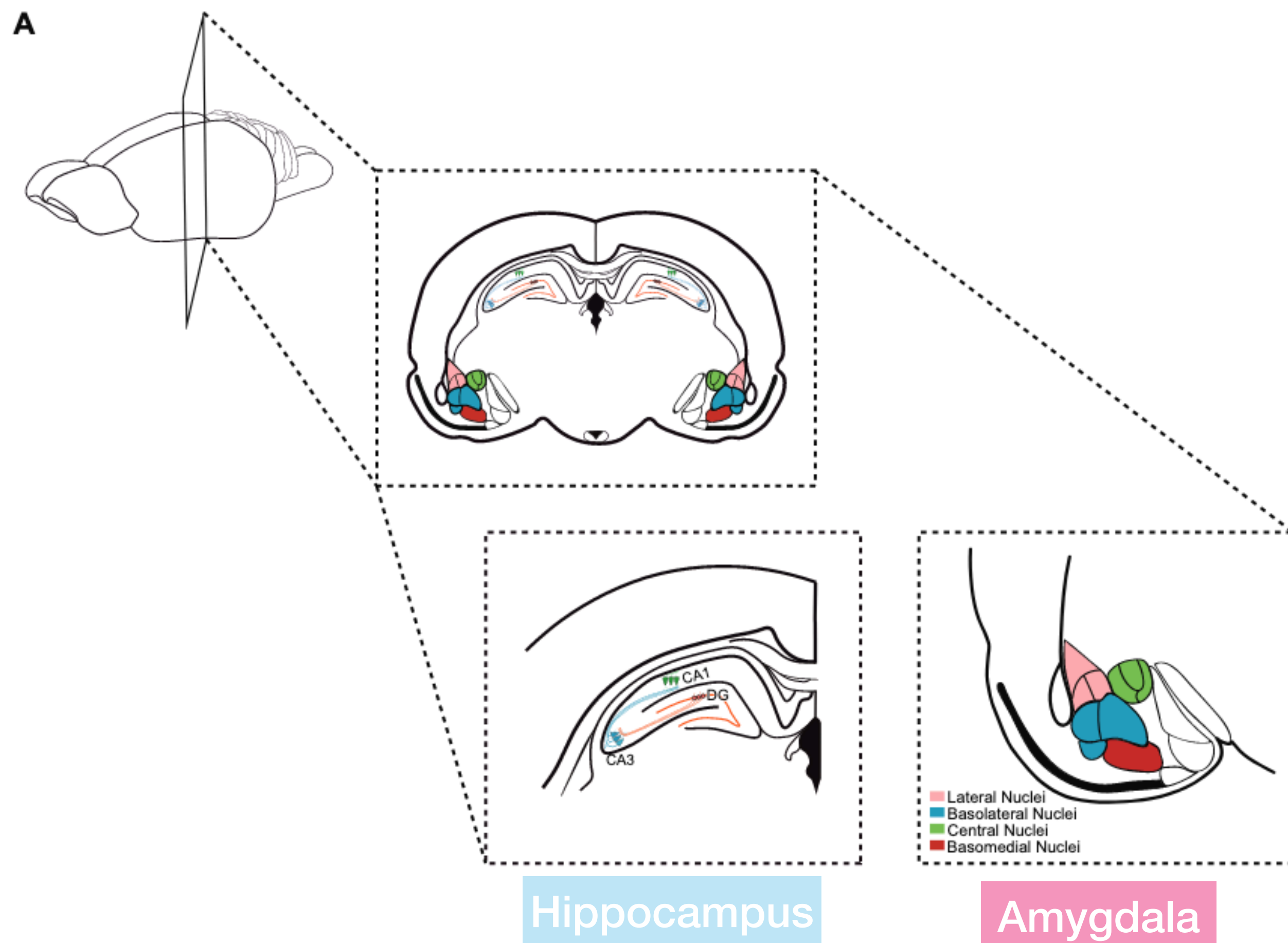
Emotional Memory

Brain regions involved : the amygdala, the hippocampus...

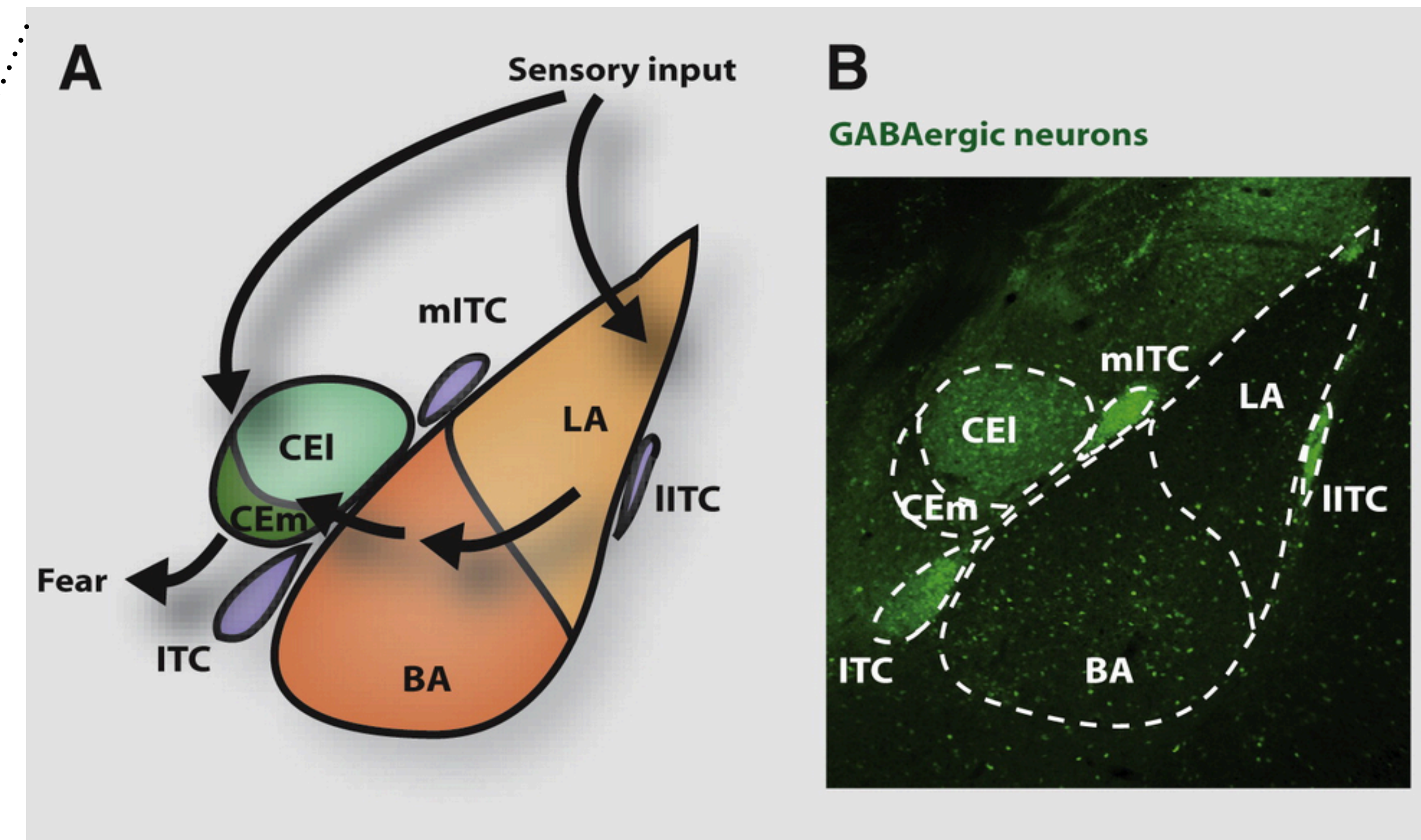


Emotional Memory

Brain regions involved : the amygdala

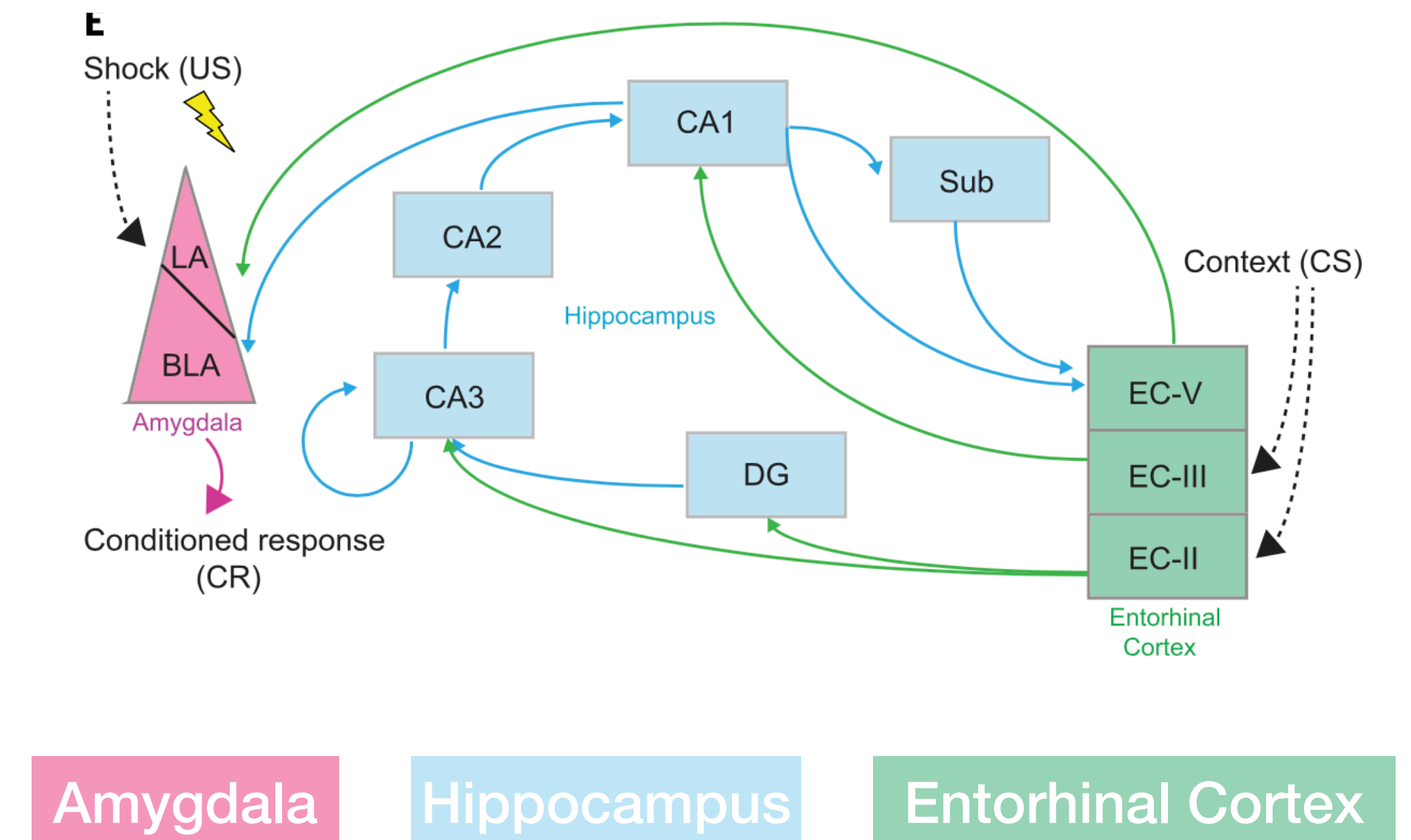
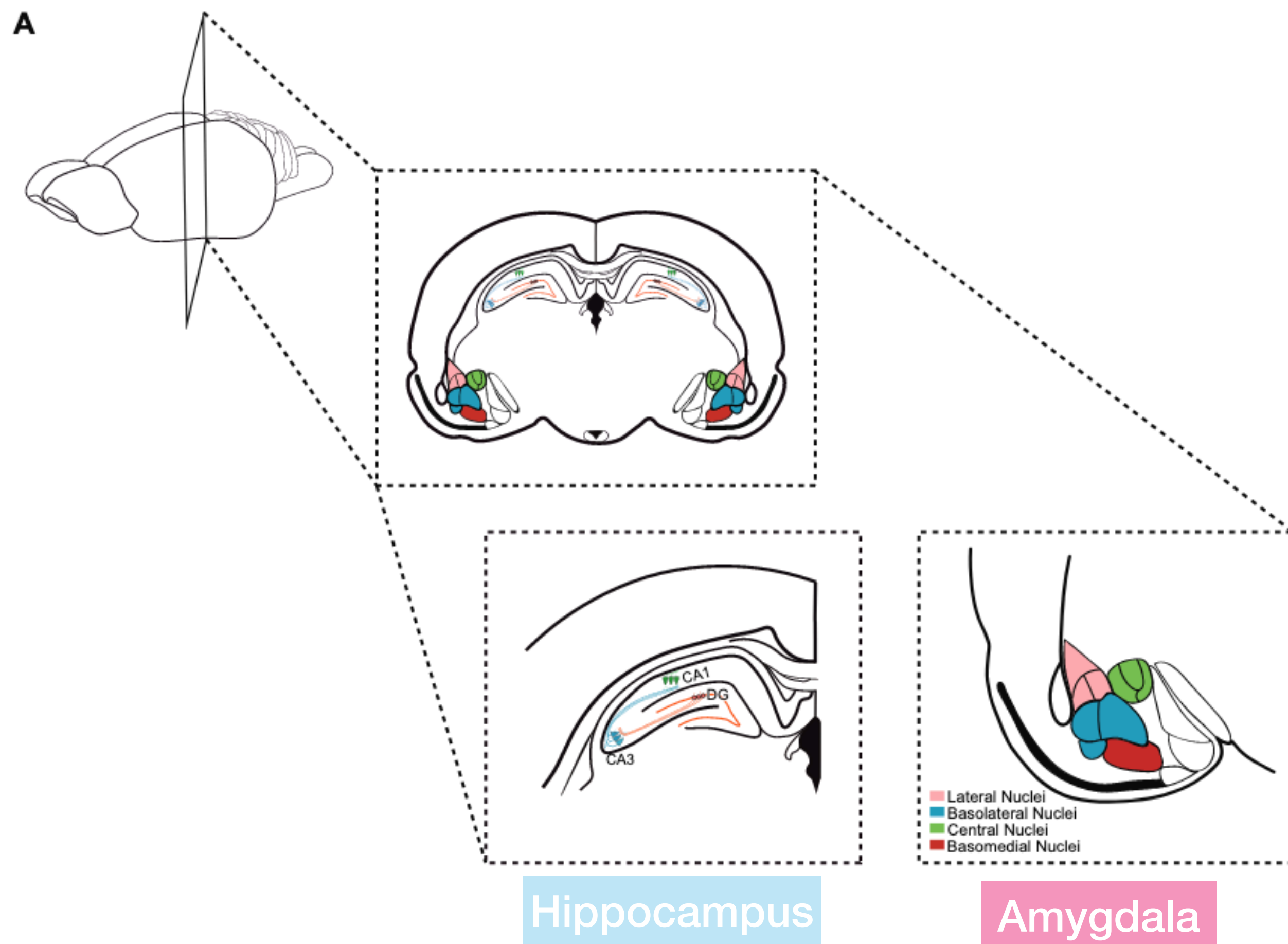


the Amygdala



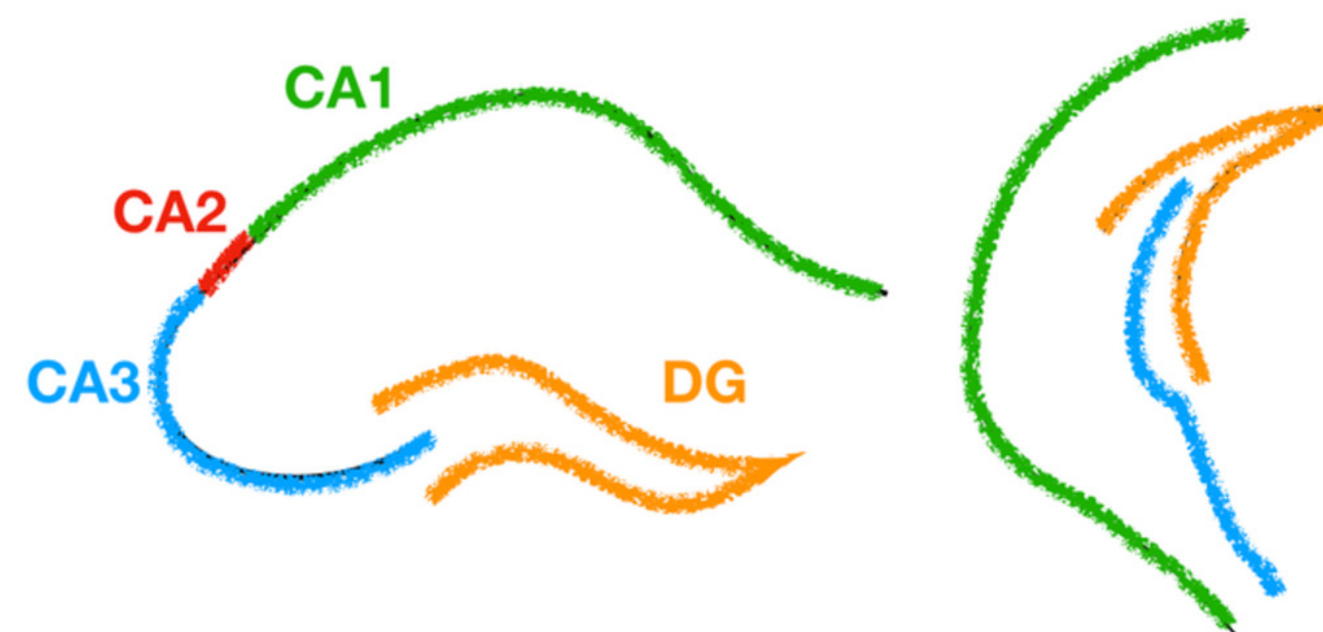
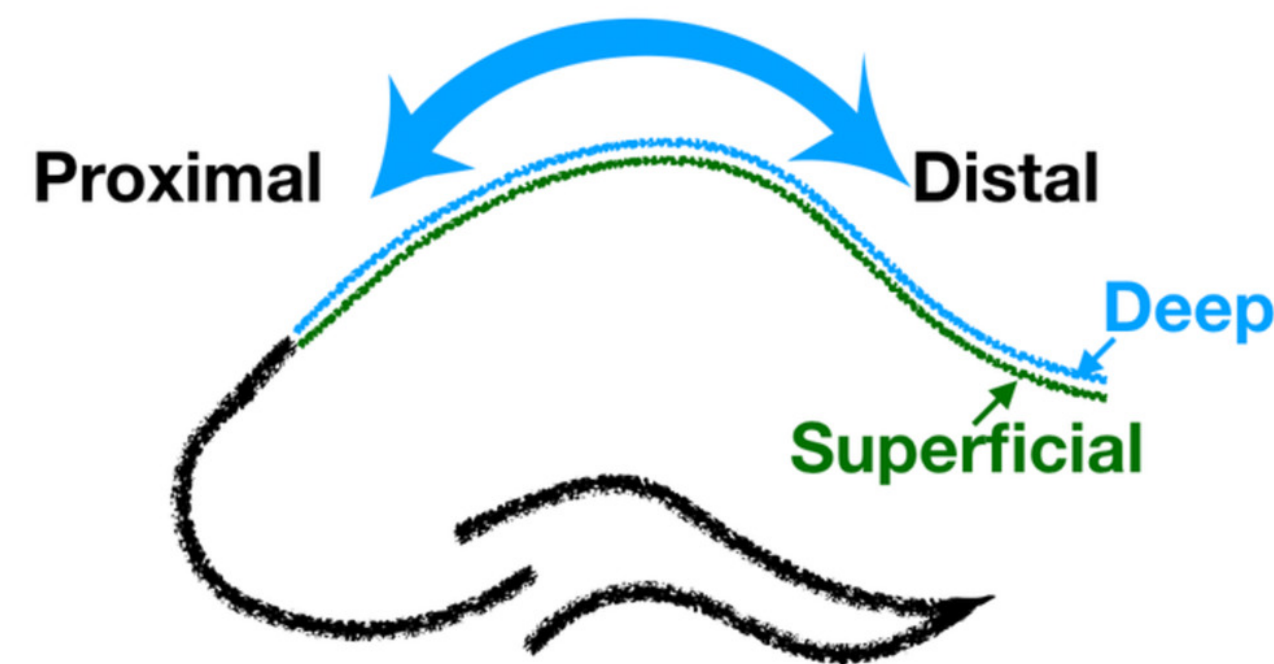
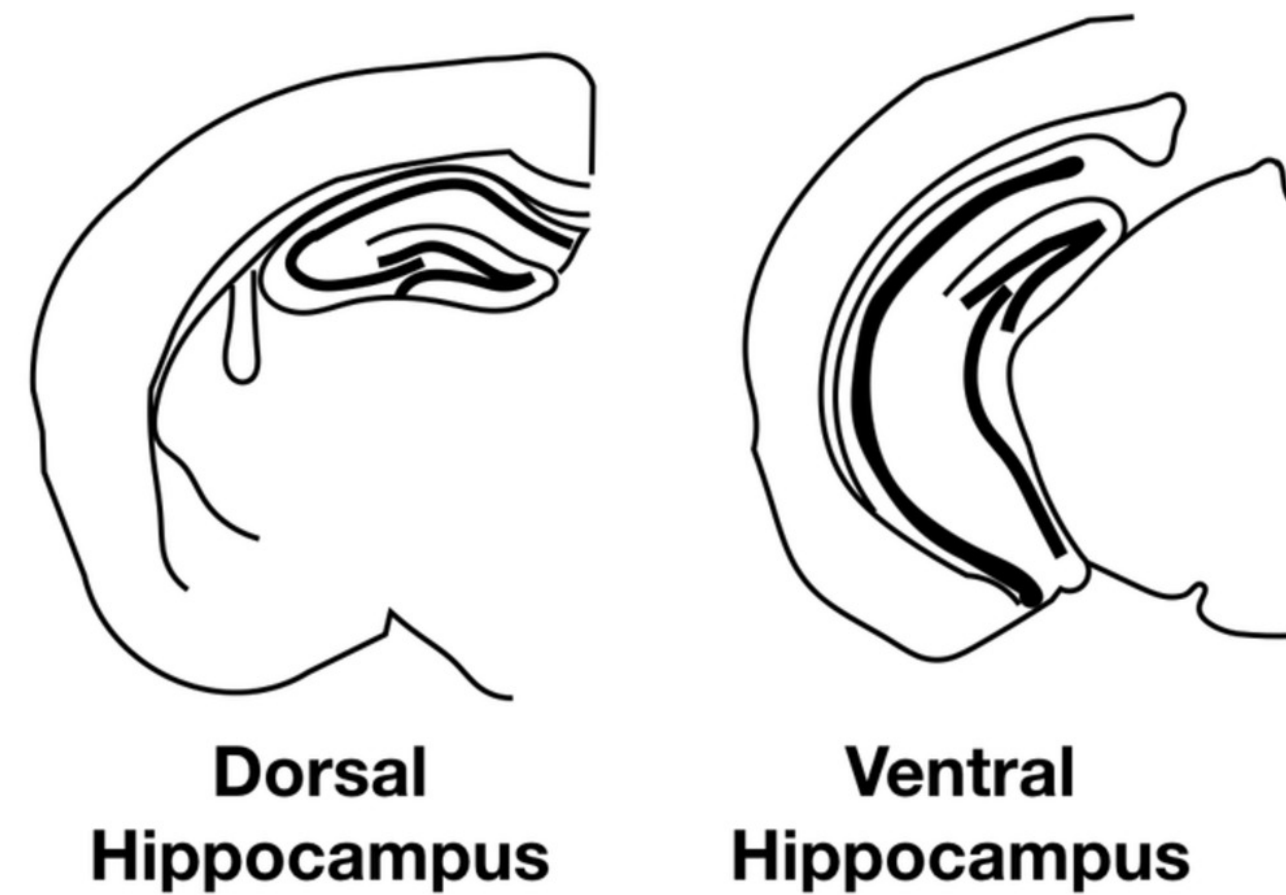
Emotional Memory

Brain regions involved : the amygdala, the hippocampus...

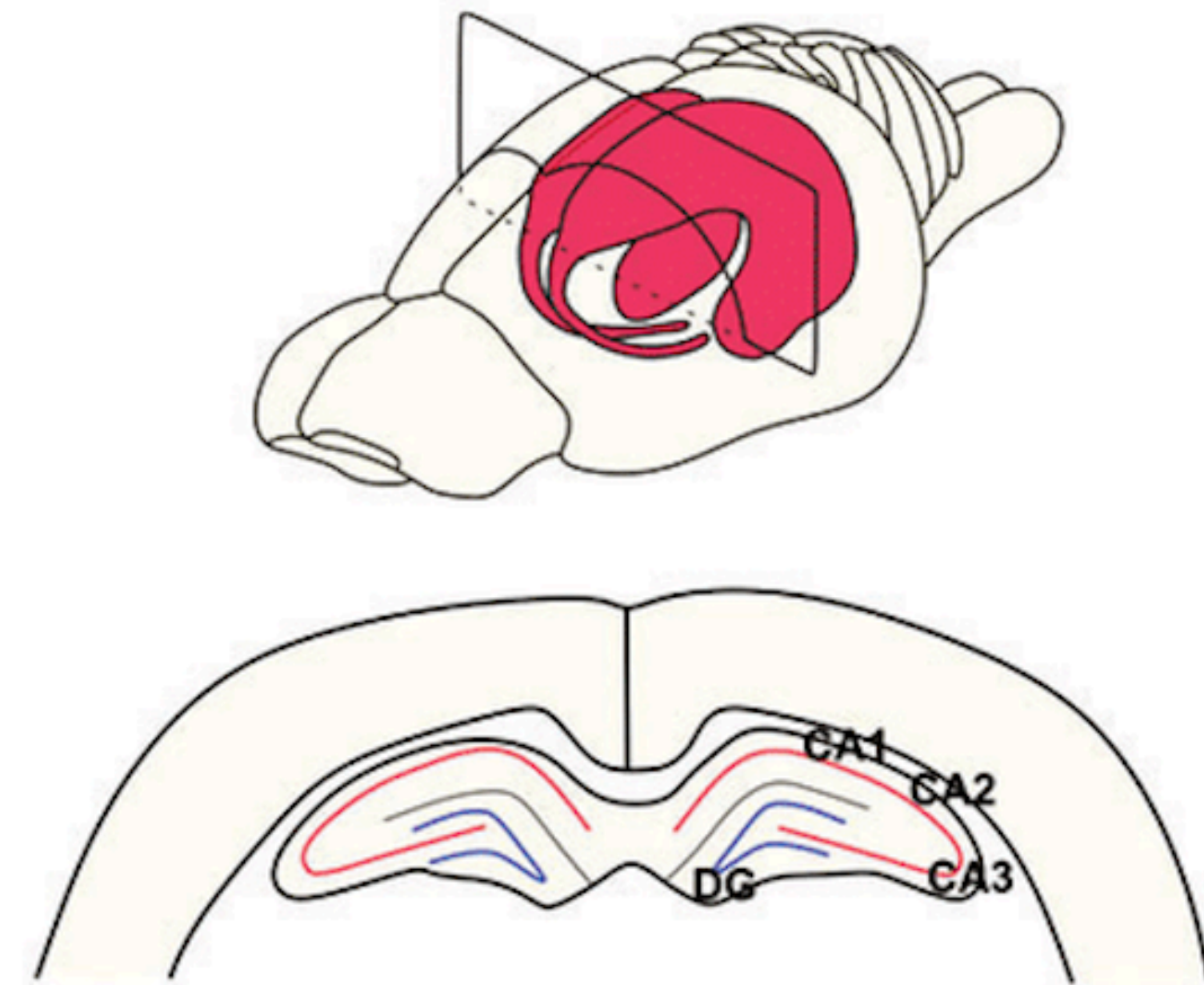


Emotional Memory

Brain regions involved : the hippocampus



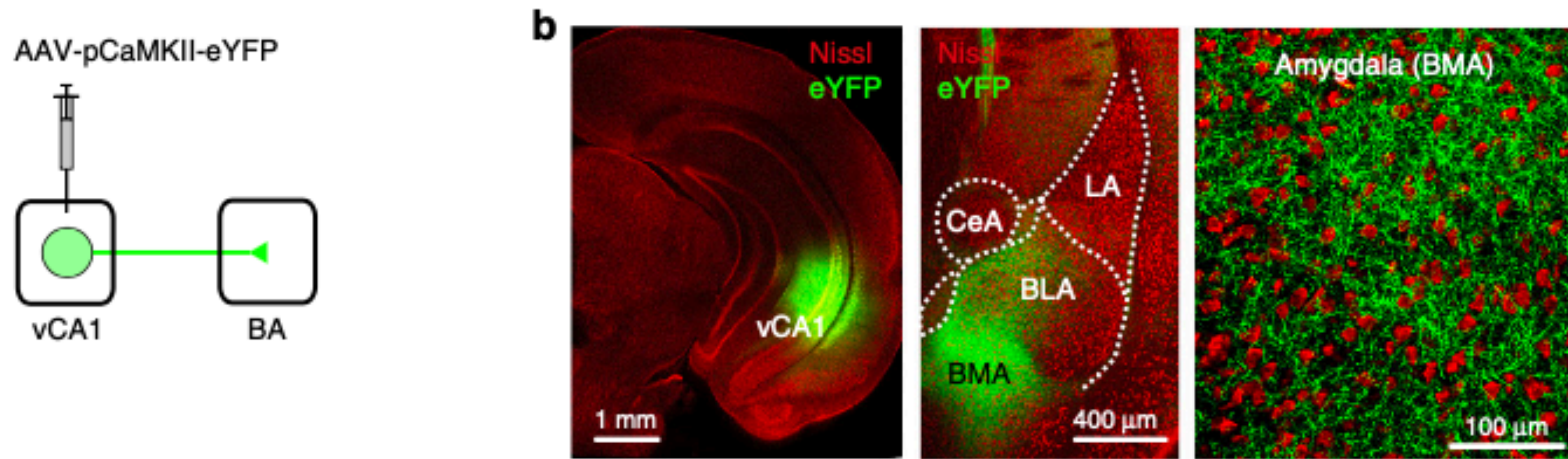
what's the role of the HPC in emotional memory?



Emotional Memory

Brain regions involved : the hippocampus

Are vCA1 pyramidal cells projecting to BA ?

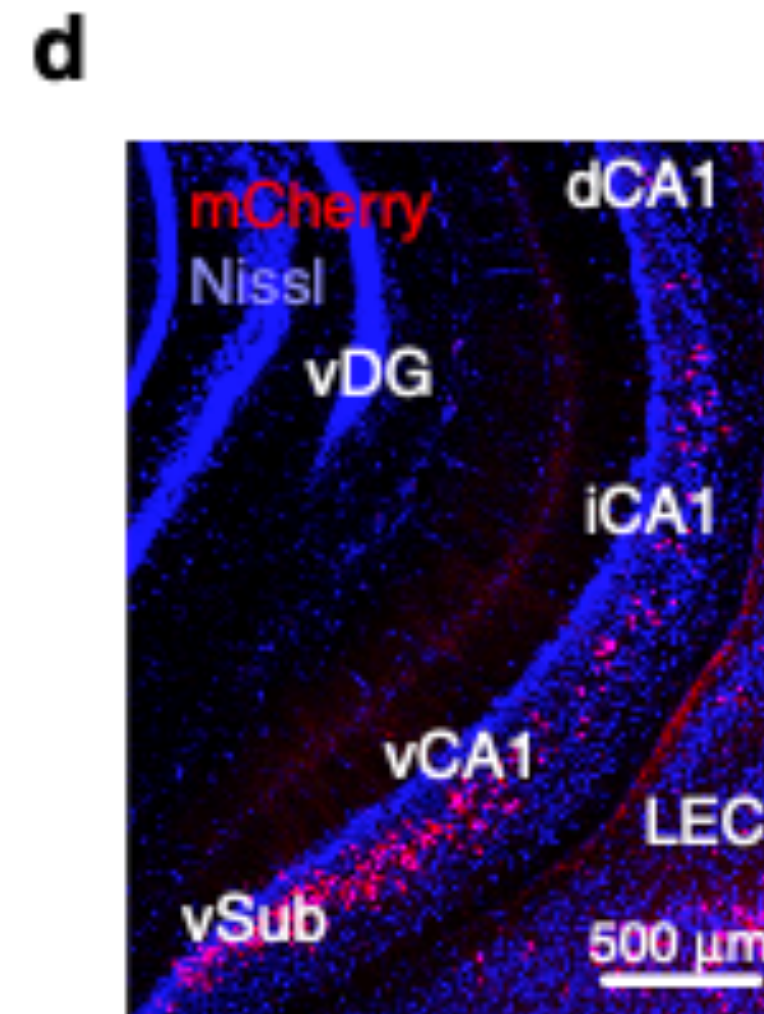
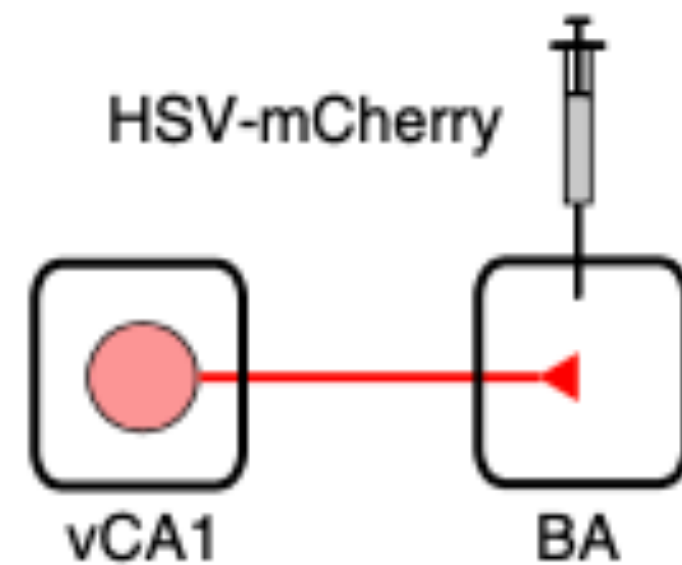


Ventral CA1 pyramidal cells project to the basal amygdala

Emotional Memory

Brain regions involved : the hippocampus

Are BA-projecting vCA1 neurons encoding context ?

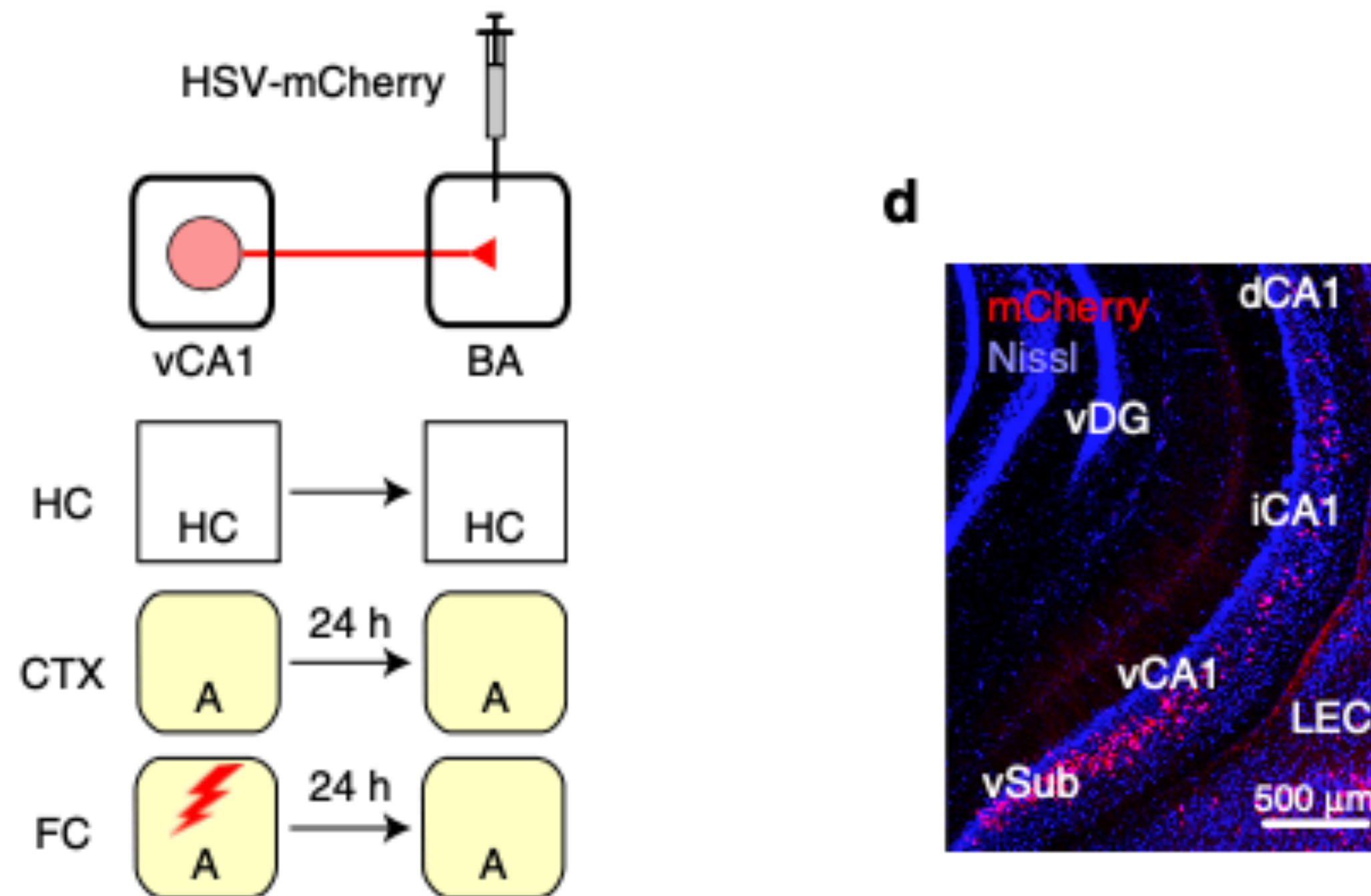


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Emotional Memory

Brain regions involved : the hippocampus

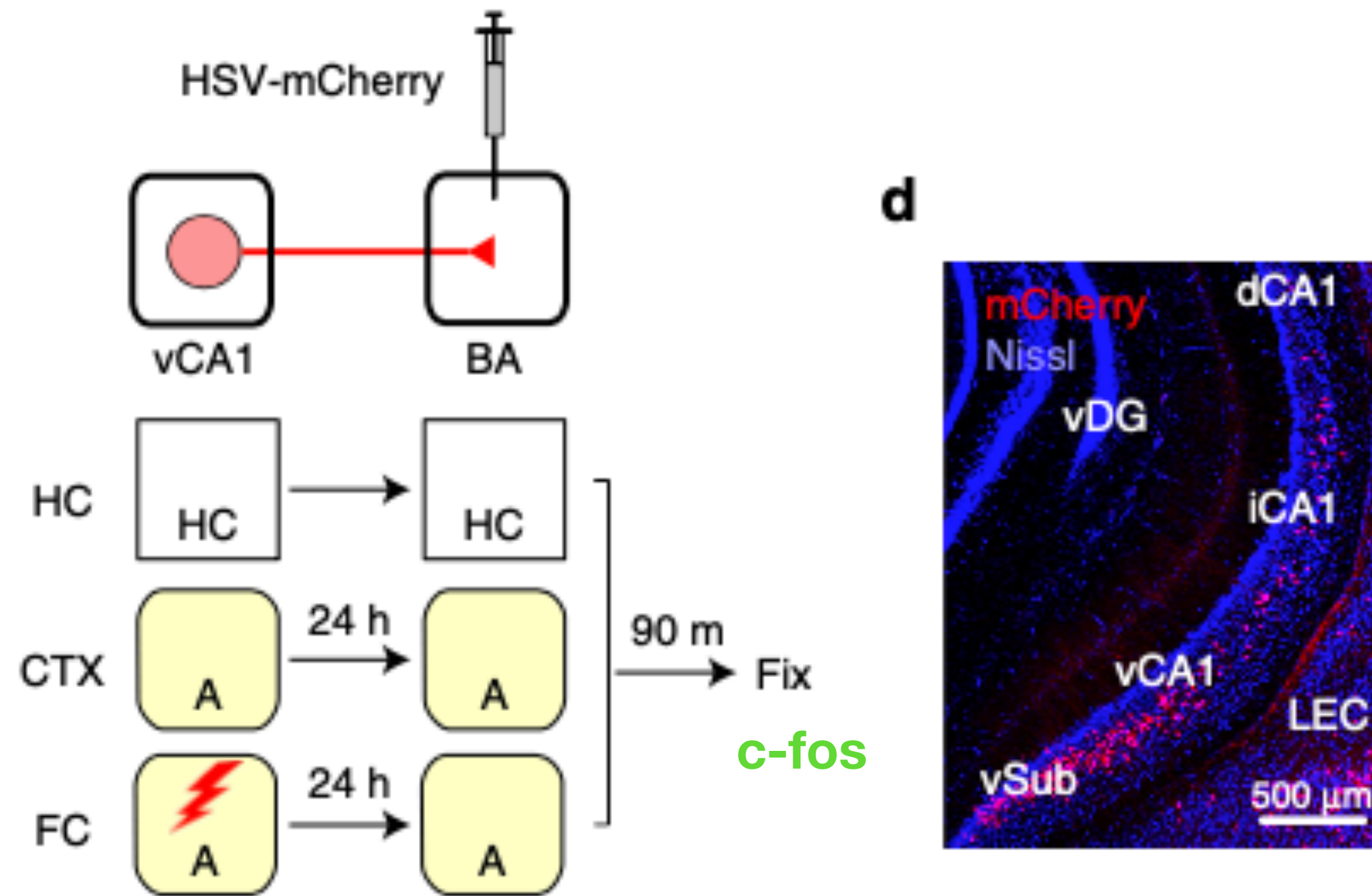
Are BA-projecting vCA1 neurons encoding context ?



Emotional Memory

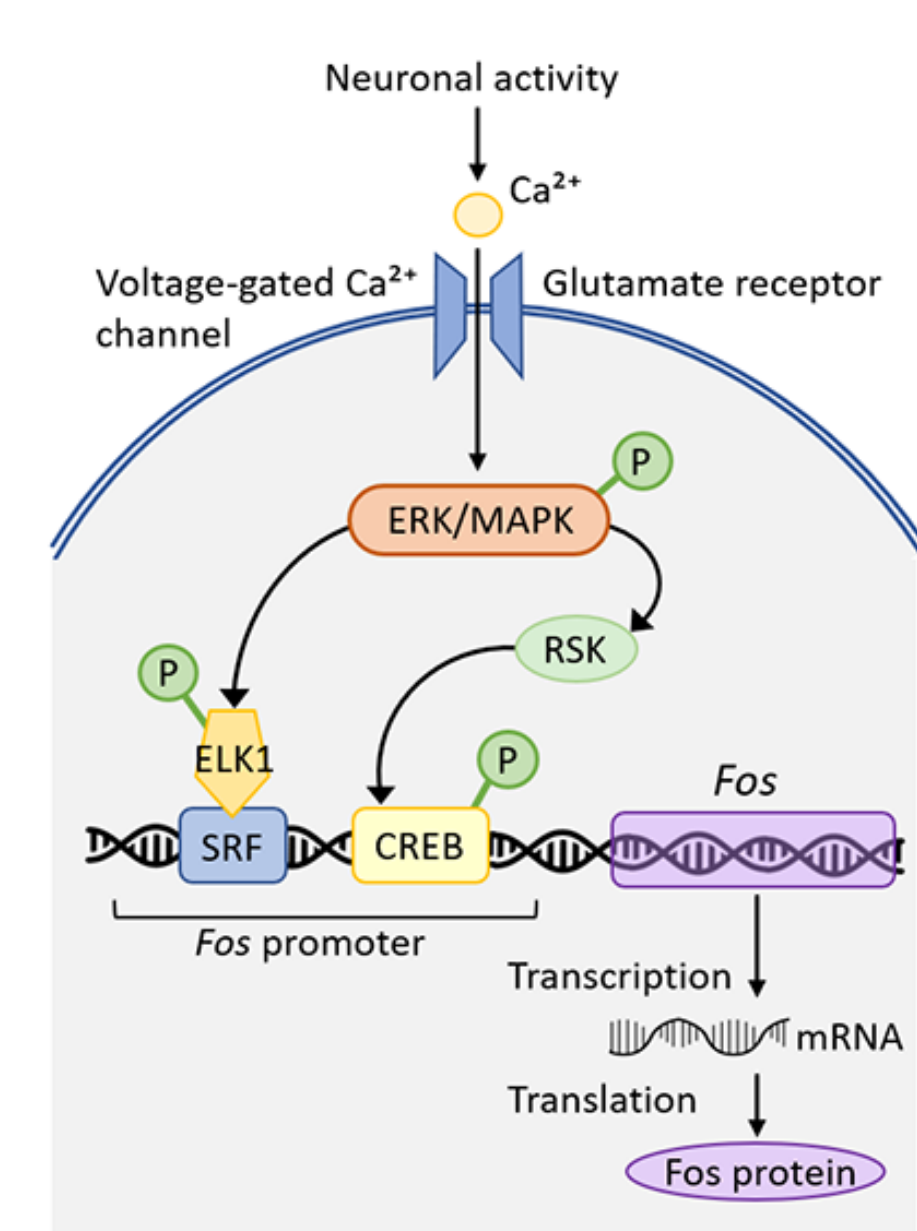
Brain regions involved : the hippocampus

Are BA-projecting vCA1 neurons encoding context ?



back to basic - IEG
Immediate Early Genes

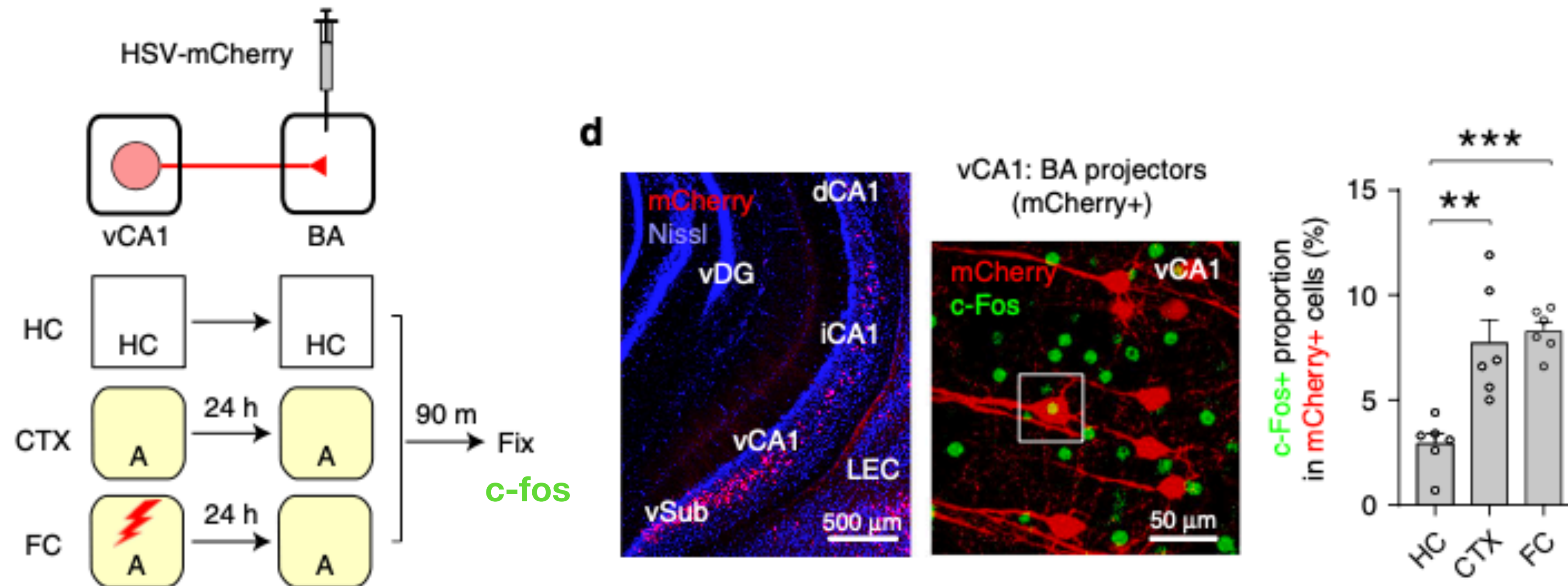
IEG represents a unique category of *transcription factors* induced in an activity-dependent manner that play a critical role in rapidly regulating gene expression.



Emotional Memory

Brain regions involved : the hippocampus

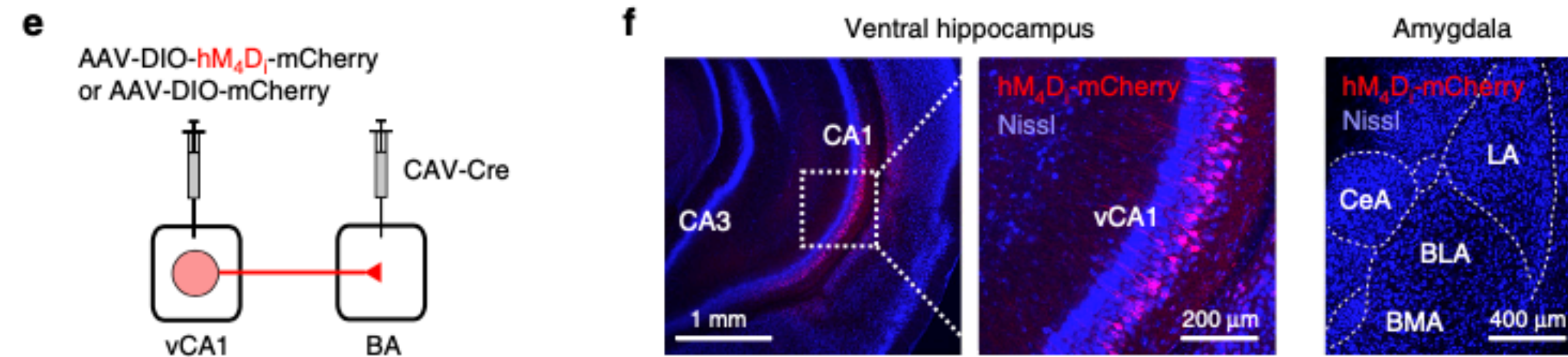
Are BA-projecting vCA1 neurons encoding context ?



about 10% of vCA1 pyramidal cells are active in a specific context

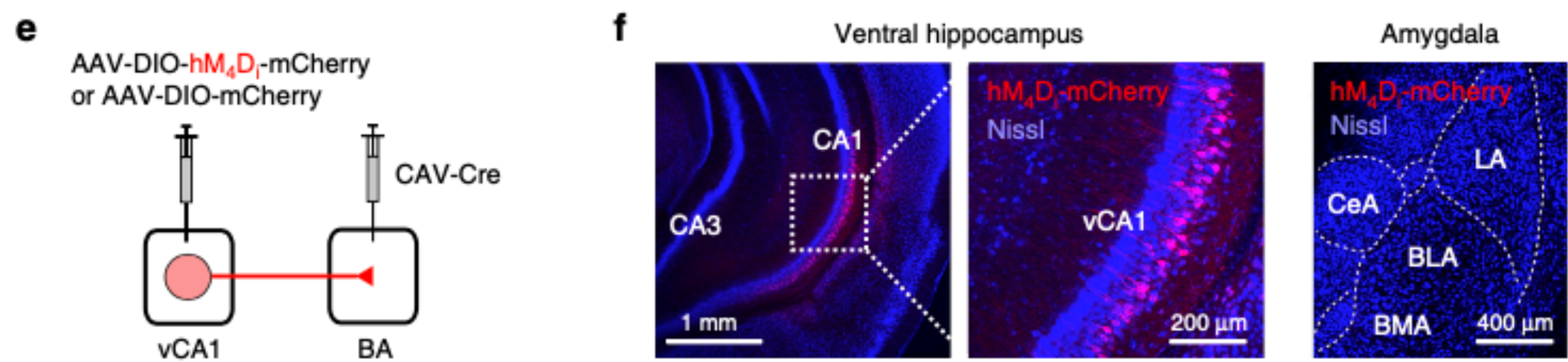
Emotional memory - Brain regions involved: the hippocampus

Is **the activity** of the vCA1 → BA pathway required for the acquisition of contextual memory?



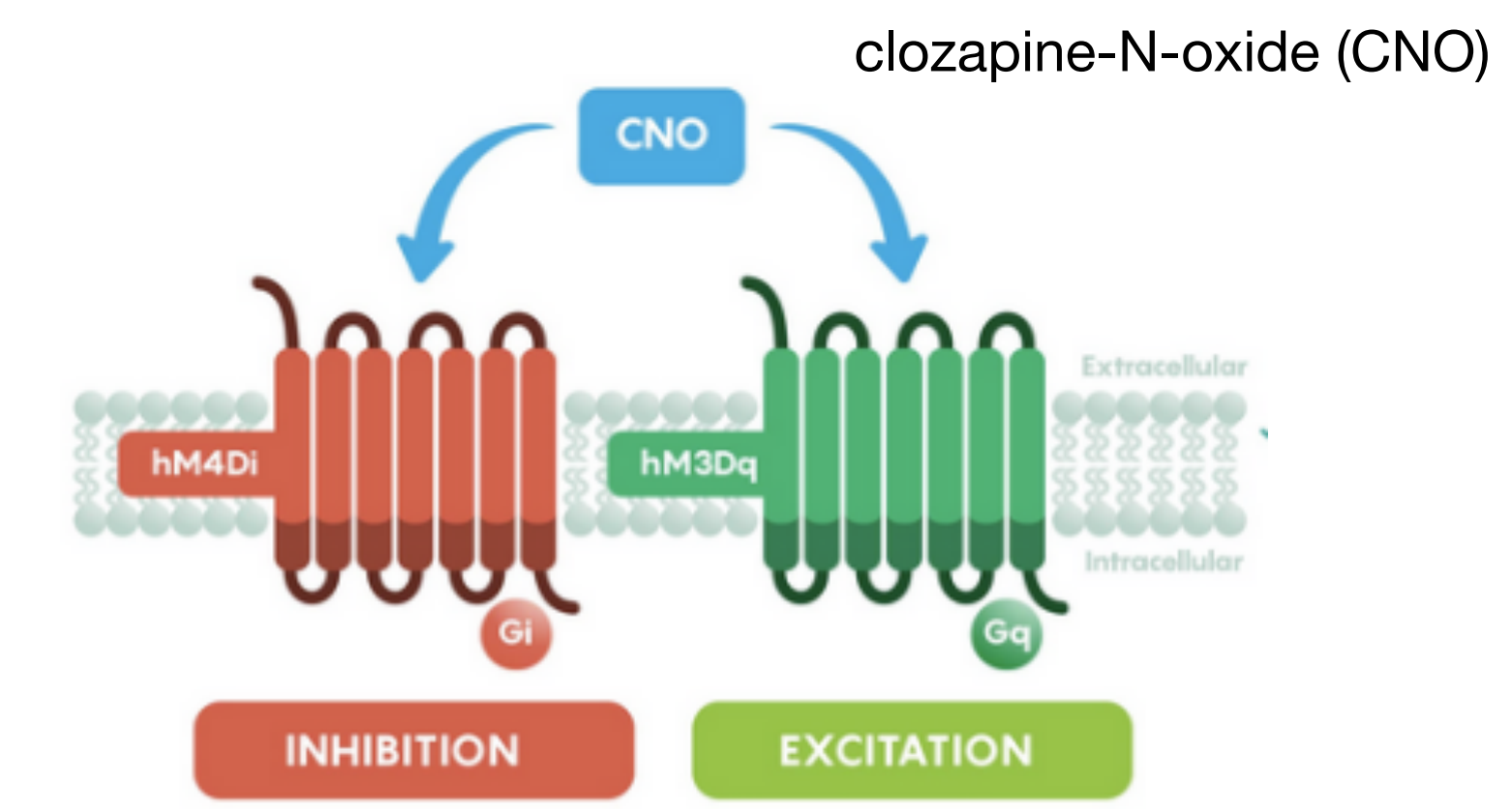
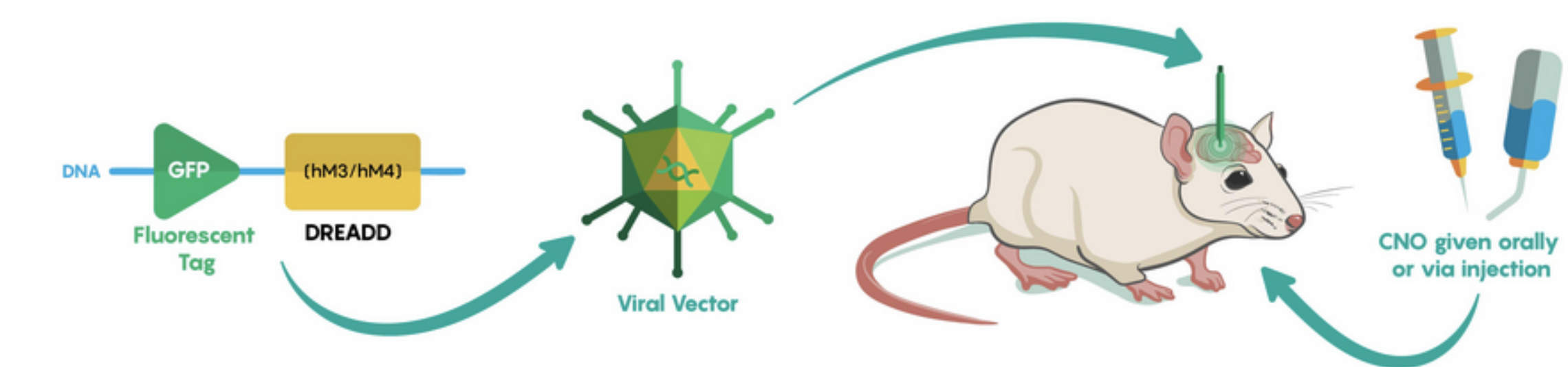
Emotional memory - Brain regions involved: the hippocampus

Is **the activity** of the vCA1 → BA pathway required for the acquisition of contextual memory?



Technique - DREADDs

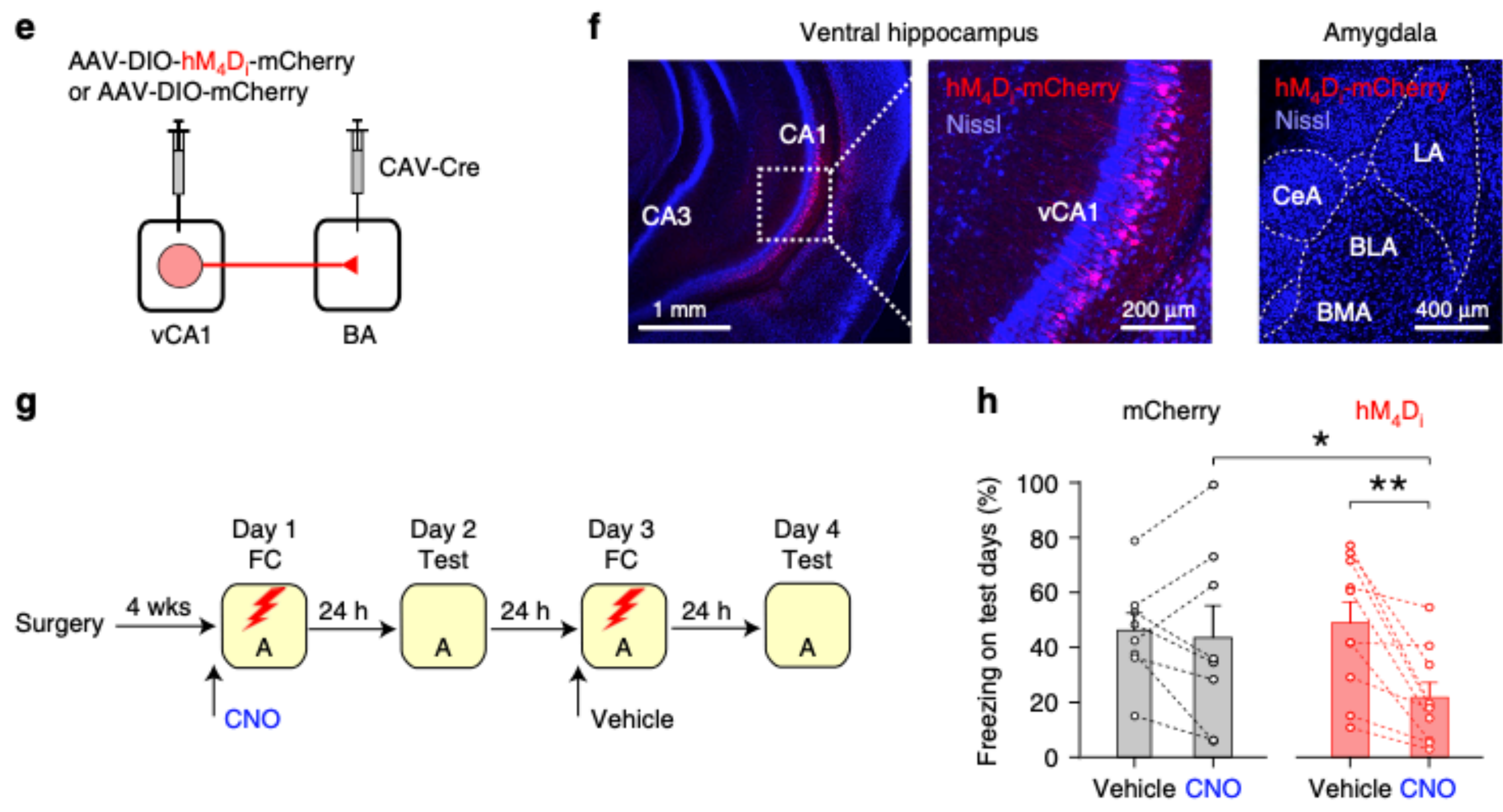
Designer Receptor Exclusively Activated by a Designer Drug



CAV-cre: Canine adenovirus used for the retrograde transduction
DIO for Double Inverted Open reading frame

Emotional memory - Brain regions involved: the hippocampus

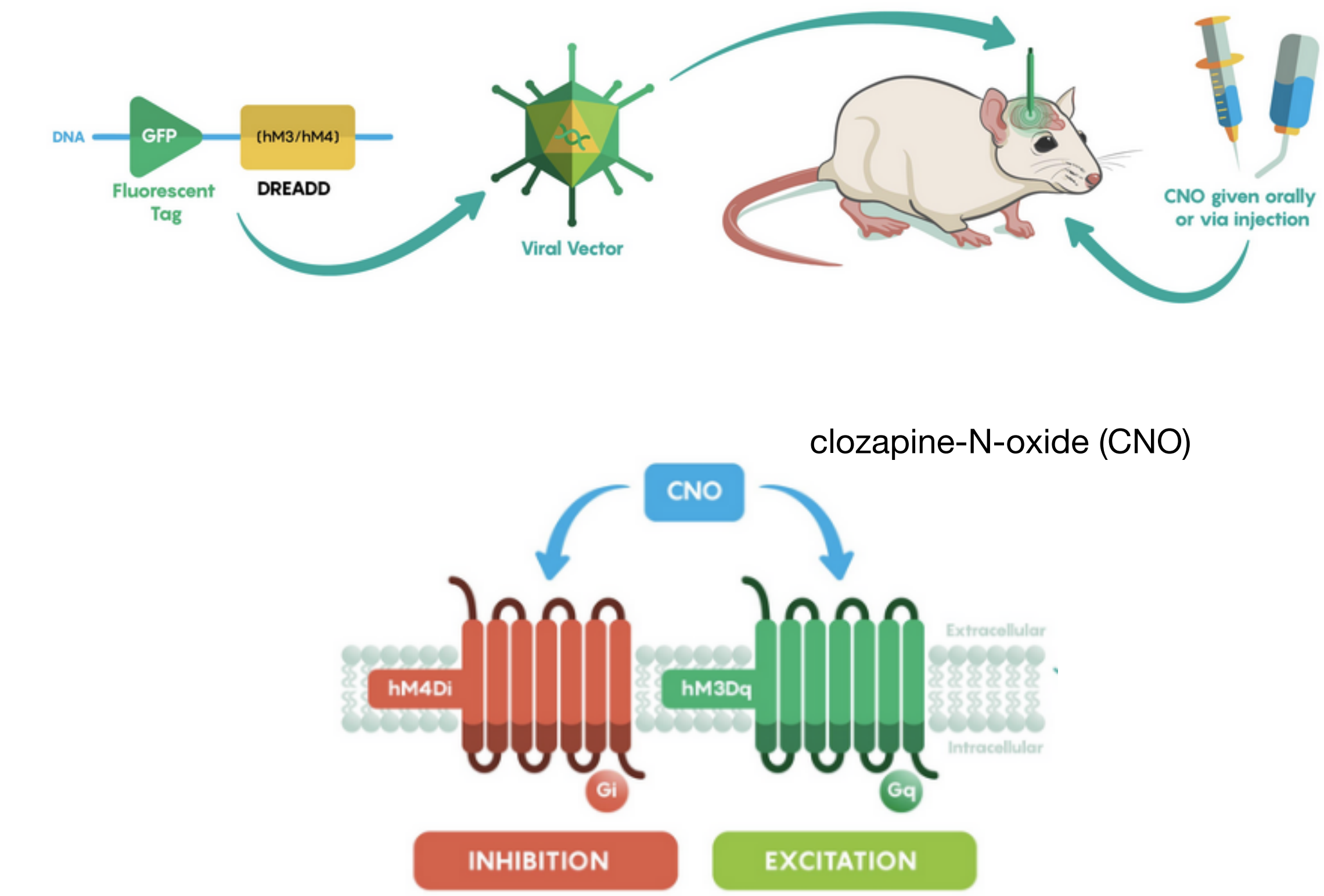
Is the activity of the vCA1 → BA pathway required for the acquisition of contextual memory?



vCA1 → BA direct pathway inhibition impaires long-term contextual memory

Technique - DREADDs

Designer Receptor Exclusively Activated by a Designer Drug

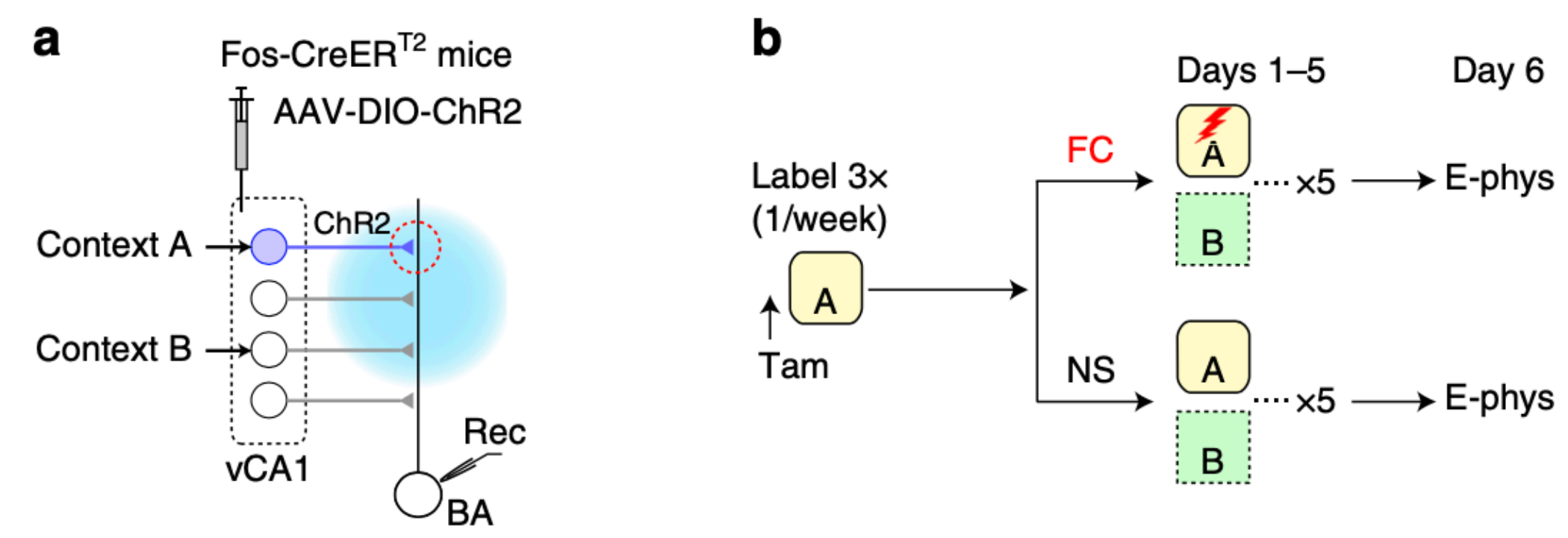


Emotional memory - Brain regions involved: the hippocampus

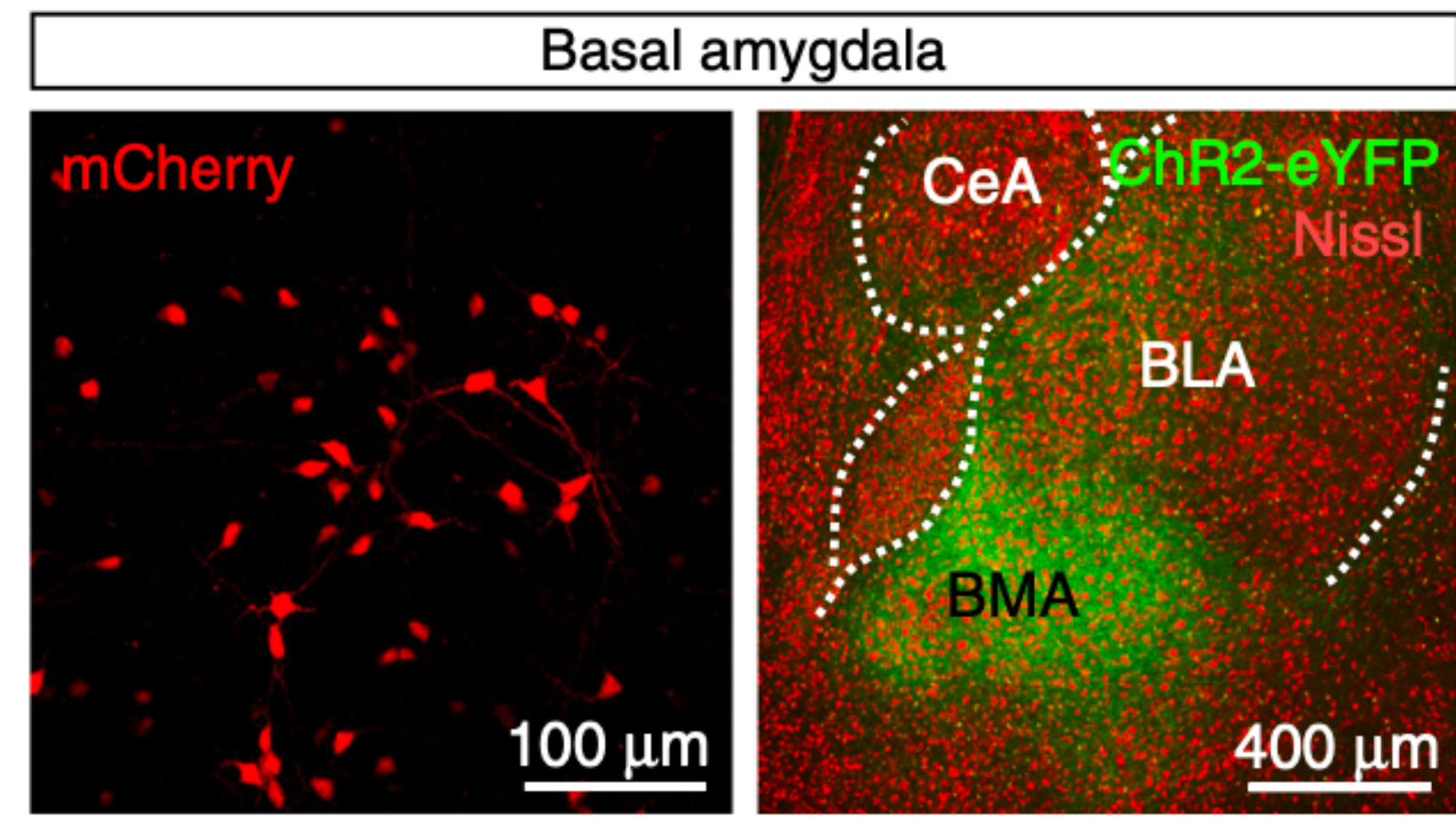
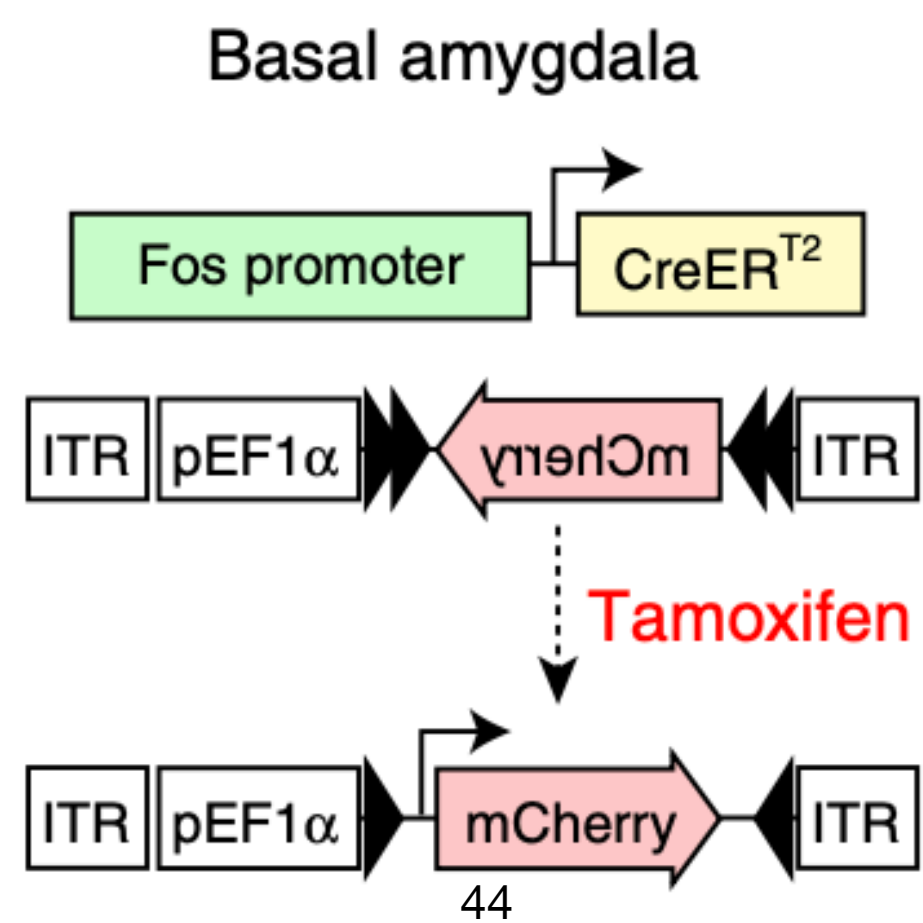
Is the activity of the vCA1 → BA pathway involved in contextual memory acquisition leading to synaptic plasticity?

vCA1 → BA plasticity

ChR2 is expressed in vCA1 activated in Context A



Technique - Genetic induction

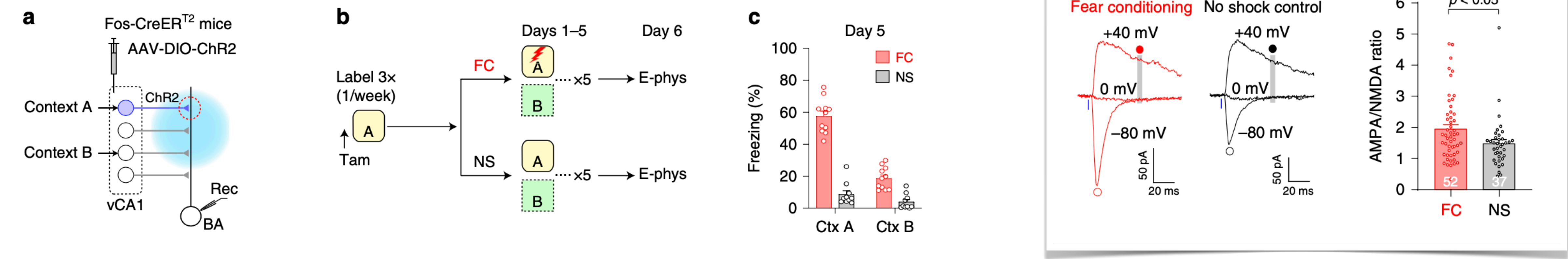


Emotional memory - Brain regions involved: the hippocampus

Is the activity of the vCA1 → BA pathway involved in contextual memory acquisition leading to synaptic plasticity?

vCA1 → BA plasticity

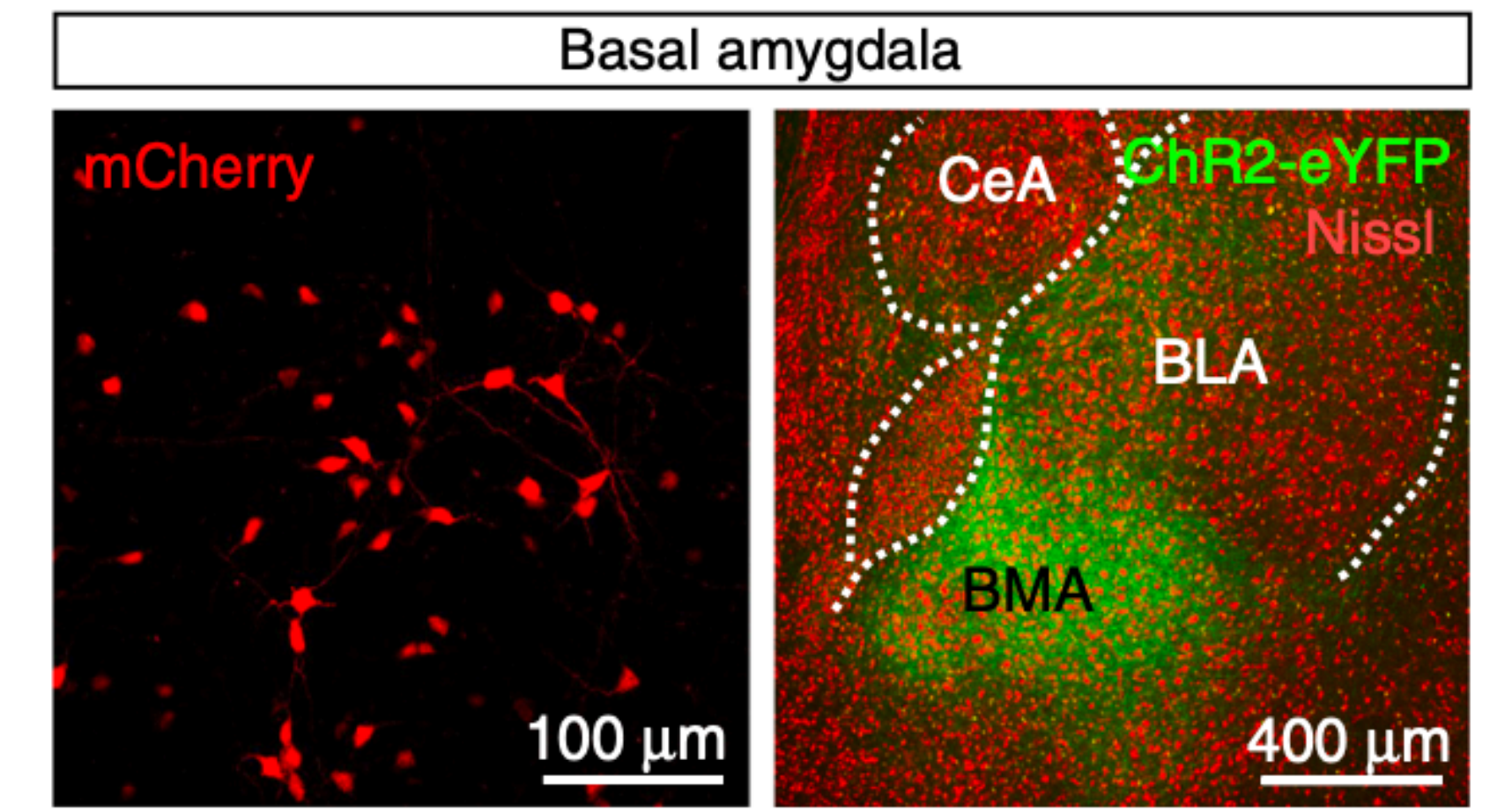
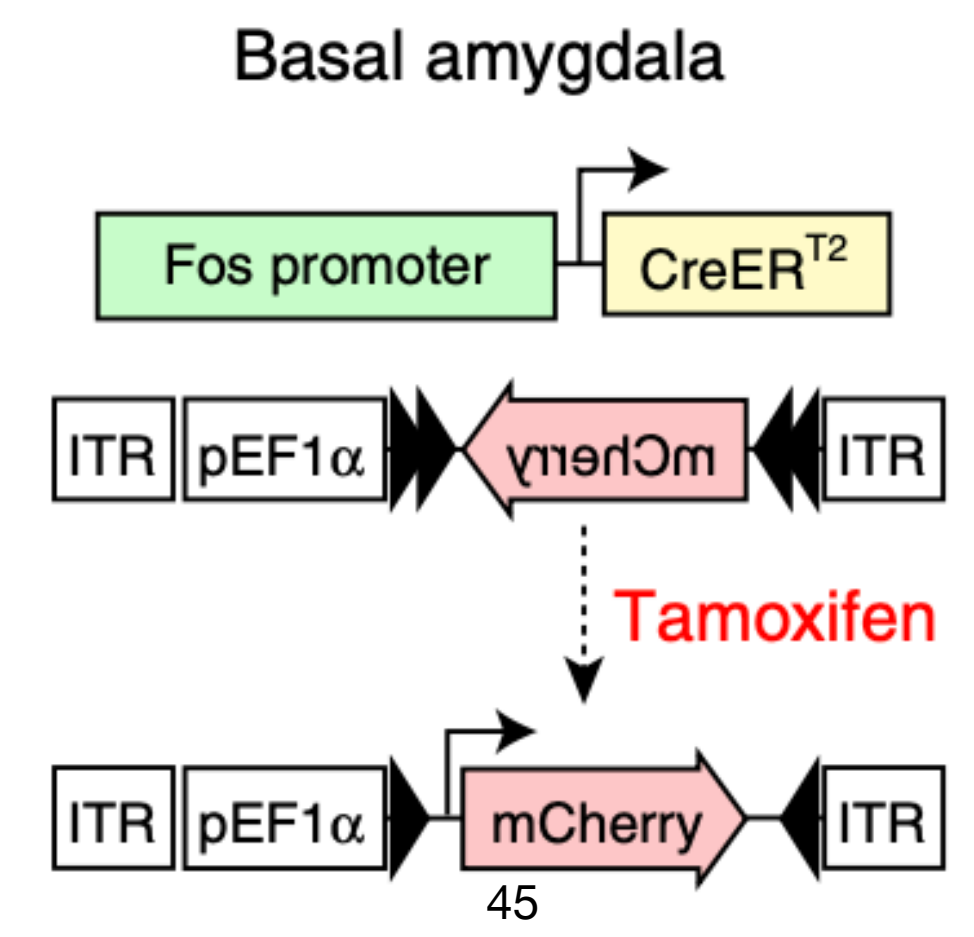
ChR2 is expressed in vCA1 activated in Context A



Technique - Genetic induction



Fos-CreERT² mice

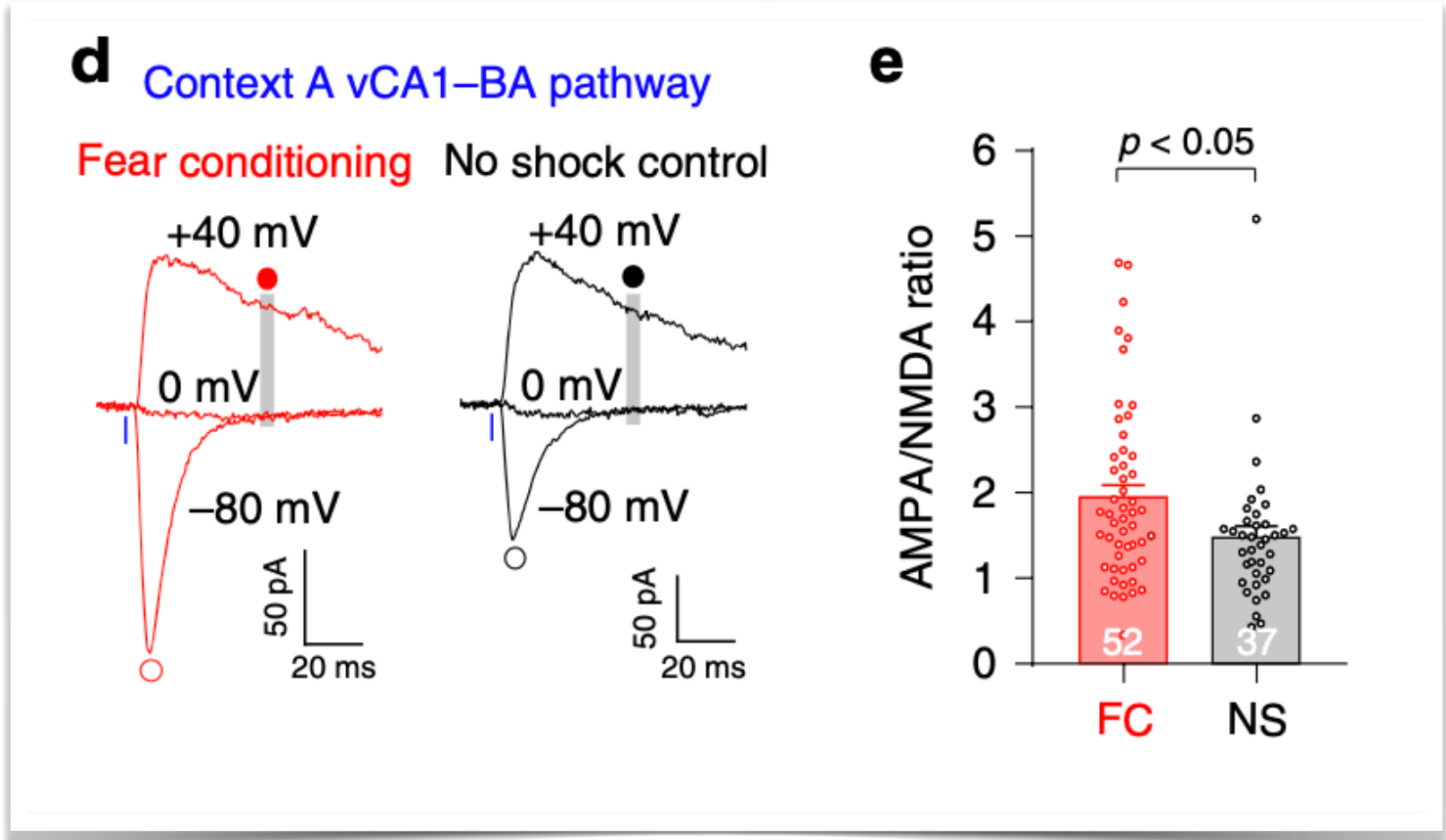
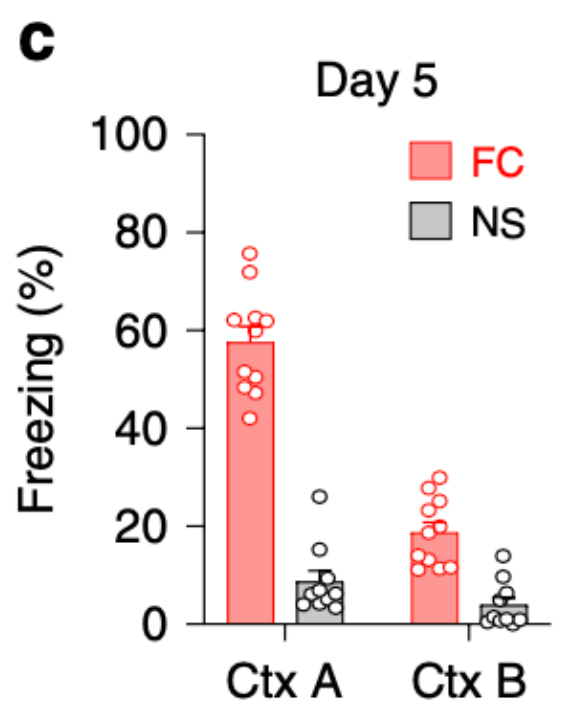
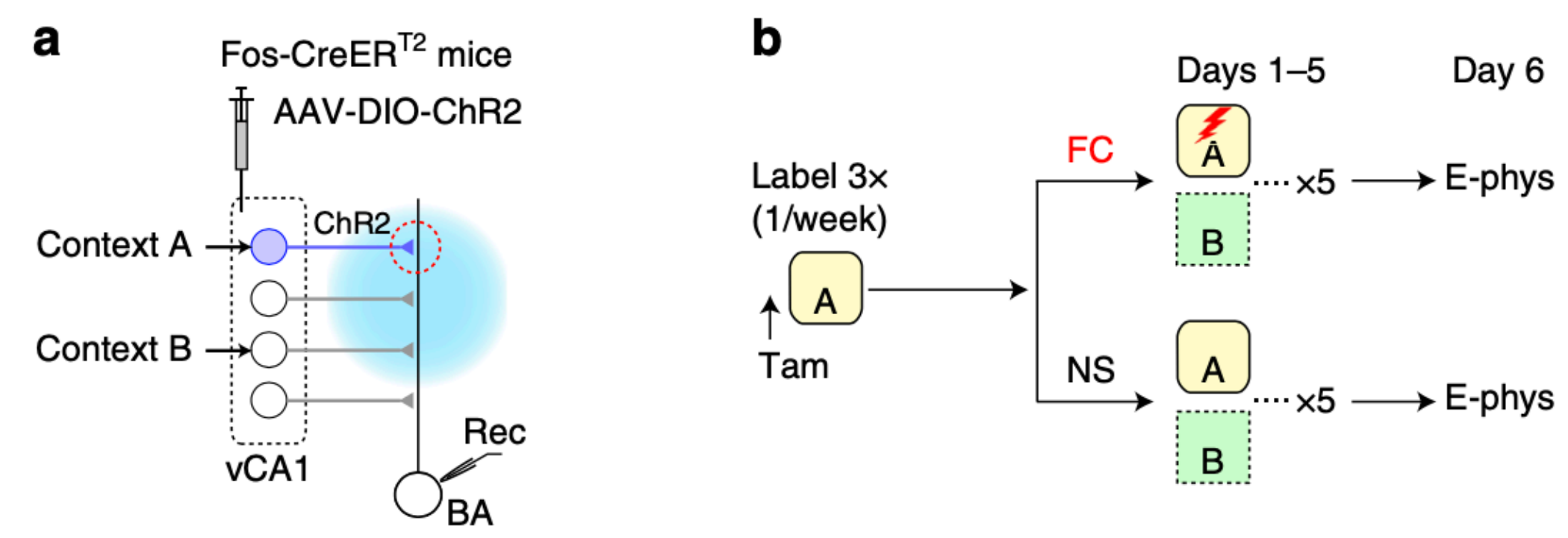


Emotional memory - Brain regions involved: the hippocampus

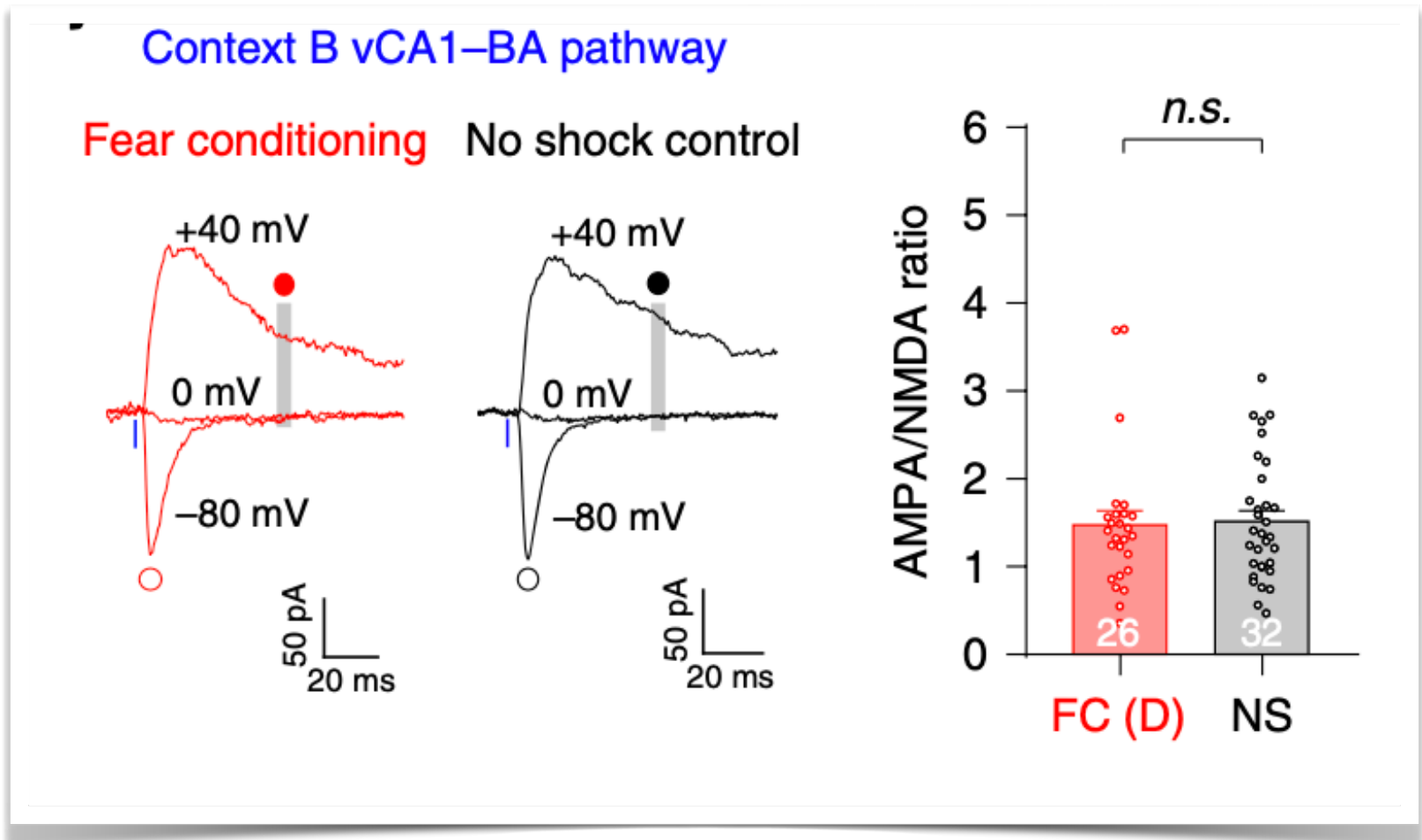
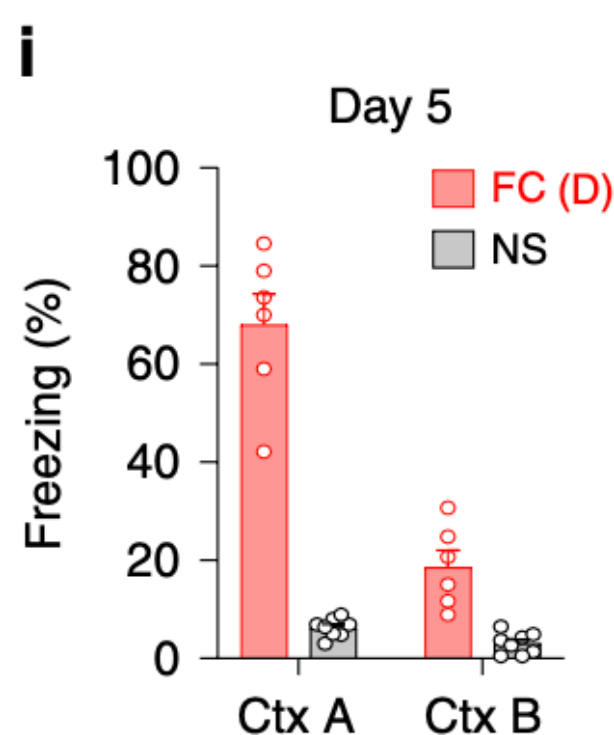
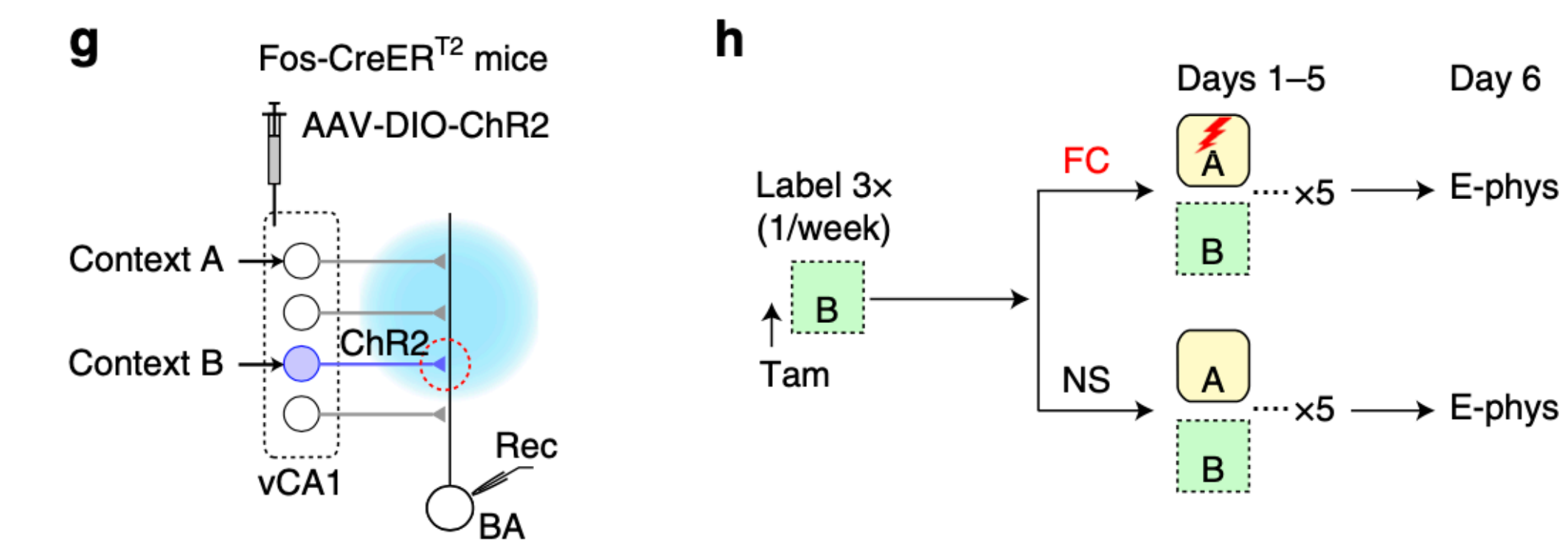
Is the activity of the vCA1 → BA pathway involved in contextual memory acquisition leading to synaptic plasticity?

vCA1 → BA plasticity

ChR2 is expressed in vCA1 activated in Context A



ChR2 is expressed in vCA1 activated in Context B

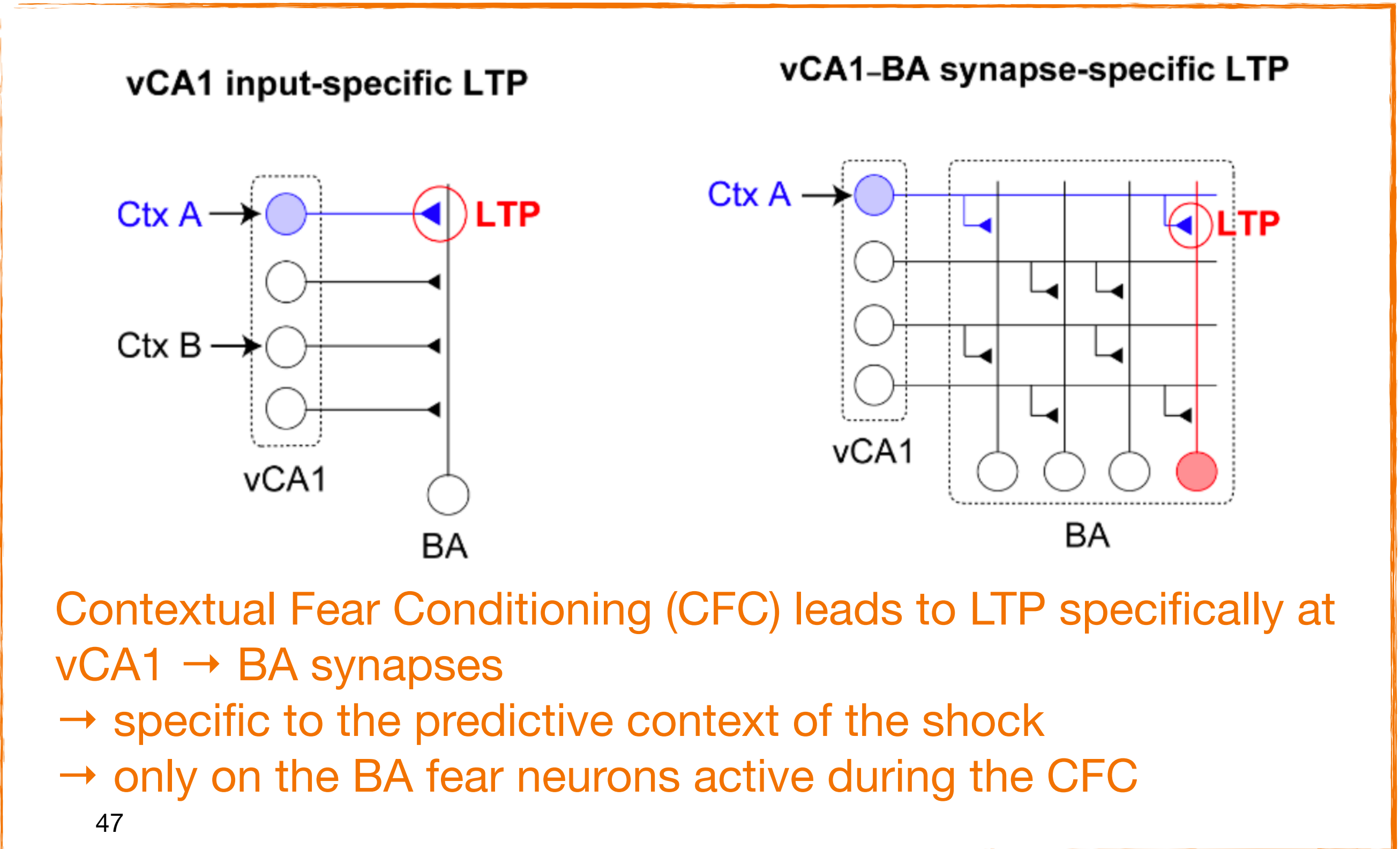
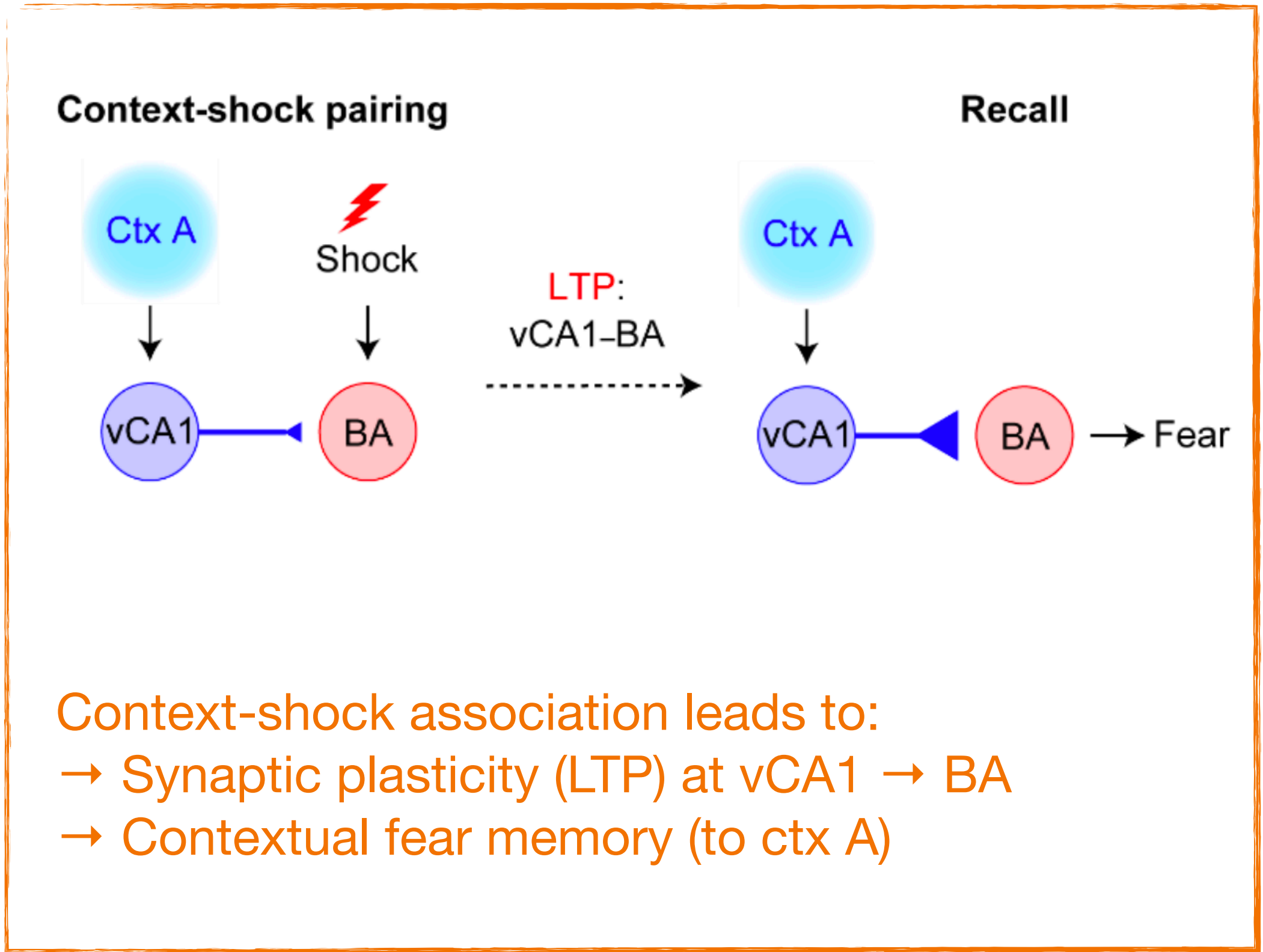
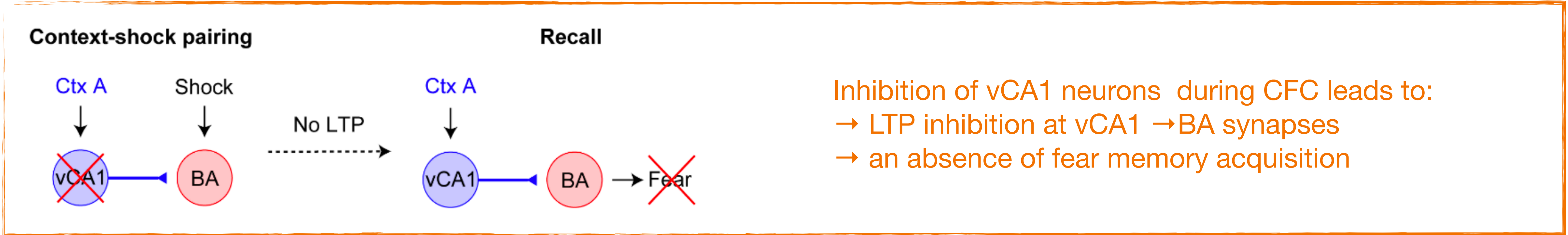


Synaptic plasticity is specifically induced at the vCA1 → BA pathway

The acquisition of a contextuel memory induces synaptic plasticity specific to the network involved

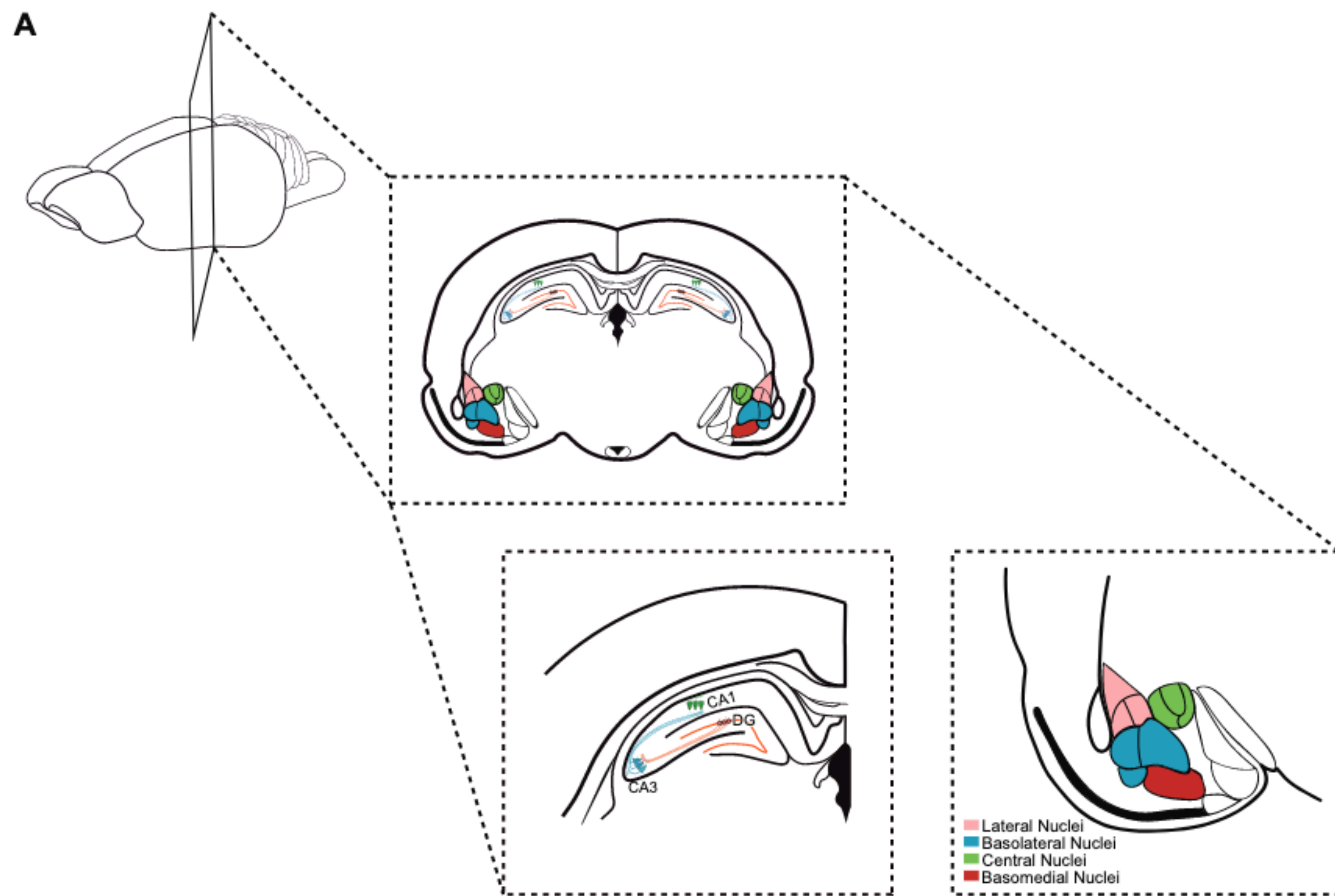
Emotional memory - Brain regions involved: the hippocampus - CCL

from Kim & Cho, 2020, Nature Com11:1382



Emotional Memory - Conclusions

Brain regions involved : the amygdala, the hippocampus...



- Identification of memories in animals
- Study of associated brain regions
- Characterisation of the neural bases

→ Understanding the dysfunctions occurring in various memory pathologies

